Evidence and Evaluation for Recursive Self-Improvement in Science

A Critical Survey

Scientific Validity, Inheritance, and Correction

Abstract

Scientific AI systems can retain data, tools, parameters, and procedures, yet improved outputs alone do not identify what these retained changes contribute or whether their scientific support survives correction and transfer. This critical survey examines these questions across life sciences, chemistry, materials, energy, and environmental science, with computational and robotic comparison cases. We distinguish scientific-output validity, feedback use, retained-capability benefit, improver effectiveness, and recursive attribution. A scientific update contract links each claim to its evidence, inherited artifact, actual consumer, comparison, and correction dependencies. A versioned configuration map separates reported outcomes from comparison design, uncertainty, and source access. Worked cases show when an interpretation should be revised, when narrower support remains, and where unknown consumption prevents further inference. Cross-domain synthesis connects acquisition, measurement validity, compatibility, and confirmation delays to evaluation choices. Common-start comparisons and a proposed randomized-formation protocol distinguish conditional improver advantages from the effects of enabling a named return path. The resulting synthesis and reporting resource support the design and interpretation of bounded scientific-improvement claims. The panel is purposive and AI-assisted; independent coding reliability, comparative usefulness, and experimental validation of the proposed designs remain open.

Keywords: recursive self-improvement; AI for Science; autonomous discovery; scientific agents; evidence provenance; experimental design.

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1 Introduction: From Scientific Feedback to Inherited Research Methods

Robot scientists connect hypotheses to experiments (King et al., 2009), mobile chemical platforms adapt experimental selection (Burger et al., 2020), and scientific agents coordinate literature, code, tools, and reports (Boiko et al., 2023; Lu et al., 2024; Yamada et al., 2025; Lu et al., 2026a). These systems increasingly retain the products of one investigation for use in another. The resulting evaluation problem has two parts: *does that inheritance improve later research, and does the evidence supporting it remain valid when its source or operating conditions change?*

The answer depends on the claimed change. Battery optimization can improve a charging protocol under a fixed acquisition procedure (Attia et al., 2020). Molecular discovery can reuse a predictor or experience package (Koscher et al., 2023a; Zhang et al., 2026c). A stronger claim concerns the procedure that generates or accepts later updates; a recursive-origin claim asks whether a return path helped produce that procedure's benefit. These require different comparisons. Our unit is therefore a versioned configuration, phase, and claim.

Scientific evidence makes inheritance especially consequential. A valid negative result, failed execution, and misleading proxy warrant different updates. A-Lab's corrected synthesis account illustrates the need to revise claims derived from earlier measurements (Szymanski et al., 2023, 2026). Yet a correction reaches an inherited method only through its actual consumption history; independent support may preserve some conclusions. The missing link is often neither a new optimizer nor another benchmark score, but an identifiable connection between evidence, retained state, and later use.

Observed progress is heterogeneous. Genuine and permuted feedback produced similar performance in tested gene-perturbation replays, whereas Cell Painting replay reported a genuine-feedback benefit (Gupta et al., 2025; Wainrib et al., 2026); differences between their configurations prevent attributing this contrast to a single cause. STOP reports both improving and degrading improvers, MetaRSI a bounded common-start advantage, and AIDE^2 an inconclusive ignition test (Zelikman et al., 2024; Tan et al., 2026; Srikanth et al., 2026). DGM's frozen-initial-modifier control tests a relevant return path, with unequal realized costs (Zhang et al., 2025b). The synthesis must explain what each comparison identifies, including unfavorable and unresolved results.

Existing accounts of RSI evaluation, scientific agents, and adaptive workflows provide essential distinctions among task performance, persistent adaptation, process revision, and provenance (Chen et al., 2026c; Han et al., 2026; Luo et al., 2026; Wei et al., 2025; Stage et al., 2025; Alnasir, 2026; Souza et al., 2025). Controlled knowledge updating also connects evidential strength and persistence to the choice and subsequent maintenance of an update (Chavira Juárez et al., 2026). We focus on a more specific analytical connection: how the scientific meaning of an improvement comparison depends on the validity and actual use of the evidence supporting inheritance. Resolving this question requires both a trace of relevant evidence use and a comparison matched to the claim.

Questions guiding the survey

1. What is claimed to improve, what is retained, and which later operation actually consumes it?
2. Which comparison and scientific endpoint distinguish that claim from additional information, search, or a changed measurement?
3. When evidence or conditions change, which conclusions should be retained, restricted, revalidated, or withdrawn, and where is the consumption history unknown?
4. Does the evidence concern a selected improver, the process that formed it, or a sustained rate of improvement?

This survey makes three contributions. First, it organizes scientific improvement claims by the evidence needed to support them, connecting scientific validity to inheritance, actual consumption, and claim-specific comparisons. Second, it provides a versioned cross-domain evidence map in which source reports, analytical judgments, and unresolved information remain distinguishable. Third, it develops evaluation guidance for separating retained-capability benefit, conditional improver effectiveness, and the contribution of a recursive return path during improver formation. The scientific update contract connects these contributions to a practical question: which conclusions remain warranted when supporting evidence or operating conditions

change? Worked cases illustrate this analysis without claiming comparative superiority, independent coding reliability, or scientific-agent recovery.

The five scientific domains expose complementary constraints: biological context, chemical execution, revisable materials characterization, delayed energy endpoints, and environmental nonstationarity. Computational and robotic cases provide additional process comparisons. Sections 2–4 establish scope and analytical rules; Sections 5–10 examine and synthesize evidence; Sections 11–13 develop evaluation and research priorities. The purposive panel supports these bounded comparisons rather than prevalence estimates. Section 2 and Appendix E document access, coding, and validation limits.

2 Scope, Review Method, and Intellectual Positioning

2.1 Scope and unit of analysis

We examine life sciences, chemistry, materials, energy, and environmental science through configurations exposing feedback use, inherited capability, process revision, or a scientific-validity constraint. Environmental coverage emphasizes atmospheric and Earth-system inference; energy coverage emphasizes devices, experimentation, and control. Robotics and computational research supply comparisons in which acquisition or improvement procedures can be inspected directly. These choices provide contrasting evidence conditions, not an exhaustive inventory of science or a ranking of disciplinary maturity.

The unit is a *versioned configuration, phase, and claim.* Several comparisons from one system are not independent studies. Background models explain a scientific interface, whereas a panel configuration must identify an inherited artifact, feedback intervention, process comparison, or validity boundary. Inclusion does not certify RSI. Positive, degraded, and inconclusive outcomes are eligible under the same rule.

2.2 What is inherited from prior work, and what is added

General RSI reviews already distinguish task performance, retention, and improved improvers (Wu et al., 2026; Chen et al., 2026b,c). Physical-AI surveys develop functional learner/improver roles, common-start and common-data comparisons, interactive acquisition, transfer, and resource accounting (Han et al., 2026; Luo et al., 2026). Scientific-agent surveys address evolving hypotheses and experimental agents; an autonomy-centered account explicitly discusses delayed and non-identifying scientific feedback (Wei et al., 2025; Zhang et al., 2025c; Duan et al., 2026).

The closest operational traditions are equally important. PROV represents use, derivation, revision, and responsible agents; truth maintenance distinguishes a lost justification from loss of all support (Moreau and Missier, 2013; Buneman et al., 2001; Doyle, 1979). Adaptive-workflow research records process changes and decision provenance (Stage et al., 2025; Alnasir, 2026; Souza et al., 2025). SkillFoundry packages scientific procedures with scope, dependencies, sources, and tests; Beaver preserves evidence provenance in scientific curation (Shen et al., 2026; Zhang et al., 2026b). Laboratory capability frameworks also distinguish learning from other dimensions of robotic proficiency (Salazar-Villacis and Benyahia, 2026). Controlled knowledge updating relates evidence, persistence, update substrates, and later maintenance (Chavira Juárez et al., 2026). Our use connects these concerns to scientific endpoints and the distinct comparisons required for feedback-use, retained-benefit, improver-gain, and recursive-origin claims. Neither structured memory, agent provenance, nor evidence-sensitive maintenance is new here; the added usefulness of this analytical specialization remains to be evaluated.

The central addition is a *joint analysis of which improvement claims remain warranted when scientific evidence or its use changes.* A process comparison alone does not establish that its endpoint remains scientifically meaningful; a derivation graph alone does not identify the effect of the process that generated an updater. The scientific update contract links those questions. It can be implemented using existing provenance representations, rather than requiring a competing ontology or new causal-identification theorem.

2.3 Shared-case analysis of the added information

Table 2 holds the published case fixed and asks what is learned by combining claim-specific evaluation with evidence validity and actual consumption. Its columns describe analytical questions, not experimental arms or independent assessor results. Prior frameworks are not assumed incapable of representing the additional information.

Table 1. Established foundations and this survey's analytical use.

Foundation	Already available	Operation in this synthesis
RSI evaluation	Claim-specific controls, common starts, full runs, transfer and costs.	Attach each comparison to the scientific endpoint and evidence version that warrant its interpretation.
Scientific agents and physical AI	Conditional validity, acquisition, persistent state and process revision.	Compare configurations by evidence conditions that determine which control is informative.
Provenance and truth maintenance	Use/derivation relations, revisions and alternative support.	Trace a changed premise to an improvement claim; stop at unknown consumption and preserve separately supported claims.
Adaptive workflows and skill libraries	Executable procedures, applicability conditions, tests and histories.	Distinguish repairing an output or library entry from showing that a better update procedure was formed.
Controlled knowledge updating	Evidence-sensitive choice of persistent state, with scope, revision and maintenance.	Connect update maintenance to scientific endpoints and the improvement claim that a comparison can support.

Note. An application-specific extension is not a demonstrated limitation of the earlier framework.

Table 2. Same-case analysis of evaluation and evidence maintenance.

Case	Narrower question	Additional information and resulting decision
A-Lab correction (Szymanski et al., 2023, 2026)	A changed assignment requires revising the published outcome summary.	Consumption of affected assignments by later learning states is unknown. Revise the summary without inferring damage to every downstream recipe or updater.
RoboChem-Flex (Pilon et al., 2026)	Prior-campaign data were retained and reused.	The inspected description does not identify transfer benefit. Withdraw that interpretation while retaining the separately supported yield and reuse record.
MolRSI memory (Zhang et al., 2026c)	A prior-memory package improves the stated target comparison.	Both arms adapt within the target. Preserve the package result; a separate wet-experimental success does not identify its effect or an isolated improver gain.
AlphaEvolve meta prompts (Novikov et al., 2025)	Disabling a co-evolved prompt component changes the tensor-decomposition outcome.	The intervention concerns a process package, not a common-start comparison of selected old/new improvers. Retain the process evidence without assigning that distinct conditional-improver estimand.
ScienceRSI transfer (Xu et al., 2026)	Verified transfer is compared with alternative learning or routing choices.	The results concern the evaluated transfer step. Cross-scale transfer and a framework diagram do not supply an observed sequence of improving improvers.

Note. These source-based analytical applications illustrate the synthesis, not its superiority or coding reliability.

Three decisions recur: *replace* an interpretation whose support changed; *retain* a narrower statement with remaining support; or *leave a downstream effect unresolved* when consumption or a comparator is missing. Section 10.7 makes these decisions inspectable at the publication and review-artifact levels. The contribution is the explicit connection between scientific support and improvement-claim interpretation, not demonstrated gains in reader accuracy or efficiency.

2.4 Retrieval, selection, and access

This is a seed-driven critical survey with a purposive panel. The historical query log and screening denominator were not retained, so the original corpus cannot be represented as a reproducible database census. The scientific-literature cutoff remains 2 November 2026.

In the R4 evidence revision on 9 November, a bounded coverage check executed six query families spanning scientific inheritance, biological feedback controls, chemical/material correction and transfer, energy optimization, environmental acquisition, and workflow provenance. The supplement retains returned entries, title/snippet screening, selected primary-source checks and dispositions. Review-triggered checks of AlphaEvolve and ScienceRSI are distinguished from query retrieval. This repairs identified omissions and documents candidate handling; it neither reconstructs the historical search nor establishes completeness across five disciplines. Appendix E specifies the rules and access gaps.

The R6 positioning review added the pre-cutoff account of controlled knowledge updating (Chavira Juárez et al., 2026) as contextual literature. It did not add panel configurations or reverify the 53 claim cards.

Versioned primary sources support effect claims. Secondary indexes are discovery routes, not substitutes for verification. Source access and the fields checked are recorded separately: historical full-text extraction is not upgraded by an abstract-only visit. A control not visible in an abstract remains undetermined. Corrections are linked to the affected source, not counted as independent experiments; later versions do not silently replace earlier evidential records.

2.5 Coding, checks, and validation limits

Coding fixes the boundary, learner-update interface and phase before naming the artifact, consumer, intervention, endpoint and resource rule. The compact map reports direction or mechanism state. A companion main-text design panel supplies comparison type, independent-run unit, confirmation boundary and uncertainty; cell records supply locators. Not extracted, inaccessible, not reported in inspected scope and inapplicable remain separate states.

The evidence resource includes source-based analytical applications, a configuration-level evidence-regime crosswalk, and an executable check of which synthesis statements depend on particular sources. R6 improves their conceptual positioning and use as reporting guidance; the underlying case-check scope remains that of the R4 evidence release. These are AI-assisted analyses, not independent human ratings, experimental replications or estimates of field prevalence. Source-removal analysis tests dependence of the assembled argument, not publication bias or search completeness. Independent assessors have not applied the frozen rules to held-out configurations; comparative reader accuracy and efficiency remain unmeasured.

Access and preservation. The edition is `RSI-Science-R6-2026-11-09`. The PDF embeds the evidence and revision bundle; the companion source archive supplies the manuscript, bibliography, code and source manifest. Earlier revision records are retained with their original status. No public repository or DOI has been assigned. Executable checks establish record consistency, not truth of the underlying scientific results.

3 A Framework for Cumulative Scientific Improvement

3.1 Claims, System Boundaries, and Functional State

3.1.1 The system boundary precedes the recursion claim

Declare the system's components, permitted modifiers, persistent state, and externally supplied capabilities. The boundary may enclose a model, agent, laboratory, or research collective, but must remain stable within a comparison. A frozen model can participate in improvement through retained tools or procedures (Zhao et al., 2023; Wang et al., 2025); an external model upgrade does not establish self-generated improvement. Human

diagnosis and external measurements remain visible inputs, even when they fall inside a larger organizational boundary. Report alternative boundaries when they change attribution rather than expanding the "self" until all progress becomes self-generated.

3.1.2 Three functional state roles

We distinguish scientific knowledge K, research capability C, and the improver U that generates, evaluates, or retains capability updates. These are roles defined by use, not disjoint modules or a sufficient dynamical state. A fitted surrogate used for prediction is ordinarily learner capability (Snoek et al., 2012; Attia et al., 2020); a retained rule governing how later capabilities are constructed can serve as an improver. Code, weights, and text can implement either role. Shared parameters may prevent component interventions, in which case the identifiable comparison concerns the compatible package.

Table 3 separates five claim families. Within feedback use, *consumption* means that a recorded observation was read or executed at a specified decision point; *influence* concerns whether changing that information changes decisions or outcomes. A trace can establish consumption without identifying influence. Mechanism and effect evidence remain separate throughout the configuration map.

Table 3. Five claim families and their evidential obligations.

Claim	Evidence needed	Main alternative
Scientific output	Claim-appropriate validation of a result.	Duplicate findings, measurement ambiguity, or a changed endpoint.
Feedback use	Consumption: a versioned read or execution trace. Influence: a valid intervention on feedback content, admission, availability, or targeting.	Availability without consumption, or an outcome change attributable to a bundled intervention.
Retained-capability benefit	A retained artifact benefits later decisions or fresh instances under a relevant comparator.	Mere reuse, more attempts, or an external upgrade.
Improver gain	A later improver produces better updates from compatible starts; report each arm's absolute gain.	Inherited solutions, unequal budgets, or a changed evaluator.
Recursive attribution	Intervention on the return path identifies its contribution to the specified benefit.	The same gain arises from the fixed external search or another changed component.

Note. A positive result in one row does not establish another.

In the map's feedback column, `M` records consumption; directional symbols summarize the intervention's result on its named endpoint. An inconclusive utility comparison neither proves non-use nor establishes that every intermediate decision was unchanged.

Persistence can be tested on fresh instances within a campaign after restoring a compatible research state; cross-project or laboratory transfer supports a broader claim. Knowledge accumulation can increase utility while leaving the research method unchanged. The five claim families are therefore neither a maturity ladder nor a cumulative score.

3.1.3 Self-correction, self-improvement, and recursive feedback

Self-correction revises an output within a task (Madaan et al., 2023). **Persistent self-improvement** is a system-involved change retained with demonstrated benefit to later research. **Recursive feedback**, used descriptively, is a return path carrying retained changes into subsequent learning. A path may exist without revising the learning procedure or improving its outcomes.

Recursive process revision requires a system-generated change to an acquisition, diagnosis, proposal, evaluation, update, or acceptance procedure that is retained and used in later learning (Srikanth et al., 2026; Zhang et al., 2026a). **Improver gain** compares the effectiveness of such procedures; **recursive attribution** identifies a named return path's contribution. A fixed external optimizer can improve a system repeatedly, and

a recursively connected system can deteriorate. We preserve authors' terminology while coding the supported claim.

3.1.4 An explicit coding procedure and boundary sensitivity

Use the following procedure before inspecting the direction of the result. Its purpose is to make interface choices explicit and repeatable, rather than to assign a unique recursion label to every representation.

1. **Fix claim, boundary, and phase.** Name the later capability whose improvement is claimed, its retention horizon, and the actors permitted to modify it. Separate construction, adaptation, and deployment when permissions differ.

2. **Fix the learner-update interface.** Record the starting artifact type, allowed information, output type, and subsequent consumer. Use the operation that produces the claimed retained capability as the primary interface; do not switch to an internal decision module because its parameters changed.

3. **Separate state from update rule at that interface.** Write $A^+ = F_\theta(A, D)$ for learner artifact A, available information D, and rule parameters θ. New observations or a refitted A under unchanged F_θ are state updates. A retained change to the mapping that generates, evaluates, or accepts A^+ is a candidate process revision. Storage format and a changed action alone do not decide the code.

4. **Trace production and consumption.** Identify who produced the retained object and an event in which it governed a later update. Repository availability is insufficient. Fixed training code can generate a revised effective rule; all outer components need not change.

5. **Assign evidence and resolve ambiguity.** Record consumption separately from intervention-supported influence, and process revision separately from gain. If two defensible interfaces yield different roles, retain both readings and identify the conclusion that changes. If an intervention breaks compatibility, report a package effect and leave component attribution unresolved.

Table 4 applies the procedure to three difficult configurations, varying the partition while keeping the reported observations fixed. These AI-assisted recodings test the consequences of declared choices; independent coding reliability remains to be measured.

Table 4. Boundary and phase sensitivity in three configurations.

Artifact and consumer	Two declared partitions	Sensitivity result
Attia et al.: fitted battery surrogate consumed by acquisition (Attia et al., 2020)	Campaign view: observations and fitted posterior are A, with fitting/acquisition in F_θ. Decision-module view: the posterior is a parameter of the next-experiment decision map.	The decision map changes, but the reported fitting/acquisition procedure does not thereby change. "Rule changed" is ambiguous unless its interface is named. Neither reading establishes improver gain.
RoboChem-Flex: prior-campaign data consumed by a later multi-task GP (Pilon et al., 2026)	Platform view: the inherited dataset is D, the GP is learner state A, and fitting/acquisition is fixed. Predictor-only view: the reused dataset is an externally supplied input; the platform that collected it is outside the boundary.	Retention and later consumption are visible at the platform boundary; self-origin is not attributable to the predictor alone. In both readings, no matched no-transfer contrast was identified in the inspected main text, so reuse does not establish benefit.
AIDE^2: AIDE_{47} harness promoted to outer improver (Srikanth et al., 2026)	Task-solving phase: the harness is research capability C. Ignition phase: it is U because it proposes edits to a common initial inner harness; the outer harness is fixed within each comparison arm.	The same artifact changes role when its consumer changes. Promotion and execution are documented, but the ignition comparison remains inconclusive. A $C \to U$ relabeling is not a positive gain result.

Note. The two readings refer to the same reported configuration. Role labels can change while the empirical claim boundary remains fixed. No new system experiment is reported.

Attia's case explains the interface rule: absorbing a posterior into a decision map changes the description without establishing revision of the campaign's learner-update procedure. RoboChem-Flex makes self-origin boundary-dependent; AIDE^2 makes role phase-dependent. None of these changes upgrades documented reuse to demonstrated improver gain. A fixed audit is compatible with a changing proposer, and explicit diagnosis is not required for a valid update.

Scientific update contract: an analysis and reporting unit

For one configuration and claim, record

$$\mathcal{C} = \big(\mathcal{B}, \tau, E, V, I, W, \Pi, J, \mathcal{D}\big). \tag{1}$$

The fields are boundary $\mathcal{B}$; phase and retention horizon τ; feedback provenance E; claim-relative validity V; inherited artifact and origin I; actual consumer W; intervention, comparator, unit, and resource rule Π; protected endpoint and scope J; and dependency/correction record $\mathcal{D}$. Record unresolved fields with reasons. This contract links the evidence needed for a claim; field completeness is not validation.

Use the contract to decide which inference the available evidence permits. Start with a named claim, trace the retained object to its consumer, and choose its comparison and independent unit using Figure 3. For RoboChem-Flex, the source reports a prior dataset consumed by a later multi-task GP, but the inspected material supplies no matched no-transfer contrast (Pilon et al., 2026). Filling the contract therefore preserves reuse while leaving retained-capability benefit unresolved. Appendix E, S6 supplies all nine fields, distinguishes source-reported consumption from executed review-level dependencies, and marks unextracted laboratory events explicitly. The output is a bounded claim and its unresolved test, not a completeness score.

3.1.5 Autonomy, recursion, and validity are distinct dimensions

Autonomy concerns who decides, recursion how retained changes return to learning, and validity whether the claim is warranted. Human-approved research can learn recursively; unattended search can use a fixed optimizer. Neither establishes scientific validity. Generalization is also claim-specific: improved molecular ranking, proof search, and tool repair do not establish causal reasoning, experimental judgment, or better question selection. State the target distribution and interfaces over which transfer is tested.

3.2 Evidence, Attribution, and Inheritance

3.2.1 Three coupled loops

The **research loop** generates observations and revises claims. The **capability loop** converts experience into retained tools, parameters, or procedures. A **recursive return path** carries a retained change into subsequent learning. Their interfaces perform different operations: validating an observation, interpreting its implications, and accepting a capability update.

Figure 1 specifies the information flow. A research policy acts on a question using available knowledge and tools; the execution record identifies samples, instruments, actions, and measurement conditions. Scientific revision can accept an observation, retain it provisionally, reject it, or retract a dependent claim. A separate development gate accepts, rejects, partially retains, or rolls back proposed capability and improver changes. An editable acceptance test belongs to the declared improver even when the diagram draws the gate separately.

A final audit evaluates frozen artifacts or complete runs on protected evidence. If its feedback later guides development, retire that evidence from the independent audit role or use a justified adaptive procedure. The architecture records dependencies; causal effects require the comparisons in Section 11.2.

3.2.2 Diagnosis as a conditional learning strategy

An unfavorable molecular assay can reflect inactivity, sample degradation, or assay failure, each recommending a different update. A discriminating control can then be more useful than retaining a rule against a chemical family. Diagnosis may use typed states, executable checks, causal models, or probabilities rather than verbal reflection. Its value depends on whether resolving ambiguity changes a consequential decision enough to justify measurement cost and delay.

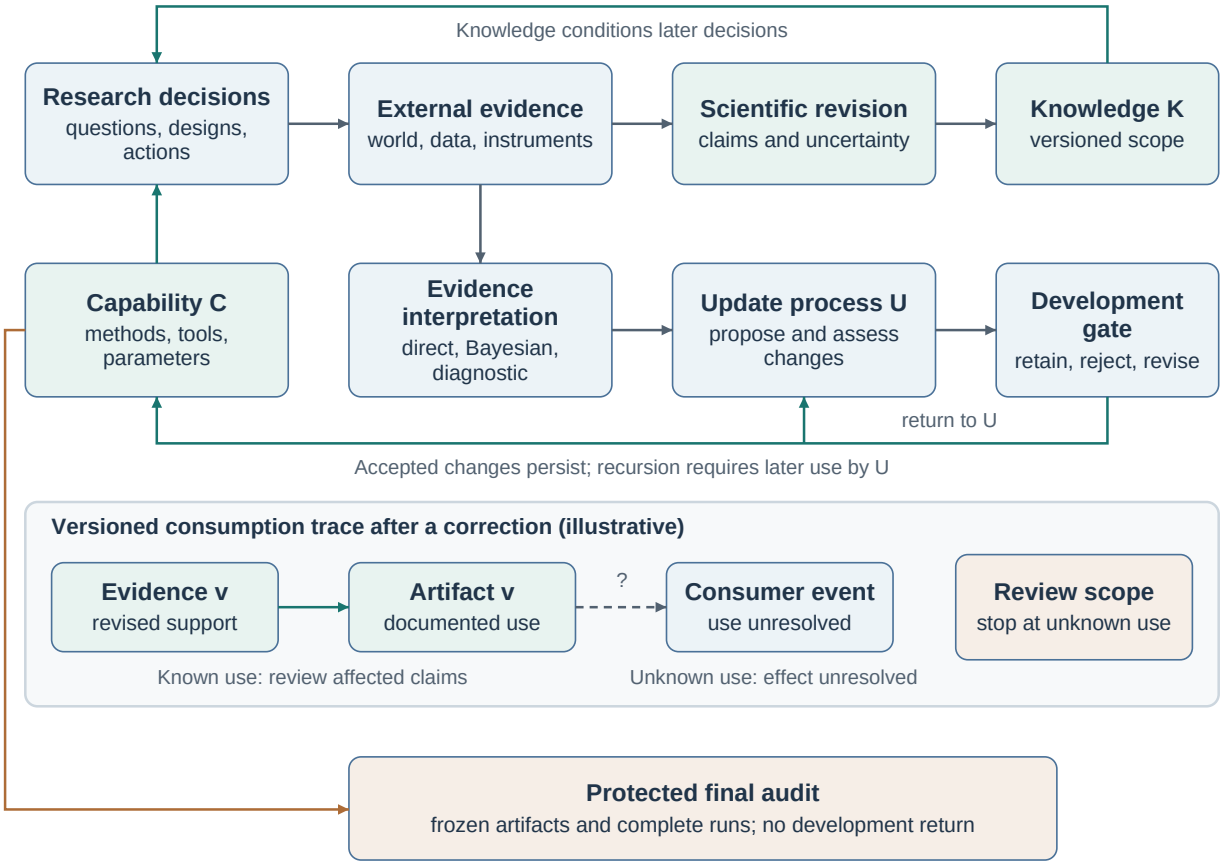


Figure 1. Evidence-to-improvement architecture and correction scope. The upper flows connect research, retained capability, and the optional return of changes to the update process. Interpretation may be direct, Bayesian, or diagnostic; explicit failure attribution is not required. The lower trace distinguishes documented consumption (solid) from unresolved consumption (dashed with a question mark). Evidence and artifact versions and the actual consumer event bound a correction review: inspect documented descendants, retain independently supported claims, and stop propagation at unknown use. Review may retain, restrict, revalidate, or withdraw a claim; a dependency does not by itself identify an update effect. The separate protected audit has no development return. These are analytical roles and an illustrative trace, not an assertion that one system implements every component.

Evidence validity, *failure diagnosis*, and *update-effect identification* answer different questions: what the measurement warrants, why the outcome occurred, and whether a change helps later research. Useful learning need not diagnose every example. Direct updating, Bayesian uncertainty, and robust policies are meaningful alternatives to explicit diagnosis and belong in its controls. A lasting causal lesson requires stronger support than a scoped predictive update.

A sound policy can also select an informative experiment whose realized result is unhelpful. Assess decision quality against the alternatives available at the time, using repeated trials, valid replay, or explicit utilities. Reflection cannot recover unrecorded causal information.

3.2.3 Evidence, knowledge, and method memory

Memory can be organized by representation, function, and lifecycle (Hu et al., 2025). Scientific memory additionally needs an epistemic type: observations record measurements under conditions, hypotheses await evidence, and methods specify procedures with a tested scope. Undifferentiated retrieval text can turn provisional explanations into apparent facts and local heuristics into universal instructions.

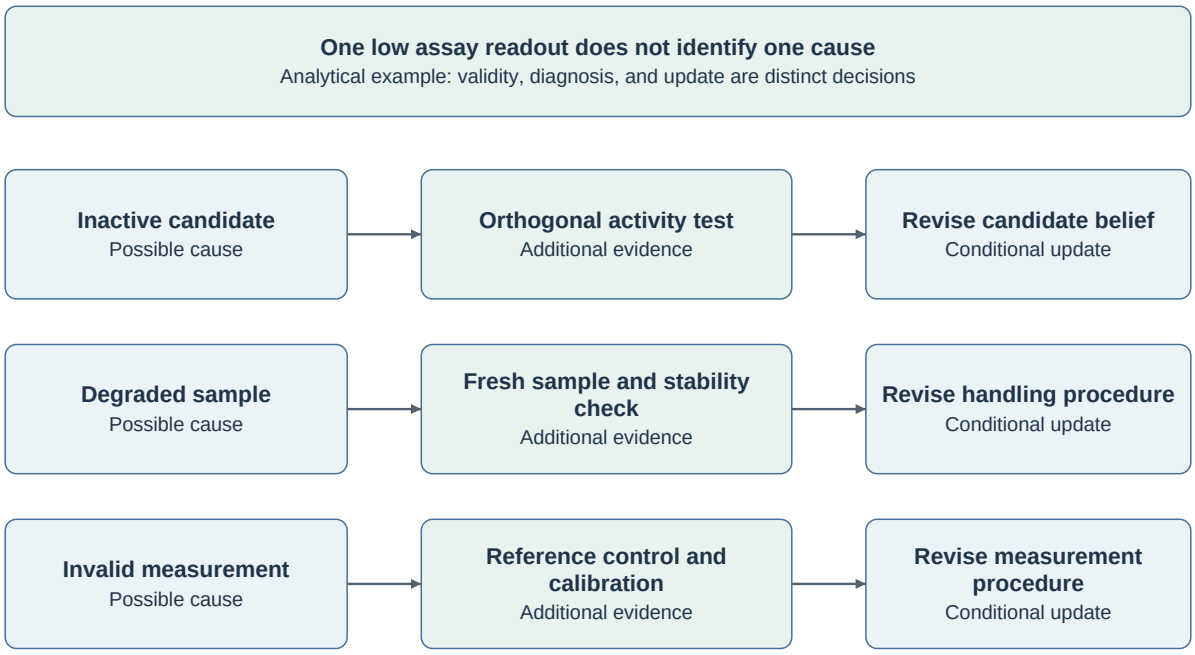


Figure 2. A hypothetical attribution problem. The same low assay readout can justify incompatible updates. Discriminating controls can help choose whether to revise the candidate belief, sample handling, or measurement procedure; unresolved explanations can instead remain probabilistic. The branches illustrate alternatives rather than diagnoses from an actual experiment; more than one cause may coexist. A later transfer test assesses the retained procedure separately from the original scientific result.

Record provenance, measurement conditions, uncertainty, dependencies, validation status, and revision triggers. For methods, include useful and failed contexts: a normalization instruction should identify the design under which it prevented batch confounding and the tested alternatives. This detail has storage and retrieval costs, which must be weighed against harmful reuse.

Truth maintenance and provenance supply the dependency principles (Doyle, 1979; Buneman et al., 2001). Their application here asks which retained artifacts consumed evidence whose status changed and which comparisons require review. A citation is not proof that a label trained a model or influenced a decision. Section 10.7 distinguishes these edges in a published correction and an executed review-artifact example. These establish scoped review obligations, not scientific-agent recovery; editing retrieved text does not demonstrate removal of its influence on parameters.

3.2.4 Multiple timescales and interface compatibility

Controller adaptation, surrogate fitting, and policy confirmation can take seconds, days, and months. A single iteration count hides latency and dependence. Instrument, model, or schema upgrades can also make a retained skill incompatible without erasing its content. Versioned interfaces and replay distinguish this from forgetting or improvement. When component swaps break those interfaces, compare complete compatible packages and leave narrower attribution unresolved.

3.3 Scientific Validation and the Scope of an Update

3.3.1 Validation must match the claim

Scientific evidence cannot be ordered along one universal ladder from weak text feedback to strong physical experiments. A machine-checked proof can be decisive for a correctly formalized theorem, while a physical experiment is irrelevant to its logical validity. Conversely, a flawless calculation establishes little about

an empirical mechanism if its model omits the relevant phenomenon. The appropriate question is what a validation procedure licenses us to conclude, under which assumptions, and with which residual uncertainty.

Table 5. Claim-relative scientific validation and update opportunities.

Feedback regime	What it can establish	Persistent update it may inform	Remaining vulnerability
Formal proof or certificate	Derivability of the encoded statement under specified rules	Proof tactics, decomposition, search policies	Wrong formalization, inappropriate axioms, specification mismatch
Executable computational checks	Correctness or performance on the executed instances	Algorithms, tools, debugging and testing procedures	Incomplete tests, hidden constraints, data leakage
Simulation	Behavior under a specified model and parameter regime	Controllers, search heuristics, candidate screening	Simulator error, omitted physics, exploitation of artifacts
Observational data	Patterns, predictions, or causal effects under explicit identifying assumptions	Measurement models, feature extraction, prediction procedures	Confounding, selection, missingness, nonstationarity
Controlled intervention	Intervention effects under randomization or a justified identification strategy	Experiment selection, protocols, mechanistic discrimination	Implementation failures, limited scope, noisy outcomes
Independent reproduction or replication	Stability of computations or findings under stated changes	Trust estimates, transfer rules, revised methodological scope	Shared data or instrumentation errors, insufficient independence

The table is a compatibility map rather than a quality ranking. The validation type and the update target must be analyzed jointly. Simulated fluid control can justify a better controller for a specified simulator without validating a real-world flow mechanism. A chemical assay can validate a candidate's activity under tested conditions without showing that the discovery system has learned a better search strategy. Formal verification can reduce uncertainty about a program's behavior while leaving scientific usefulness unresolved.

4 Mechanisms for Learning How to Discover

4.1 Foundation Models as Components of a Learning Research System

Scientific foundation models supply representations, proposals, approximate simulation, and priors, as illustrated by protein prediction, design, and materials screening (Jumper et al., 2021; Watson et al., 2023; Merchant et al., 2023). Using these capabilities does not itself revise the research process. Conversely, retained tools can improve a system built around a frozen model (Zhao et al., 2023; Wang et al., 2025). Architectures and training regimes vary, so "foundation model" is not an adaptation or evidence category (Choi et al., 2025).

Four interfaces organize their research roles: *representation* maps observations into features; *proposal* generates hypotheses, candidates, or programs; *surrogacy* predicts consequences and allocates measurements; *orchestration* selects tools and coordinates execution (Dauparas et al., 2022; Merchant et al., 2023; Kusne et al., 2020; Boiko et al., 2023). One model may implement several. The analysis follows the interface receiving feedback, the accepted change, and its later consumer.

These roles alter evidence acquisition as well as prediction. Molecular models determine which compounds reach measurement (Stokes et al., 2020; Koscher et al., 2023a), biological representations constrain proposable sequences (Abramson et al., 2024; Madani et al., 2023), and materials predictors allocate computational and experimental scrutiny (Merchant et al., 2023; Szymanski et al., 2023). Battery-lifetime prediction can accelerate selection before the final endpoint arrives (Attia et al., 2020). Consequently, performance on the evidence a system selected does not alone establish superiority over alternative acquisition policies (Kusne et al., 2020; Iwasaki et al., 2026).

Table 6 applies the update contract to mechanism targets: input evidence, acceptance, retained artifact, later use, and a claim-specific comparison. Development evidence selects updates; protected audit evidence assesses the resulting claim (Cawley and Talbot, 2010; Dwork et al., 2015b). The rows can coexist in one system and do not define exclusive method families.

For example, an assay result can refit a predictor under a fixed selection procedure (Koscher et al., 2023a). Revising the procedure that generates or accepts later predictors changes a different object. Search-space revision and successive-improver comparisons make this distinction concrete (Iwasaki et al., 2026; Srikanth et al., 2026). The interface rule in Section 3.1.4 determines which operation is being compared.

Table 6. Update contracts by mechanism target.

Target	Input, accepted change, later use, and critical comparison
Experience use	Measured outcomes or execution traces enter records or retrieval rules. A validity and relevance check governs retention; later research or update generation reads the saved material. Compare additional information with a changed memory procedure, and check for obsolete or misapplied lessons (Zhao et al., 2023; Zhang et al., 2025a).
Predictive or generative behavior	Measured or generated data enter parameter updates. An endpoint-specific validation test selects a checkpoint; later proposals or judgments use it. Compare under matched access to validated information, including retention and transfer. Parameter updates may be broadly reusable but difficult to trace or withdraw (Zelikman et al., 2022; Shumailov et al., 2024).
Evidence acquisition	Past measurements and costs inform a selector or schedule. Development validation retains a policy used to choose later observations. Compare from compatible starting states with acquisition costs included. Selecting the next measurement is a research capability; the selector is an improver component only when it governs later capability updates (Attia et al., 2020; Iwasaki et al., 2026).
Update generation	Search histories, failures, and evaluations inform a revised prompt, program, or learned modifier. An acceptance test retains a version that produces later capability changes. Trace its execution and compare earlier and later improvers from common starts; broader edit scope also raises verification and compatibility costs (Zhang et al., 2025b; Srikanth et al., 2026).
Feedback and acceptance	Outcome records and detected evaluator failures inform revised tests, reward models, or retention gates. Later search uses the accepted evaluator, while a protected audit checks the scientific endpoint independently. A higher score under a changed judge does not establish better science; stronger checks also consume evidence and expert effort (Cawley and Talbot, 2010; Szymanski et al., 2023, 2026).

Note. Each row identifies an experience-to-use path and the comparison needed to interpret its benefit.

Function and carrier impose different trade-offs. Explicit records support provenance and targeted correction at retrieval cost (Packer et al., 2023; Hu et al., 2025); parameters influence many decisions but make individual premises harder to withdraw (Shumailov et al., 2024); executable procedures expose interventions but require compatible tools and schemas (Coley et al., 2019; Steiner et al., 2019). Acquisition adaptation also pays training and confirmation costs (Kusne et al., 2020; Attia et al., 2020). The preferred carrier therefore depends on validity, expected reuse, and correction cost. The following subsections identify separable comparisons and the cases where only a compatible package can be evaluated.

4.2 Memory and the inheritance of research experience

MemGPT provides an architecture for retention and access under limited context (Packer et al., 2023); ExpeL extracts reusable language insights (Zhao et al., 2023); Agent Workflow Memory retains action sequences (Wang et al., 2025). These carriers preserve different objects: an experimental transcript records evidence, a summary encodes interpretation, and an executable workflow prescribes action. Storage alone establishes neither subsequent consumption nor benefit.

Scientific memory must preserve the distinction. Measurements need sample, instrument, protocol, and uncertainty; explanations need evidence and alternatives; procedural lessons need a tested scope. A low assay readout should not become a general instruction to reject a compound family while degradation or assay failure remains plausible. The memory can instead retain the observation, competing explanations, and a proposed discriminating test separately. Rejection, qualification, and correction are legitimate acceptance outcomes.

ReasoningBank distills strategies from trajectories judged successful or unsuccessful and combines memory with further test-time experience (Ouyang et al., 2025). Scientific use additionally requires checking what those labels mean. An inactive candidate, unsuitable protocol, and faulty instrument imply different actions. Record which memory was retrieved or executed, and distinguish the benefit of extra experience from the benefit of changing how experience is selected, compressed, or applied.

MemEvolve searches implementations of encoding, storage, retrieval, and management while experience accumulates (Zhang et al., 2025a). This exposes a possible return path into the memory procedure. To test that procedure, compare earlier and later mechanisms from compatible starts with common experience and downstream information. A complementary contrast fixes the mechanism and varies content. Inseparable content and procedure support a package comparison. General agent-benchmark gains do not validate the scientific claims stored in memory.

Explicit records make premises easier to locate and revise but incur retrieval and maintenance costs. Summaries reduce those costs while potentially losing diagnostic distinctions; executable workflows act directly but require compatible tools and instruments. A useful report connects source experience, acceptance, retained object, actual consumer, measured benefit, and conditions for revision or withdrawal.

4.3 Synthetic data and persistent parameter updates

STaR uses answer correctness to select or repair generated rationales before training (Zelikman et al., 2022); Self-Instruct generates and filters instruction examples (Wang et al., 2022). The model contributes to its training material, while selection rules and seed information remain consequential. A generated scientific explanation is a hypothesis or training aid, not an empirical observation without supporting measurement.

Scientific adaptation should label measured, simulated, inferred, and generated content separately. Acceptance checks validity for the intended use, duplicate or leaked evaluation targets, and a prespecified development outcome. The saved checkpoint becomes an inherited research artifact through later prediction, proposal, diagnosis, or another named operation. Self-Refine instead revises a current output with a frozen model (Madaan et al., 2023); parameter learning changes behavior across episodes and can also retain error or interfere with earlier capabilities. Distributional risks from generated training data depend on mixture, selection, and grounded inputs (Shumailov et al., 2024).

Compare adapted and unchanged models with matched access to validated information and a common inference budget, using independently specified later tasks. Report retention, transfer, and interference alongside the development score. An information-matched control distinguishes the adaptation route from access to additional facts; a compute-matched control distinguishes it from extra inference. If parameters and other retained components cannot be separated coherently, evaluate the package.

Parameters are useful when many later decisions can reuse experience without retrieving large records. Explicit records offer advantages when premises are uncertain, frequently corrected, or require precise withdrawal. The choice depends on reuse and maintenance demands. In both cases, preserve the path from laboratory evidence to accepted training examples, saved parameters, and later research use. These are requirements for scientific adaptation; the cited general-purpose training studies do not themselves establish their fulfillment.

4.4 Prompts, programs, and feedback-generating components

DSPy optimizes compositional language-model pipelines (Khattab et al., 2024), TextGrad propagates textual feedback through computation graphs (Yuksekgonul et al., 2025), and GEPA diagnoses traces and combines prompt candidates through Pareto-based selection (Agrawal et al., 2025). These methods make feedback actionable at the level of instructions, connected modules, or candidate populations. Pipeline improvement does not establish improvement of the optimizer that produced it.

Textual criticism can supply hypotheses and counterexamples absent from a scalar score; the term "gradient" does not give it the mathematical properties of a derivative. Test proposed changes rather than accepting a persuasive diagnosis. Likewise, a per-rollout advantage need not survive accounting for generation, reflection, and validation costs.

Promptbreeder evolves mutation prompts as well as task prompts (Fernando et al., 2023); Retroformer trains retrospective feedback using environment rewards (Yao et al., 2023); Self-Rewarding Language Models use model judgments in iterative preference optimization (Yuan et al., 2024). These place adaptation in proposal, diagnosis, or evaluation. Scientific use requires testing empirical discrimination, beyond feedback persuasiveness, judge agreement, or instruction performance.

An evolving internal judge can accelerate search while a protected external audit assesses its results. Independence concerns information flow and measurement, not model names: two judges sharing an untested assumption can agree without supplying independent support.

4.5 Automated design and the evolution of search spaces

Neural architecture search learns a controller from validation performance (Zoph and Le, 2016); population-based training adapts models and hyperparameter schedules (Jaderberg et al., 2017); AutoML-Zero searches programs including learning algorithms (Real et al., 2020). These expand the optimized object while retaining designed representations, budgets, and selection. Finding a better learning algorithm need not improve its discovery process.

Scientific counterparts include experiment selectors, analysis programs, instrument-control routines, and coordination protocols. Their discovery can be valuable under fixed outer search. Process-revision claims require identifying the learned component that later produces, assesses, or retains capability changes; improver gain requires the corresponding comparison. Successive candidate "generations" are not sufficient evidence.

POET co-generates environments and solvers and transfers solutions between them (Wang et al., 2019), suggesting how research systems could construct informative intermediate problems. Environmental novelty, however, does not ensure scientific importance: an endogenous curriculum can stay inside a narrow solvable family. Scientific transfer therefore requires externally meaningful target questions and evidence that intermediate solutions help address them. Finite demonstrations do not establish indefinitely continuing progress.

4.6 Experimental policies and the value of information

Bayesian optimization uses accumulated evaluations to guide costly search (Snoek et al., 2012); Entropy Search values observations for information about the optimizer (Hennig and Schuler, 2012); Deep Adaptive Design learns a mapping from past observations to subsequent designs (Foster et al., 2021). A learned policy can change experimental choices without continuing to revise its learning procedure.

These methods separate candidate improvement, uncertainty reduction, and acquisition-policy learning. Measurements near a promising candidate can advance optimization while discriminating little among explanations; an unfavorable outcome can eliminate a hypothesis. Report these gains and physical measurement costs separately. Fresh campaign instances test scoped retention, whereas changed tasks or conditions test wider transfer. Table 4 distinguishes changing acquisitions under one policy from revising the policy itself.

4.7 Composing mechanisms without obscuring attribution

Observations can enter memory, guide program revision, generate experiments, and become adaptation data. Final performance then combines more information, extra search, inherited capability, and possibly a better improvement rule.

Choose interventions that preserve the ingredients required by the claim: fix content while changing retrieval, preserve a candidate pool while changing selection, or compare proposal rules on fresh tasks. State which parts remain compatible and which resources are matched. The outcome is more reliable knowledge, or comparable knowledge at lower cost, within a declared target scope; component attribution requires more than a better final composite system.

4.8 Contrasting Update Contracts and Reported Process Evidence

Reflexion retains feedback for subsequent attempts, whereas Voyager retains executable skills (Shinn et al., 2023; Wang et al., 2023). Both expose a consumer of prior experience. Their contracts concern retained-content benefit; a larger memory or skill library does not alone identify a better memory-building or skill-learning rule. SEAL instead generates adaptation data and instructions consumed by parameter learning (Zweiger et al., 2025). The proposal generator and the adapted model must be distinguished even when they share an implementation. These contrasts specify mechanisms, not a single scientific-validity hierarchy.

STOP: positive and degrading improver comparisons. STOP applies an improver program to its own implementation. Its GPT-4 experiment reports better downstream meta-utility across five independent STOP runs, with evaluation on 50 held-out parity instances; improvement is not monotonic in every run. The smaller-model experiments use 25 runs and show aggregate degradation. A selected later improver also transfers to five additional tasks (Zelikman et al., 2024, §§5.1–5.3). This is relevant positive and contrary improver evidence, not merely editable code. The selected-snapshot transfer result should not be reinterpreted as the expected outcome of independently rerunning the entire improver-generation process on each new task. The endpoint is computational problem solving, not empirical scientific discovery.

DGM: a return-path control under unequal realized cost. DGM compares its evolving agents with *DGM w/o self-improve*, in which the initial meta-agent remains the modifier while descendants still populate the search. The complete system performs better in the reported comparisons. This is a directly relevant intervention on who generates later modifications; fixed archive and parent-selection machinery do not remove that return path (Zhang et al., 2025b, §§4.3–4.4, App. E.1). However, reported SWE-bench costs are approximately US$22,000 per complete run versus US$10,000 for a baseline run. The contrast is therefore evidence about the implemented process packages, not a matched-total-cost estimate or a common-learner comparison of isolated improver snapshots. Neither omission nor unconditional certification is appropriate.

MetaRSI: a bounded common-start result. MetaRSI forks the same released system for a second term, comparing the original and meta-updated improvers under the stated same term budget and sealed splits. Reported average gains are 4.9 and 7.2 percentage points. Appendix F specifies five independent outer seeds per condition covering complete terms, and separates improvement and meta-training ledgers (Tan et al., 2026, §5.4, App. F). This supports a reported positive conditional improver comparison; it is not evidence of no relevant test. The 2.3-point contrast is neither a new result calculated by this survey nor proof that meta-training cost is repaid. Its code and closed-form endpoints do not establish open-ended empirical discovery, and the snapshot contrast alone does not isolate a particular recursive information path.

Taken together with AIDE^2's inconclusive common-start test, these configurations require a balanced conclusion: improvement mechanisms have yielded positive, degrading, and uncertain effects under different conditions. The next question is which information paths, task distributions, interfaces, and budgets account for those differences. A fixed audit is compatible with all of these outcomes.

5 Life Sciences: Learning Across Biological Contexts

5.1 Biological models and experimental feedback

Biological models connect existing data to candidate selection. Evo 2 supports genome-scale sequence prediction and design, while scGPT supplies representations for cell annotation, integration, and perturbation prediction (Brixi et al., 2026; Cui et al., 2024). Their scientific role depends on the decision consuming the output and the experimental endpoint subsequently measured.

Protein design illustrates the gap between useful representations and biological endpoints. RFdiffusion generates structures; AlphaFold 3 predicts biomolecular interaction structures (Watson et al., 2023; Abramson et al., 2024). Neither structural plausibility alone establishes expression, binding, or function. Assays must address the intended endpoint and preserve cell state and measurement conditions, since validated activity need not transfer across them.

Experimental results can update a property predictor, candidate-selection rule, or reusable procedure while the underlying biological model remains fixed. These are separately testable changes. Their value depends on

the bottleneck: when confirmation is slow, selection may matter more than generation; when an assay is a poor proxy, improved proxy prediction may leave biological utility unchanged.

5.2 Evaluating the use of experimental feedback

An agent may select promising biological candidates using prior knowledge without benefiting from the experimental results it receives. Completing successive rounds of selection therefore does not, by itself, establish that feedback improves its performance. A controlled test can compare corresponding agents receiving genuine or permuted outcomes while keeping the backbone, task, candidate space, and selection budget fixed. A performance advantage under genuine feedback supports a contribution from the information contained in the outcomes. Permuted feedback is a control on that information, rather than a random-selection baseline; similar performance across the two conditions does not establish that the agent never processes feedback.

The interpretation of such a comparison depends on how the agent obtains evidence. Biomni coordinates biomedical tools through retrieval-based planning and code execution (Huang et al., 2026), while CellVoyager autonomously analyzes existing single-cell data (Alber et al., 2026). In these workflows, immediate feedback concerns computational execution and analysis. BioDiscoveryAgent instead evaluates sequential perturbation selection by revealing previously measured outcomes from existing experimental datasets (Roohani et al., 2025). This retrospective replay allows feedback to be controlled without conducting new experiments. Robin illustrates a prospective return path: literature and analysis agents develop therapeutic hypotheses, human researchers execute experiments and supply domain-specific instructions, and experimental results inform subsequent investigations. Its dry age-related macular degeneration case connects candidate selection to increased retinal pigment epithelium phagocytosis in vitro and follow-up analysis; the findings remain preclinical (Ghareeb et al., 2026). Experimental validation must also be distinguished from evidence of learning: the Virtual Lab's human-assessed nanobody designs demonstrate useful scientific outputs, but do not by themselves establish that retained experience improves later investigations (Swanson et al., 2025).

Retrospective replay has enabled direct tests of feedback benefit, with different results across configurations. Gupta and colleagues compare a no-tool BioDiscoveryAgent with a corresponding agent receiving permuted outcomes and report broadly similar discovery counts (Gupta et al., 2025). Wainrib and colleagues report a benefit from genuine feedback in a Cell Painting perturbation-selection task (Wainrib et al., 2026). Table 7 records the configurations, experimental units, and outcomes needed to interpret these findings.

Table 7. Genuine and permuted feedback in biological perturbation selection.

Configuration	Controlled comparison	Observed result and scope
Gupta et al., Table 1 (Gupta et al., 2025)	No-tool BioDiscoveryAgent; five gene datasets; five rounds of 128 selections; five runs per condition. Genuine versus permuted feedback.	For Claude 3.5 Sonnet on IL2, mean cumulative hits were 59.4 versus 57.6. This is one configuration-specific example, not an aggregate across datasets or a test of all tool-augmented agents.
Wainrib et al., § 3 and § 4.6 (Wainrib et al., 2026)	Sonnet 4.6 ICL-EF; ten Cell Painting features; ten rounds of 100 selections; ten replicate campaigns per feature and condition. Genuine versus within-batch-permuted feedback.	Mean feature-level hits were 29.3 versus 18.4; the reported paired feature-level test gave $p = 0.003$. Outcomes were retrieved from JUMP data, not newly measured.

These results support configuration-specific conclusions about feedback benefit. The similar means reported for the Gupta comparison do not constitute a formal equivalence test. Conversely, Wainrib and colleagues' positive result concerns paper-defined screening hits, identified by a reported $p < 0.05$ for the target morphological feature, rather than independently confirmed biological discoveries. Differences in backbone, task structure, candidate space, feedback representation, and tool access prevent the cross-study contrast from identifying which factor accounts for the different outcomes. Retrospective replay also omits aspects of prospective experimentation, including execution failures, measurement delays, and batch variation. Its contribution is to test whether informative outcomes improve selection performance under specified conditions. Establishing

that the agent retains a better research method, transfers it to new biological contexts, or improves the procedure that generates subsequent updates requires separate comparisons.

5.3 What biological research systems inherit

Biological inheritance can preserve observations, executable tools, selection strategies, or parameters. An assay table supplies information; a tool saves implementation; a diagnostic rule changes later inquiry. The comparison must identify which of these objects crosses the task boundary.

STELLA makes two of these objects explicit: a library of reasoning templates and an expanding collection of tools (Jin et al., 2025a). In the inspected arXiv v1, a successful workflow can be distilled into a template, and a tool-creation agent can add executable resources to the Tool Ocean, including database queries and calls to biological models. That is inheritance of templates and tools. The reported evidence does not show inheritance of the full research state, or a measured change in the rule that decides which template or tool is kept. The quantitative support is a test-time curve on question-answering benchmarks. Figure 2B varies the trial budget from one to nine times the baseline: accuracy on Humanity's Last Exam (Biomedicine) rises from 14% to 26%, and accuracy on LAB-Bench LitQA rises from about 52% to 63%, each averaged over three runs. Because the number of trials changes, these curves do not isolate a better updater at matched cost. A chemoresistance workflow illustrates how a template might be saved for a similar later problem; it is not a separate transfer test with that template frozen and the budget matched. The benchmarks are database and literature questions, not prospective experiments. The architecture states that human experts and wet-lab results can enter the loop, but the reported curves do not show that they did. The separately titled bioRxiv v2, posted in January 2026, reports expanded biological and robotic demonstrations (Jin et al., 2026). Our access to that revision covers metadata and abstract claims, so the v1 attribution limit should not be extended to all later experiments; conversely, later product claims should not be imported as audited evidence of inherited capability. The version distinction matters because the question is which retained artifact caused a later gain, not whether a system bearing the same name produced useful results.

ScienceBuddy isolates a narrower object. In its fixed-model experiment, instruction and skill text are adapted over 288 conversations in 24 batches, while the task model (Qwen3.5-4B), the execution loop, the tool interface, and the reflector remain fixed. The initial and selected harnesses are then compared on a separate validation set: validation accuracy rises from 31.1% to 51.1% (Xue et al., 2026). This supports retained procedural text on the tested tasks, which cover literature reading, database judgments, protocol troubleshooting, and gene and variant assessment. It does not establish a 20-point gain in biological discovery, or isolate every contributing edit. Accuracy along the adaptation stream is a different measurement and is not this comparison. In the coupled design, learned parameters return to later harness adaptation. A fixed reflector does not by itself rule out a changing effective update policy; establishing improved improver efficacy requires a separate comparison from common starting states. That coupled package contrast is not the result used here.

Transfer tests should name the changed endpoint, sample composition, or measurement process. Comparing factual memory, procedures, and complete histories from compatible starting states separates their contributions. Fresh instances in a continuing project test persistence; new cell systems or laboratories test external scope. If a procedural advantage vanishes after factual information is equalized, the evidence supports informational benefit within that comparison.

5.4 Translating inherited procedures into prospective biological value

A prospective agenda connects feedback use, retained procedure benefit, and biological transfer around one declared phenotype. The first experiment freezes an experiment-selection procedure after adaptation on source campaigns and evaluates three arms: the reference procedure with target-context information only; the reference procedure with source-campaign factual memory; and the adapted procedure with the same factual memory. All arms use the same tools, target-task information, and acquisition and confirmation budget. Their interventions may diverge because the resulting trajectories are part of the procedure's effect. Prespecified strata separate source-like instances, a biological shift such as cell state or cohort, and a measurement shift such as batch or instrument.

The primary endpoint is confirmed phenotype utility under common assays, independent confirmation, and a deadline. Secondary outcomes include invalid selections, uninformative assays, scientist review, redesign,

repeats, and time. The three arms separate factual inheritance from retained procedural benefit (Ghareeb et al., 2026; Jin et al., 2025a; Xue et al., 2026).

The second experiment tests failure attribution after a negative or inconclusive assay. Candidate causes include an inactive intervention, degraded sample, invalid assay, or insufficient power. Compare an outcome-only updater, an uncertainty-aware updater that retains the explanation set, and a diagnosis-aware updater that can request discriminating controls. All receive the same initial record, available actions, and evidence budget; independent controls later adjudicate the cause. Primary outcomes are the next intervention, downstream confirmed utility, and the frequency with which ambiguity becomes a persistent invalid rule. Diagnostic measurements, delay, and expert review enter the resource total. Diagnosis is predicted to add the most value when plausible causes imply different consequential actions and discrimination value exceeds its cost, as illustrated in Figure 2.

A subsequent common-start comparison of earlier and revised policy-learning procedures would test improver gain, using identical adaptation evidence and budgets with protected final confirmation. This is distinct from the preceding information, procedure, and diagnosis contrasts.

6 Chemistry: Learning from Reactions and Executable Procedures

Chemical research links molecular proposals to executable procedures and measured outcomes. Its distinctive inheritance objects include reaction records, predictors, and preparation methods; their scientific value depends on product identity and measurement context.

6.1 Chemical foundation models as interfaces to experimental decisions

Chemical interfaces combine planning, representations, and execution. Coscientist connects instructions and retrieved documentation to laboratory operations (Boiko et al., 2023); molecular encoders and task-specific predictors support candidate selection.

ChemBERTa-2 encodes canonical SMILES for molecular regression or classification, comparing masked-token pretraining with regression of 200 computed RDKit descriptors. Scaffold-split fine-tuning shows task-dependent transfer; structure-only inputs leave assay and synthesis context unspecified (Ahmad et al., 2022).

Uni-Mol pretrains molecular and protein-pocket models on atom types and coordinates using masked-atom and coordinate-recovery objectives. It supports property, conformation, and ligand–pocket distance prediction, with dependence on conformation quality. Binding geometry remains distinct from measured affinity, selectivity, and activity (Zhou et al., 2023).

Task-specific models connect molecular representations to experimental selection, as illustrated by Granda's reactivity model (Granda et al., 2018). Koscher's platform combines a hierarchical graph-completion generator pretrained on ChEMBL and fine-tuned on curated dye molecules with property predictors retrained from experimental feedback (Koscher et al., 2023a). The generator's initial transfer learning and the predictors' subsequent experimental updates serve complementary roles within the same workflow, providing a concrete link between pretrained capability and learning during discovery.

Execution connects these predictive capabilities to physical operations. AI-assisted synthesis planning and chemical programming translate proposed chemistry into laboratory procedures (Coley et al., 2019; Steiner et al., 2019). Steiner's Chempiler combines procedural instructions with a graph of available hardware to generate device commands, enabling execution through compatible modules. Procedure files preserve reaction, workup, isolation, and purification steps. These explicit artifacts provide a basis for reusing and revising experimental methods alongside the models that guide candidate selection.

These interfaces locate the editable objects in the following cases: predictive models and executable procedures, together with experience used to propose or assess their later revisions.

6.2 Closing the loop from experimental measurements to molecular design

Chemical experimentation supplies both useful products and observations that improve subsequent decisions. In Burger and colleagues' mobile-chemist study, Bayesian search guided 688 experiments and identified photocatalyst formulations with sixfold activity relative to the initial formulations (Burger et al., 2020). This illustrates how a fixed search method can turn sequential measurements into a substantial chemical benefit.

The learning mechanism becomes clearer when the retained observations, updated model, and next experiment can each be identified.

Granda and colleagues provide such a data–model–experiment chain (Granda et al., 2018). A support-vector machine trained on 72 expert-labelled mixtures assigns reactivity labels from nuclear magnetic resonance (NMR) and infrared (IR) observations. A separate linear-discriminant-analysis (LDA) model is refitted to accumulated reaction records and ranks unperformed combinations, starting from 90 random platform experiments. The retained records and LDA predictor thus redirect subsequent exploration. The 100 retrospective simulations in their Figure 3c assess exploration and prediction on previously labelled reactions. For prediction accuracy, the authors' simulation notebook uses random binary predictions in the random-search arm, without fitting an LDA predictor; this is a different comparison from freezing the learned predictor during a campaign. Automated reactivity detection is complemented by product-level follow-up: chemists established the products, and a later correction revised the structure and mechanism assigned to compound 22 (Granda et al., 2019). Retaining such corrections alongside reaction labels is essential for learning chemically meaningful relationships.

Koscher and colleagues connect generation, multistep synthesis, purification, and measurement (Koscher et al., 2023a). This account retains the earlier full-text extraction; the R3 access check recovered an institutional abstract, not the publisher full text (Appendix E). A pretrained graph-completion generator and synthesis planner propose accessible dye-like molecules. Message-passing ensembles predict absorption and partition coefficients, and a random forest predicts photooxidative degradation. Scheduled or campaign-triggered retraining adds measurements to the initial data while retaining hyperparameters; updated predictions and uncertainties guide later design.

Across three exploration rounds, the platform realized 294 previously unreported molecules from 720 selected targets spanning four scaffolds. Their Figure 5 reports cross-validated mean absolute errors decreasing from 52 to 11 nm for absorption maximum, from 1.3 to 0.7 for $\log K_{\mathrm{ow}}$, and from 0.30 to 0.21 for $\log_{10}(k_{\mathrm{deg}}/\mathrm{s}^{-1})$. Assay availability differs: these results draw on 245 absorption spectra, 294 partition measurements, and 211 degradation-rate measurements. The gains characterize the explored scaffold families; the authors report little change in prediction across general chemical space. In a subsequent exploitation study, the platform realized nine of 14 selected benzophenothiazinone derivatives. Absorption and photostability predictions guided useful candidates, while partition coefficients were systematically underpredicted by about two log units, revealing the importance of ionization and measurement conditions. These results establish local predictive improvement and its use in discovery under a human-configured retraining procedure. A matched comparison of retrained and frozen predictors during discovery would be needed to isolate retraining's contribution to discovery efficiency. Operators supply materials, recover hardware faults, and perform off-platform structural follow-up where required.

RSI-Anything extends the comparison to agent-guided reaction-condition search. In two Suzuki–Miyaura searches, its paper reports best observed gas chromatography (GC) yields of 93.0% versus 87.5% after 20 evaluations, and mean best observed flow-screen yields of 96.15% versus 82.16% after 40 evaluations, relative to Bayesian optimization (Cui et al., 2026). The later chemical release provides a distinct, inspectable configuration: offline sequential replay over a 321-condition pool, with 100 initializations and eight initial observations included in the 40-observation budget. Calibrated predictions and the expected-improvement rule remain fixed while feedback updates the incumbent and available candidates (Gen-Verse, 2026). This release makes feedback-guided selection explicit; it should be interpreted separately from the paper's two reported comparisons and from online model adaptation.

For reuse, reaction records must distinguish reactivity, product identity, isolated yield, and assay-derived properties. Conditions, analysis versions, and human interventions determine which observation can revise a predictor or procedure.

6.3 From property adaptation to transferable chemical methods

ADMET-EvO and MolRSI illustrate two computational inheritance objects: endpoint-specific predictive recipes and experience guiding the search for those recipes (Zhou et al., 2026; Zhang et al., 2026c).

ADMET-EvO organizes this activity around endpoint contracts specifying assay context, measurement semantics, provenance, and evaluation partitions. Its planner proposes interventions in training data, molecular

representations, and predictive models. Evaluation records supported, rejected, and inconclusive outcomes, which condition subsequent interventions; selected recipes are frozen before final testing (Zhou et al., 2026). The system retains both a predictor and intervention evidence: evaluation outcomes inform which changes merit further investigation.

ADMET-EvO reports a reduction in cumulative fitting time from 2.25 to 0.62 hours (source-rounded values), with a source-reported reduction of 72.2%, with 20 of 22 endpoints meeting the stated non-inferiority criterion (Zhou et al., 2026). The comparison evaluates score-only and compute-aware recipe selection, so it concerns selected-recipe fitting efficiency rather than total research cost or improvement between successive improvers. Numerically matching public artifacts use fresh validation; protocol differences and incomplete run-level correspondence are detailed in Section S5 of the review-methods appendix (OTeam-AI4S, 2026a).

MolRSI makes the inherited experience explicit. Solution memory retains executable configurations and development scores; refinement memory records modifications and outcomes; execution memory retains failures, repairs, and resource information. These records inform workflow selection, revision, and execution under fixed search equations (Zhang et al., 2026c). An executable recipe supplies capability C, while experience used to propose or assess later recipe changes can also serve an improver function U.

For antibacterial prediction, experience from 25 independently searched source tasks is pooled before evaluation on six target endpoints. The prior-memory comparison holds the library, backbone, evaluation protocol, and 20-attempt target allowance constant, and both arms continue learning within the target task. With prior experience, mean area under the precision–recall curve (AUPRC) on held-out test data was higher on all six antibacterial targets (their Figure 2e). The comparison tests inheritance of solution, refinement, and execution experience as a bundle; their individual effects remain coupled. Candidates are evaluated across five data splits within each workflow search, rather than five independent agent searches.

Configuration reuse offers a complementary route to useful transfer. MolRSI's zero-search transfer refits a selected source workflow on target data, removing target workflow search while retaining fitting and inference. In a separate *Acinetobacter baumannii* case, target validation selects among inherited workflow configurations. These results show how prior computational investment can supply practical starting points for a new endpoint. The target benchmark also defines the chemical scope of this evidence: 38,583 of 39,274 distinct supplied SMILES strings (98.2%) occur in all nine antibacterial and cytotoxicity datasets. This is exact string matching without additional chemical-structure standardization; the overlapping endpoint evaluations alone do not establish transfer to a new compound collection (Zhang et al., 2026c).

At the chemistry–biology interface, MolRSI transfers experience from 14 miniprotein tasks to ranking candidate binders to peptide–major histocompatibility complexes (pMHCs). Under matched target-search conditions, AUPRC on the held-out MART-1 pMHC benchmark is reported as 0.0172 with prior memory and 0.0125 without it. This is a bundle-level comparison under the reported search conditions, without an estimate of between-search variability. Prospective binder experiments provide a separate test of the resulting ranking workflow: the experimental ablation removes ranking, rather than prior memory, and measured outcomes are not fed back into policy updates (Zhang et al., 2026c). The computational and experimental results consequently address complementary questions about inherited experience and the utility of the resulting designs.

Table 8 preserves these different return paths. The unresolved comparison is which experience improves later experimental allocation and diagnosis across chemical settings.

6.4 Learning transferable experimental and diagnostic methods

Two tractable questions concern inherited workflow refinement and diagnosis of failed experiments.

For workflow refinement, source reaction campaigns can provide both reusable solutions and records of how those solutions were developed. Comparisons among a fixed acquisition baseline, solution-only inheritance, and solutions accompanied by refinement and execution records would test the additional value of this experience. Source observations, initial target observations, available tools, retrieval allowances, and resource caps should be matched; shuffled edit–outcome associations can test whether informative links drive the benefit. A further comparison can give both arms the same raw history and vary how it is organized or used.

Table 8. Retained state and evaluated contributions in chemistry.

Configuration	Retained state and later use	Demonstrated contribution and scope
Mobile chemist (Burger et al., 2020)	Activity measurements and Bayesian search state guide subsequent formulations.	Experimentally improved formulations within the declared search space.
Granda et al. (Granda et al., 2018)	Accumulated reactivity records refit LDA and redirect experiment selection.	Reactivity-model adaptation and redirected exploration; retrospective evaluation.
Koscher et al. (Koscher et al., 2023a)	Accumulated measurements retrain property ensembles used in subsequent molecular design.	A physical discovery loop with lower cross-validated prediction errors in explored scaffold families.
ADMET-EvO (Zhou et al., 2026)	Evidence verdicts guide interventions; selected recipes are frozen for evaluation.	Evidence-guided predictor construction and reduced selected-recipe fitting cost; computational endpoints.
MolRSI, prior-memory comparison (Zhang et al., 2026c)	Solution, refinement, and execution records guide target-workflow search.	Prior-experience transfer at a common target-search allowance; component effects remain coupled.
RSI-Anything, chemical release (GenVerse, 2026)	Yields update the incumbent and available candidates under frozen predictions.	Offline feedback-guided selection under a fixed predictor and acquisition rule.

Where compatible interventions are possible, versions of the workflow-selection and refinement procedure retained before and after source-stage adaptation can be evaluated from common target starting workflows, with the same kinds of within-target updates permitted. Measuring success within a budget and the resources needed to obtain independently confirmed products would connect method learning to useful chemical output. Independent repetitions of complete campaigns, with prespecified treatment of budget exhaustion and censored outcomes, would assess its reliability; held-out reaction families would test its scope.

For diagnosis, corrected product assignments, calibration errors, and execution failures can become training experience for procedures that identify future faults. Every comparison arm should receive the same adjudicated source corrections. Compare record correction alone, correction plus predictor retraining, and the same correction and retraining plus a system-generated diagnostic revision. Recording human edits and freezing the learned procedure before assessment on fresh faults makes the origin and later use of the revision explicit. Reduced false acceptance, fewer missed valid outcomes, and lower recovery effort would demonstrate scientific value; changes of instrument or analysis protocol would provide further tests of transfer.

Both designs protect independent confirmation, replacing it whenever results return to development. Costs include source experience, target search, confirmation, materials, downtime, computation, and scientist intervention; compatible whole-package baselines remain useful when component effects cannot be separated.

7 Materials: Learning Across Structures, Processes, and Properties

7.1 Generative and predictive models in a multiscale evidence chain

Materials workflows connect composition, structure, processing, and properties. Models can propose structures, approximate energies, allocate higher-fidelity calculations, or select preparation conditions; each interface supplies a different target for feedback.

M3GNet learns energies, forces, and stresses from relaxation data, supporting reusable interatomic prediction; its authors propose additional DFT data through active learning as a future extension (Chen and Ong, 2022). MatterGen instead jointly generates atom types, coordinates, and lattice parameters under property conditions (Zeni et al., 2025). Combining generation and surrogate screening can save calculations, but shared data and approximations can correlate their errors. Agreement between them is not independent confirmation.

The experimental interface adds variables that a crystal representation does not necessarily specify. A candidate must be connected to a preparation route, and the resulting sample must be characterized under conditions relevant to the claimed property. For MatterGen's experimental proof of concept, 32,768 candidates generated for four bulk-modulus targets were computationally filtered to 75, from which experts selected four for attempted synthesis; one yielded a sample identified by Rietveld refinement as approximately 98% compositionally disordered $TaCr_2O_6$ and 2% Cr_2O_3 (Zeni et al., 2025). The generated target was ordered $TaCr_2O_6$ conditioned on 200 GPa. Nanoindentation at four locations on the same pellet, together with a DFT-derived Poisson ratio of 0.30, gave estimated bulk moduli averaging 158±11 GPa, with a maximum of 169 GPa. The authors use this maximum as their best estimate because they hypothesize that local non-compactness lowers some measurements. This connection between generated candidates and physical materials identifies the information that a subsequent learning cycle should retain: the preparation route, the relation between ordered and disordered structures, secondary-phase content, the measurement unit and the assumptions used to infer the target property.

A useful record therefore names the represented object, predicted quantity, consuming decision, and revising evidence. A verified DFT structure can enter potential training; a disordered experimental product requires its preparation and measurement context before revising a structure–property label.

The bottleneck determines the relevant outcome: verified-candidate cost when calculation is scarce, characterized-sample yield when preparation or identification is limiting, and retained accuracy under composition or processing shifts. A stronger generator alone may merely increase unresolved candidates.

7.2 Computational learning and physical experimentation close different loops

GNoME supplies direct evidence of an implemented computational learning loop. New DFT calculations enter model training, and updated models support further exploration; newly stable structures also become starting points for later candidate generation. With the structural architecture fixed after initial development, six active-learning rounds reduce the structural model's formation-energy mean absolute error from 21 to 11 meV per atom, while the fraction of structurally screened candidates verified as stable rises from below 6% to above 80% (Merchant et al., 2023). Evaluation groups all structures with the same reduced composition into either training or test data, using an 85:15 split. This tests prediction on held-out reduced compositions, while permitting shared elements. The improved filter converts accumulated calculations into more productive subsequent DFT selection. Model training, available structures and data volume evolve together in this trajectory. Comparing independent complete learning histories from matched initial data and resource limits would test the value of the updating procedure.

Physical laboratories close a different part of the evidence chain. A-Lab connects synthesis planning, robotic execution and X-ray diffraction (XRD) interpretation, using the products of unsuccessful recipes to select follow-up experiments (Szymanski et al., 2023). Its active-learning stage improved target yield for nine materials, including six for which the initial literature-inspired recipes gave zero yield. These local improvements sit within a campaign whose confirmed outcome was revised from the original 41/58 to 36/57: one target was removed because it had been included in training data, and four of the remaining reported successes became inconclusive after manual diffraction reanalysis (Szymanski et al., 2026). The revised accounting is therefore 36 confirmed, four inconclusive and seventeen not obtained. The manual reanalysis was external, post hoc validation and did not contribute to the decisions made during the original campaign. Confirmed target presence also permits impurity phases. The engineering contribution and the revision of scientific labels together show why retained synthesis records should distinguish the original automated decision, subsequent confirmation and phase purity.

The synthesis of $CaFe_2P_2O_9$ provides a complete local update case (Szymanski et al., 2023, Supplementary Note 6). Four initial recipes produced glassy, low-yield samples at 1100 °C. Testing them at 800 °C exposed $Ca_3(PO_4)_2$ and $FePO_4$ intermediates, from which little thermodynamic driving force remained to form the target. ARROWS^3 recorded the inferred pairwise reactions and proposed alternatives expected to avoid this pathway. It retained a distinct route using CaO, Fe_3O_4 and $(NH_4)_2HPO_4$, rather than repeatedly pursuing precursor combinations that converged to the same intermediates. At 900 °C this route formed $CaFe_3P_3O_{13}$, an intermediate with a larger calculated driving force to react with CaO; at 1000 °C the target yield reached 75%, satisfying the campaign's stopping threshold of greater than 50%. Here, a failed experiment changed

a retained reaction record, that record changed later selection, and a subsequent experiment improved the measured outcome.

This reaction memory also supports avoiding redundant experiments: A-Lab accumulated 88 pairwise reactions, and reported search-space reductions of up to 80% when many precursor sets formed the same intermediates (Szymanski et al., 2023). Its useful range is governed by identifiable conditions. Intermediate phases must be sufficiently resolved, the pairwise approximation must describe the reaction sequence, and reused observations must correspond to compatible synthesis conditions, including atmosphere.[1] Melting can permit reactions among more than two phases, while overlapping diffraction peaks can obscure which reaction occurred (Szymanski et al., 2023, Supplementary Note 11). These conditions locate where further characterization or a richer reaction description can improve the next experimental decision.

Table 9. Materials evidence across computation, synthesis, and characterization.

Study	Evaluated unit and result	Return path and human role
GNoME (Merchant et al., 2023)	Formation-energy MAE: 21 to 11 meV/atom; structural screening hit rate: below 6% to above 80%.	DFT results retrain the model and enlarge the structure pool for later screening; architecture and evaluation rules are researcher specified.
MatterGen (Zeni et al., 2025)	One of four expert-selected candidates yielded a sample dominated by disordered $TaCr_2O_6$.	Experts select candidates and conduct experimental confirmation; the denominator is the selected experimental set.
A-Lab and correction (Szymanski et al., 2023, 2026)	36/57 targets confirmed; four inconclusive and seventeen not obtained; nine targets improved by active learning.	Observed products update reaction records and subsequent recipes; manual diffraction reanalysis externally validates outcomes after the campaign.
CAMEO (Kusne et al., 2020)	On a pre-synthesized 177-point library, iteration 19 at about 10 h located the largest optical band-gap difference ΔE_g; full characterization required over 90 h.	Active learning directs XRD measurements; an expert fits pre-acquired optical spectra during each cycle.
Attia et al. (Attia et al., 2020)	Cross-domain reference: charging protocols evaluated with early-life estimates and later validation.	A protocol-response model directs later experiments; faster evaluation and more efficient selection provide distinct gains.

CAMEO links active learning to materials characterization and property optimization at a synchrotron (Kusne et al., 2020). In its demonstrated campaign, it selected measurements on an already synthesized composition library, incorporated XRD-derived phase information, and requested expert fitting of pre-acquired optical spectra to estimate the optical band-gap difference ΔE_g at each selected composition. The resulting phase and property estimates guided later measurements, concentrating optimization on promising phase regions after the phase-mapping convergence criterion was reached. Iteration 19 located the largest reported ΔE_g in about 10 h, compared with more than 90 h for all 177 composition points. Preparation and optical acquisition preceded this search. The 100-run comparisons for each strategy were simulations conducted after data collection. This makes measurement selection and interpretation central: an effective discovery loop can improve knowledge of a prepared material library. Likewise, battery experimentation separates faster candidate evaluation from more efficient candidate selection (Attia et al., 2020). These two improvements reduce research time through distinct mechanisms and should be measured separately.

These cases cannot share a raw "discovery count": predicted structures, attempted syntheses, assigned phases, and working devices have different denominators. Time to a declared confirmation endpoint exposes whether an upstream gain removes a downstream bottleneck.

[1] ARROWS documentation.

7.3 Inherited materials knowledge depends on its measurement and reference state

An inherited energy estimate depends on its computational reference; a phase assignment on characterization; a functional property on the sample and test conditions. Omitting these dependencies can change the meaning of an otherwise accurate retained statement.

Three adaptation targets follow: prediction, allocation across evidence fidelities, and diagnosis of unsuccessful preparation versus unresolved characterization. Lower prediction error does not establish that an extra calculation or measurement repays its cost; a biased low-fidelity score can concentrate search in its least reliable region.

The A-Lab correction illustrates why evidence maintenance must extend beyond the paper record (Szymanski et al., 2023, 2026). Manual reanalysis left some previously successful phase assignments inconclusive. The corrected article identifies this reanalysis as external validation of synthesis outcomes, separate from the evidence used for autonomous decisions. This boundary case motivates a future maintenance layer: if a revised label supplied training data or a reusable synthesis rule, inspect the affected artifacts and test their remaining support. Exact parameter-level reversal may be unavailable; targeted reevaluation can still determine which capabilities remain supported.

Link structures to calculations, preparation attempts, characterization artifacts, and interpreted claims. A revised phase assignment need not erase useful process observations. Selective review should preserve conclusions with independent support and request further characterization when competing identities imply different actions; unresolved identity should remain unresolved.

Transfer of a structure or successful route reuses completed work; transfer of an allocation policy may reuse a research method. Compare these interventions from compatible starts, alongside a whole-system baseline, on fresh instances and new chemical or preparation regimes.

7.4 Learning when to calculate, synthesize, or resolve an ambiguous sample

M1: Does learned evidence allocation improve confirmed materials outcomes? The distinguishing materials question is how a research policy allocates evidence across fidelity levels. A study could learn a policy for choosing among surrogate evaluation, higher-fidelity calculation, synthesis and additional characterization. After development, freeze the initial and adapted policies and cross them with initial and adapted predictors: unchanged, policy-only, predictor-only and coupled adaptation. Start each evaluation campaign with the same initial structures, initial evidence, available actions and resource envelope. Comparing policies at a fixed predictor tests decision quality; the predictor-only arm tests gains from prediction, and the coupled arm tests their practical combination. The arms may acquire different evidence as their decisions unfold; this difference is part of the allocation policy's effect, whereas unequal starting data would confound it.

Repeat independent complete campaigns, paired by starting conditions and blocked by laboratory or instrument. Fresh instances test persistence; held-out chemical families and preparation regimes test transfer. Computational endpoints require fixed references and stability criteria. Reserved characterization tests identity in an existing specimen; route reproducibility requires independently prepared batches. Repeated scans measure precision, not preparation reproducibility. Protect final confirmation from selection and keep unresolved samples separate.

Count target-specific claims confirmed by the deadline against the assigned panel, not just selected successes. Charge development, computation, synthesis, instrument occupancy, characterization, and expert review. Report reference-task performance and attempted chemical coverage to detect gains from choosing easier targets. H1-style allocation benefit is constrained or refuted if a sufficiently precise comparison excludes the prespecified gain over predictor-only adaptation, or internal scores rise while confirmed outcomes reproducibly fall.

M2: Can identity revision selectively repair inherited capability? After a campaign, supply all arms with the same corrected phase assignment or reference calculation. A-Lab motivates the identity-revision case (Szymanski et al., 2023, 2026). Compare source-record correction, full maintenance of a declared component set, and dependency-guided review with targeted reevaluation of that set. Prespecify full maintenance, such

as retraining a predictor or rebuilding a recipe library. Charge the expert dependency reference; a dependency requires inspection, not automatic withdrawal.

Use independent correction-and-recovery histories and keep each correction with its descendants within one development or evaluation partition. Prespecify affected artifacts and independently supported controls, then test fresh compositions and preparation conditions. Reanalysis can revise a recorded identity; independent batches are needed to test an inherited route. Preserve process information whose interpretation survives.

Measure obsolete identity recommendations, retained prediction or preparation utility, recovery time, and confirmed outcomes. Charge tracing, adjudication, new characterization, retraining, and lost instrument opportunities. Dependency-guided maintenance is useful only if it reduces obsolete carryover without exceeding declared capability-loss and cost limits. Failure to meet those bounds, or cheaper and better full maintenance, defeats its proposed advantage.

8 Energy: Learning Under Delayed Feedback and System Constraints

8.1 Prediction, experiment design, and control have different learning targets

Energy research separates forecasts, response models, experimental selection, and control. Charging consequences can take many cycles to observe; a controller can enable an experiment without learning a transferable scientific explanation. These objects should not all be called "energy foundation models.

TimesFM includes electricity data in pretraining and transformer-temperature series in downstream evaluation (Das et al., 2024). Reusable forecasting therefore does not imply that every energy benchmark is unseen, or identify effects of new tariffs or charging policies. Intervention claims require suitable models, designs, or new identifying evidence.

A more domain-specific forecasting layer has emerged. WindFM is an open research preprint for probabilistic wind-power forecasting, pretrained on multivariate data from more than 126,000 sites in the WIND Toolkit and evaluated on unseen sites, including an out-of-distribution evaluation in Inner Mongolia (Fan et al., 2025). DeepWind, published in *Energy* in 2026, reports pretraining on approximately 560 billion time-series points from diverse wind sites and evaluates zero-shot and few-shot transfer across eight datasets in WindBench (Wang et al., 2026). These works target cross-site forecasting and uncertainty estimation; their transfer results do not by themselves show that a system learns from the outcomes of its own physical experiments or improves its research procedure.

EnergyFM is a domain-specific family for smart-meter analytics. Its Energy-TTM and Energy-TSPulse models are pretrained on large-scale building-meter data and evaluated on load forecasting, anomaly detection, and appliance classification (Arjunan et al., 2026). GridSFM addresses a different target: it is a pretrained surrogate for AC optimal power-flow solutions across grid topologies and operating scenarios, with predictions also usable as warm starts for conventional solvers (Yang et al., 2026b). These models extend reusable representations or approximate computational solutions beyond forecasting, but neither capability should be equated with autonomous experimental learning. SolarFM is best described more cautiously as a spatiotemporal irradiance-forecasting framework that incorporates a time-series foundation model for zero-shot forecasting alongside task-specific encoders; the name alone does not establish that the complete SolarFM framework is itself a broadly pretrained foundation model (Shan et al., 2026).

The battery and fusion cases below distinguish experimental learning from reusable prediction and control. Their different feedback locations determine what state persists and what comparison is needed.

8.2 What actually changes in three experimental loops

Attia et al. budget 192 cells for 100-cycle tests in four batches, using an early-life predictor and Bayesian optimization; their four-round analysis retains 185 cells after seven exclusions. The reported 16-day search excludes predictor training and subsequent validation. This separation matters: early prediction changes evidence latency, while acquisition changes which protocols receive evidence. The posterior is updated during the campaign; the inspected procedure does not revise the optimizer itself (Attia et al., 2020). A scientifically useful retained result would include the relationship between early and final outcomes, not only the selected protocol. The update chain is measurement during 100 cycles → pretrained lifetime prediction →

Table 10. Energy feedback and retained objects.

Case	Evidence and decision	What changes or persists	Boundary of the evidence
Charging (Attia et al., 2020)	Early-life measurements guide protocol selection	Gaussian-process posterior changes; early-life predictor is pretrained	Four experimental batches are one optimization campaign
Clio (Dave et al., 2022)	Conductivity guides electrolyte selection	Conductivity observations enter Bayesian optimization	Pouch-cell confirmation is separate; cell filling involved a human operator
TCV (Degrave et al., 2022)	Plasma measurements guide coil commands	Trained control policy is reused	Runtime state feedback does not update deployed weights

Note. Campaign updates and runtime feedback are different mechanisms; none of these comparisons establishes an improved improvement procedure.

Gaussian-process posterior → later protocol selection. The retained response model changes; retraining of the early-life predictor or improvement of the acquisition procedure is not established.

Clio couples robotic conductivity measurements to the Dragonfly Bayesian-optimization planner, conducting 42 evaluations in two work-days. Within the specified single-salt/ternary-solvent design space, the reported maximum conductivity is 13.7 mS/cm. The six-fold acceleration is estimated through simulated optimization against random search, not through replicated competing physical campaigns. Seven candidates were selected for follow-up cell testing; the robotic optimization loop and pouch-cell validation are distinct stages, and a human operator manually transferred electrolyte into the dry pouch cells. The cell tests support practical relevance against a prespecified baseline, but their outcomes are not reported as feedback that updates Dragonfly (Dave et al., 2022). The additional scientific question is whether this connection remains predictive across device conditions; optimization of the inexpensive property alone cannot answer it.

Degrave et al. instead train a plasma controller in simulation and deploy the resulting policy on TCV without further tuning its policy-network weights. Their training includes limits reflecting the simulator's valid operating region (Degrave et al., 2022). This is physical validation of a learned control capability. Its success can expand experimental access, but the route from improved access to improved scientific learning still needs a measured link.

Thus "experiments saved" hides three distinct changes: label latency, assay throughput, and physical access. Compare cost per confirmed endpoint decision and uncertainty remaining at the deadline.

8.3 Retaining proxy validity and device failures across campaigns

Energy records should connect candidate, measurement procedure, early prediction, final endpoint, stopping reason, and device history. A scalar winning-protocol reward cannot distinguish a failed hypothesis from assay failure, ageing, or operating-condition shift. This is a proposed record, not an established capability of every case.

Four transfer populations require different holdouts: physical devices, manufacturing or experimental batches, operating conditions, and material chemistries. Success on one does not establish the others; random splits of correlated measurements from a campaign are insufficient for such claims.

Stopping based on interim performance can informatively censor lifetime endpoints. A stopped run is not necessarily a failure: safety intervention, equipment fault, deviation, and prespecified futility have different meanings. Retain timing, reason, interim evidence, and endpoint status. Censoring-aware analysis requires assumptions; a confirmation subset continued to the final endpoint can test proxy validity. Early-life predictions and conductivity remain distinct from cell-life and device validation (Attia et al., 2020; Dave et al., 2022).

Diagnosis locates the update: batch-specific miscalibration implicates prediction, redundant sampling with calibrated uncertainty implicates acquisition, and failed external validation implicates the proxy. Check calibration and endpoint agreement within the declared transfer strata.

Cross-campaign reuse also creates an accounting choice. A historical corpus is an inherited asset whose acquisition cost can be amortized over many campaigns; it should not be presented as free evidence in an end-to-end comparison. Report both the marginal cost of the next campaign and the accumulated cost of producing the reusable asset. Sharing the same starting corpus isolates a method comparison, whereas allowing each arm to generate its own history evaluates the combined learning-and-acquisition system. Both are meaningful, with different estimands.

8.4 Two tests that separate faster search from better scientific learning

Both proposed studies use independently assigned complete campaigns, matched initial information and resource envelopes, protected confirmation, and confirmed device outcomes. Account for cells, compute, instrument time, calendar time, and human intervention.

Proxy repair versus acquisition improvement. A proposed battery study would compare four configurations in a factorial design: fixed or recalibrated early-life prediction, crossed with a reference or experience-updated acquisition rule. Each configuration starts from the same historically available corpus and receives the same total resource envelope, including a prespecified budget for delayed confirmation. Final cycle-life outcomes are withheld from model calibration, policy development, and candidate selection. Selected candidates and reference protocols receive the same prespecified validation, with safety and protocol-deviation rules fixed in advance. The primary outcomes are confirmed cycle-life performance and the cost per independently confirmed useful protocol; calibration, uncertainty coverage, and selection diversity diagnose the mechanism. The comparison should be repeated across independently assigned complete campaigns, with cells and protocols treated as nested observations rather than independent campaigns. If early stopping depends on interim performance, stopped observations must retain their censoring status and reason. This design can falsify the claim that a faster search reflects better scientific learning if the advantage disappears when judged against final cycle-life outcomes or after confirmation costs are included.

Transfer of a correction, rather than a winning candidate. A second proposed study would retain a discrepancy model linking a low-cost assay to a device-level endpoint, then evaluate it on a prespecified held-out shift. The shift should be named in advance—for example, a new device, manufacturing batch, or operating condition; a chemistry change is a separate and more demanding transfer claim. Compare (i) reuse of the original predictor, (ii) reuse of the corrected predictor, and (iii) re-estimation from scratch under a matched information, compute, and confirmation budget. The held-out endpoint must remain unavailable during correction development, and the evaluation should report calibration as well as endpoint error and confirmed decision utility. Reusing the old winning electrolyte or protocol alone is a separate baseline, because transferring a candidate is not the same as transferring a predictive correction. Successful transfer would support inherited scientific capability; it would not by itself establish improver gain. That stronger claim requires a further common-start comparison showing that the inherited procedure makes subsequent corrections more effectively. The controller, assay, predictor, acquisition rule, and update procedure should therefore remain separately versioned throughout the study.

9 Environment: Learning Under Nonstationarity and Limited Intervention

9.1 Reusable prediction and the missing observation decision

Atmospheric prediction and coastal sampling illustrate evolving environments, uneven observations, and limited intervention; they do not represent all environmental science. The comparison distinguishes updating a field estimate from learning where to observe or how to improve either capability.

Aurora supplies a clear foundation-model example: a pretrained Earth-system model is adapted to multiple forecasting tasks. Its air-quality comparison uses CAMS analysis as the reference; it matches or exceeds CAMS forecasts on 74% of the evaluated targets. This is a fraction of targets, not a 74% reduction in error, and the reference is an assimilated data product rather than an independent direct measurement (Bodnar et al., 2025). The result supports reusable predictive capability with a specified reference, without identifying the consequences of an emissions intervention.

NeuralGCM combines physical dynamics with learned parameterizations. In the reported seasonal-cycle experiment, 35 of 37 initializations complete two-year simulations without instability, with prescribed historical

sea-surface temperature and sea ice (Kochkov et al., 2024). These initializations test forward-simulation stability; they are not independent repetitions of a learning campaign. A hybrid simulator can be an important scientific component without being a general-purpose foundation model or a recursively improving system.

9.2 Forecast verification, conditional simulation, and active sensing

Table 11. Environmental feedback and validation.

Case	Feedback or reference	Retained or updated object	Validation boundary
Aurora (Bodnar et al., 2025)	Analysis products and forecast targets	Pretrained and fine-tuned weights	Forecast skill relative to specified products
NeuralGCM (Kochkov et al., 2024)	Training observations; prescribed boundaries	Learned parameterizations	Conditional trajectories, including stability
Adaptive AUV (Berild et al., 2024)	Salinity measurements select later waypoints	Updated spatial posterior; fixed selection criterion	Simulated repetitions plus one paired field mission

Note. Updating a field estimate differs from changing the observation-selection rule.

Berild et al. combine an ocean-model-derived spatial prior with in-situ salinity measurements and an uncertainty-based waypoint criterion. Their evaluation includes 100 simulated trajectories per model and a single approximately 1.5-hour two-vehicle field mission, comparing adaptive sampling with a preplanned route. A pre-survey informs a human adjustment of the salinity threshold. The observed deep-water discrepancy is approximately 3.5 g/kg, with the ocean model predicting higher salinity (Berild et al., 2024). This case demonstrates physically grounded adaptive acquisition and exposes a model–environment mismatch; it does not demonstrate improvement of the acquisition algorithm across missions.

The comparison also clarifies an experimental tradeoff. A policy optimized to locate a front need not supply the broad coverage required for a regional inventory. Reporting lower uncertainty only over the adaptively visited region can reward the policy for avoiding difficult areas. An evaluation should therefore state the scientific target before selecting a route, and distinguish uncertainty about that target from the completeness of the environmental map. If both matter, they require separate endpoints or an explicitly justified utility.

Analysis products, prescribed boundaries, and in-situ sensors supply different reference evidence. Forecast accuracy need not identify an emissions effect, and a corrected salinity map need not explain the discrepancy. The scientific claim must specify whether it concerns description, prediction, or causation.

9.3 What should survive a changing environment and observing system?

Persistence requires diagnosing what changed. Yesterday's posterior may be stale, while a persistent sensor-offset correction may transfer; mistaking an oceanographic event for that offset can instead create error.

We propose distinguishing three sources of change before updating a model: the physical process, the observation process, and the institutional data product. A new sensor, assimilation scheme, quality-control rule, or data-release version can alter the apparent target without an equivalent change in nature. Provenance must therefore accompany the inherited state. For analyses assembled from multiple institutions, recording which product version supplied training, initialization, selection feedback, and final verification is necessary to diagnose shared errors. Agreement with a common upstream product should not be counted as independent corroboration.

Adaptive acquisition adds another complication: the system helps decide what becomes observable. A later learner trained only on targeted measurements may perform well near previously detected fronts while losing accuracy elsewhere. A proposed archive should preserve both measurement values and the observation policy that produced them, including inaccessible regions and discarded readings. Randomized exploration or a separately maintained reference network can provide information about regions the learned policy neglects; neither is guaranteed to be feasible in every deployment.

Keep observing-system transfer separate from environmental-regime transfer: instruments and data products may change without equivalent physical change, or physical relationships may change under stable instruments. The diagnosis determines which retained correction remains warranted.

9.4 Testing useful acquisition under drift and limited field budgets

Separate learning from access to better observations. A proposed coastal-monitoring experiment would compare fixed and adaptive routes, and reference and updated inference models. Where compatible, replay each acquired dataset through both inference models to estimate the value of the model update conditional on evidence. The live comparison estimates the combined value of acquisition and inference. Replayed data cannot reveal what an unvisited location would have shown, so a held-out sensor network or scheduled reference transect is needed for external assessment. Predeclare both front-location accuracy and regional coverage, and repeat complete missions across environmental conditions; 100 simulated routes cannot substitute for 100 field deployments.

Test whether a retained correction improves the next learning episode. In a second proposed design, an initial campaign produces a discrepancy model and a revised observation rule. On subsequent held-out missions, compare inheritance of neither object, the discrepancy model alone, the rule alone, and both. Match initial information and resource limits while allowing routes to diverge. The primary endpoint is externally verified uncertainty reduction at a declared cost; reporting should include missed regions and observation failures. This design can reveal negative transfer when a correction encodes a transient event. Evidence that an inherited rule generates more informative subsequent data would support acquisition-policy improvement. A recursive claim additionally requires documenting how the preceding system produced that rule and demonstrating its benefit as an improver from controlled starting states.

10 Cross-Domain Synthesis: What Transfers and What Breaks

10.1 An Evidence-Regime Crosswalk for the Configuration Panel

A disciplinary label does not identify the observation available to an updater. Chemistry includes both replayed molecular labels and newly measured reaction outcomes; robotics includes construction feedback, simulated training, and physical evaluation. We therefore add a configuration-level crosswalk to the five-claim map. The two instruments answer different questions: the map records which claim a comparison supports, while the crosswalk records the evidence conditions under which that support was obtained.

The accompanying `evidence/evidence_regimes_r4.json` and CSV cover the same 35 configurations as Table 13. Each row links its claim cards and primary-source locators to seven fields: evidence timing, reported resources, intervention access, measurement validity, retained artifact, human role, and confirmation endpoint. Source-card facts are preserved separately from our interpretive grouping and inference boundaries. Unknown latency, complete cost, and human input remain unknown. In particular, a computational evaluator is not assumed cheap or instantaneous, and a campaign's duration is not substituted for the latency of one observation. This is an AI-assisted secondary coding resource, not an independent source replication or a validated prediction of which architecture will work best.

The groups are navigation aids, not disjoint physical laws: the row-level fields retain mixed evidence, such as simulated training followed by real-hand tests. The source-removal checks in Appendix E, S9 examine how selected conclusions depend on particular sources. They cannot repair purposeful sampling or turn the panel into a systematic estimate of the field.

10.1.1 A claim map with an explicit design panel

Table 13 is a navigation map, not an evidence ranking. Its 35 configurations and 175 cells refer to 53 versioned claim cards (39 in the panel; 14 domain-only). Tables 14–16 expose the design, repetition unit, confirmation boundary and uncertainty of every directional F/C/U/R row. A positive sign therefore remains visibly distinguishable from a precise or isolated effect. The sample is purposive; neither row counts nor symbol frequencies estimate field prevalence.

Table 12. Evidence-regime coverage of the selected panel.

Evidence setting	Configuration IDs	Required interpretive distinction
Computational execution or scoring (17)	U01–U08, U13, U23–U25, U27, U32–U35	Execution tests, DFT references, and model judgments certify different endpoints; their common digital format is not a quality grade.
Previously measured outcomes (4)	U09–U10, U21–U22	Replay controls information access; it does not include original measurement delays or prospective execution failures.
Physical campaigns (6)	U11–U12, U14–U16, U26	Feedback, final confirmation, and later external reanalysis may occur at different times and affect different decisions.
Field observation (1)	U17	Sampling changes what is observed while the environment evolves; a simulated trajectory is not an additional field mission.
Robot training and evaluation (7)	U18–U20, U28–U31	Simulation, hardware trials, human construction, and final auditing must be distinguished by phase.

Note. Groups describe the coded configuration and can contain several disciplines. Counts describe this panel only; no field prevalence is inferred. Full row-level facts and locators are supplied in the crosswalk.

Table 13. Five-claim configuration map.

Configuration	S	F	C	U	R	Qualification
STOP / GPT-4 U01 (Zelikman et al., 2024)	NA	M	NA	+	M	Five full runs; computational scope.
STOP / smaller models U02 (Zelikman et al., 2024)	NA	M	NA	-	M	Model-dependent degradation.
DGM / modifier control U03 (Zhang et al., 2025b)	NA	M	NA	NA	P	Unequal-cost path-control package.
MetaRSI / term-two fork U04 (Tan et al., 2026)	NA	M	NA	+	M	Same start; formation cost separate.
AIDE^2 / ignition U05 (Srikanth et al., 2026)	NA	M	NA	?	M	Three runs per arm; inconclusive.
ScienceBuddy / harness U06 (Xue et al., 2026)	NA	M	+	ND	NA	Fixed-model harness contrast.
HyperAgents / transfer U07 (Zhang et al., 2026a)	NA	M	+	ND	M	Package, not isolated meta-agent.
Organon / AHC026 U08 (Ahmed et al., 2026)	NA	M	NA	ND	M	One trace; concurrent revisions.
Perturbation replay / Gupta U09 (Gupta et al., 2025)	NA	?	NA	NA	NA	Not an equivalence result.
Cell Painting replay U10 (Wainrib et al., 2026)	NA	+	NA	NA	NA	Specific retrospective feedback test.
Mobile chemistry / campaign U11 (Burger et al., 2020)	+	M	NA	ND	M	Activity, not optimizer comparison.
Koscher / predictor reuse U12 (Koscher et al., 2023a)	NA	M	M	ND	M	Historical full text; R3 abstract recheck.

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Table 13 (continued)

Configuration	S	F	C	U	R	Qualification
ADMET-EvO / recipes U13 (Zhou et al., 2026)	NA	M	+	ND	M	Recipe fitting, not total cost.
A-Lab / corrected targets U14 (Szymanski et al., 2023, 2026; Leeman et al., 2024)	+	M	NA	ND	M	Target realization, not novelty.
Attia / battery campaign U15 (Attia et al., 2020)	NA	M	NA	ND	M	Posterior update under fixed rule.
Clio / conductivity U16 (Dave et al., 2022)	+	M	NA	ND	M	Conductivity, not device utility.
AUV / adaptive mission U17 (Berild et al., 2024)	NA	M	NA	ND	M	One field mission.
Eureka / reward search U18 (Ma et al., 2024a)	NA	+	M	ND	M	Reward versus reward generator.
ENPIRE / construction U19 (Xiao et al., 2026)	NA	M	M	NA	NA	Human-assisted setup phase.
ENPIRE / policy phase U20 (Xiao et al., 2026)	NA	M	M	ND	M	Fixed verifier; editable training.
MolRSI / antibacterial memory U21 (Zhang et al., 2026c)	NA	M	+	ND	M	Prior-memory package; one search per target.
MolRSI / pMHC memory U22 (Zhang et al., 2026c)	NA	M	+	ND	M	Same target budget; run variance unknown.
SwarmRSI / graph ablation U23 (Meng et al., 2026b)	NA	M	+	ND	M	Graph organization and reuse as a package.
MASS / workflow transfer U24 (Lee et al., 2026)	NA	M	+	ND	ND	Final executor fixed; search roles change.
DiscoveryRSI / generations U25 (Zhong et al., 2026)	NA	M	+	ND	M	Automated quality; novelty declines.
RoboChem-Flex / transfer U26 (Pilon et al., 2026)	NA	M	M	ND	M	Reuse documented; benefit undetermined.
GNoME / active-learning rounds U27 (Merchant et al., 2023)	NA	M	+	ND	M	Predictor and data change together.
RoboCat / self-generated data U28 (Bousmalis et al., 2023)	NA	M	+	ND	M	Data-package benefit; limited run variance.
ADR / matched large policies U29 (Akkaya et al., 2019)	NA	M	+	ND	M	Large-arm compute match; one training run.
VLAW / two rounds U30 (Guo et al., 2026)	NA	M	+	ND	M	Real input matched; synthetic data differ.
ASPIRE / frozen library U31 (Lu et al., 2026b)	NA	M	+	ND	ND	Held-out use; adaptation frozen.
ScienceBuddy / coupled loop U32 (Xue et al., 2026)	NA	M	M	ND	M	Model and harness co-adapt.
AlphaEvolve / meta prompts U33 (Novikov et al., 2025)	NA	M	NA	P	P	One component contrast; U/R attribution unresolved.

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Table 13 (continued)

Configuration	S	F	C	U	R	Qualification
ScienceRSI / transfer recipe U34 (Xu et al., 2026)	NA	M	+	NA	NA	4B checkpoint; no repeated recursive cycle.
ScienceRSI / feedback targeting U35 (Xu et al., 2026)	NA	+	NA	NA	NA	Gate/placement ablations; one checkpoint.

Legend. S: scientific output; F: feedback use; C: retained capability; U: improver comparison; R: recursive-origin contribution. Symbols summarize bounded claim states; the design panels and cell records supply their evidential scope. +/−: reported positive/degraded direction, not a common evidence grade; ?: tested but superiority inconclusive; M: mechanism or reuse without benefit attribution; P: a positive process-package/component contrast with unresolved attribution. In U, P is not an identified old/new-improver gain; in R, P is not a cost-isolated origin effect. R=M requires a named retained object consumed by later learning. ND: undetermined in this extraction; NA: not assessed in this row. Neither code establishes paper-wide absence. A-Lab's S=+ concerns corrected target realization, not novelty.

Reading the comparisons. STOP and MetaRSI report positive conditional improver directions, while smaller-model STOP degrades and AIDE^2 remains inconclusive. These observations reject a universal claim that self-application helps, but do not estimate its average effect across systems. The R4 primary check confirms MetaRSI's common-start design and five outer seeds; the inspected Section 5.4, Figure 11(c), caption and Appendix F do not supply a contrast interval. STOP's one-SE bands, AIDE^2's min–max bands and MetaRSI's printed domain means are different uncertainty objects.

AlphaEvolve now has an explicit configuration instead of a pending narrative reference. Its tensor-decomposition ablation enables or disables prompt co-evolution; it does not apply frozen old/new prompts to the same starting objects. The improver and recursive-origin columns retain this single component-package contrast: disabling prompt evolution also removes its return path, but fixed-prompt/context matching and absolute costs are not extracted. These are two bounded readings of one comparison, neither an old/new improver effect nor an isolated return-path effect (Novikov et al., 2025). ScienceRSI adds a retained-checkpoint comparison and a distinct feedback-targeting comparison. Its selective and dense-transfer arms both include task rewards; its gate and placement ablations keep the joint objective. The reported 4B results are single selected-checkpoint evaluations, not repeated recursive generations (Xu et al., 2026).

MolRSI's prior-memory contrasts, SwarmRSI's graph package, MASS's common final executor and DiscoveryRSI's sequential checkpoints answer different questions. The latter's quality gain coexists with lower novelty and importance. The panels retain these distinctions instead of pooling unlike effect sizes or treating all positive symbols as validated gains. Historical source readings remain historical: the new panels reorganize their recorded evidence and do not imply fresh primary verification of every row.

Table 14. Improver and process-package comparisons.

Study and comparison	Run unit	Confirmation	Uncertainty and inference
STOP / GPT-4 U01 *Positive direction* Snapshot improvers on a common task start	5 complete STOP searches; 50 test instances are separate	Independent LPN instances unused for improvement	One-SE bands; exact curve values not digitized. Conditional improver direction, not formation-cost or universal benefit.

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Table 14 (continued)

Study and comparison	Run unit	Confirmation	Uncertainty and inference
STOP / smaller models U02 *Degraded direction* Same probe, smaller base models	25 complete searches per model	Held-out LPN instances under the same task protocol	One-SE bands; declining means. Model configurations are separate, not 50 pooled replications.
Darwin Gödel Machine U03 *Process-package contrast* Initial modifier retained versus evolving modifier	Whole search; independent reruns not extracted	Benchmark package trajectories; final-audit separation not fully extracted	80 iterations in each arm; about $22k versus $10k. The path-package effect does not isolate an equal-cost return-path effect.
MetaRSI / term-two fork U04 *Positive direction* Original versus revised improver from the same S1	5 complete-term seeds per condition	Sealed split after freezing; Appendix F prohibits within-term audit use	7.2 versus 4.9 points; no contrast dispersion/CI in inspected section, figure or appendix. Formation cost separate.
AIDE^2 / ignition U05 *Inconclusive* Two outer agents applied to the same inner AIDE47	3 complete 50-step searches per arm	Private grade selects incumbents; no external final audit of each seed	0.780 versus 0.782; bands are min-max, not CI. Superiority and sample efficiency remain inconclusive.
AlphaEvolve / meta prompts U33 *Process-package contrast* Meta-prompt component enabled versus disabled	3 independent seeded searches	Tensor-decomposition objective also drives search; no separate prompt-transfer audit extracted	Intra-target SD, not contrast CI. One ablation: frozen old/new improver and isolated return-path effects are unidentified; fixed prompt/context matching and absolute costs not extracted.

Note. Study names and row IDs link to Table 13 and its source-located cell records. Outcome direction is distinct from design, repetition, confirmation and uncertainty.

Table 15. Feedback interventions.

Study and comparison	Run unit	Confirmation	Uncertainty and inference
Gupta et al. / perturbation replay U09 *Inconclusive* Real versus permuted biological feedback	5 complete replay campaigns; 5 rounds of 128	Retrospective IL2 outcomes; one model/configuration	59.4 versus 57.6 hits; contrast uncertainty not extracted. Similar means do not establish equivalence.
Wainrib et al. / Cell Painting replay U10 *Positive direction* Genuine versus batch-permuted labels	10 campaigns per arm per feature; 10 features	Retrospective outcomes; inference pairs feature summaries	29.3 versus 18.4 hits; paired feature-level p=0.003. Does not establish why another replay differs.
Eureka / reward search U18 *Positive direction* Reward reflection removed from fixed search	5 searches per task; 5 rounds of 16 candidates	Simulated tasks; final-policy trials are not new reward-search runs	Reported score drops 28.6% without reflection. Uncertainty for this contrast not extracted; other policy CIs are not substitutes.
ScienceRSI / feedback targeting U35 *Positive direction* Verification gate / supervision placement ablations	One selected 4B checkpoint evaluated per arm	Internal-dev selection precedes external tests; reported overlap filtering	Full versus gate-removed: 55.0/59.3 versus 50.6/57.7 on two science groups. No seed SD or paired CI; feedback direction only.

Note. Study names and row IDs link to Table 13 and its source-located cell records. Outcome direction is distinct from design, repetition, confirmation and uncertainty.

Table 16. Retained-capability comparisons.

Study and comparison	Run unit	Confirmation	Uncertainty and inference
ScienceBuddy / harness U06 *Positive direction* Selected harness versus initial harness; model fixed	Validation first responses; independent adaptation reruns not extracted	Separate validation tasks	31.1% to 51.1%; contrast intervals not extracted. Retained harness-package benefit, not an isolated edit or improver effect.
HyperAgents / transfer U07 *Positive direction* Whole transferred hyperagent under frozen-meta probe	5 reported runs; selected source agents	Target validation selects descendants; test grades score them	Median 0.00 [0.00,0.13] versus 0.63 [0.54,0.63], bootstrap 95% CIs. Different initial implementations prevent an isolated improver comparison.
ADMET-EvO / recipes U13 *Positive direction* Compute-aware versus score-only retained recipes	22 endpoints; full-agent reruns not extracted	Protected post-freeze recipe readout; protocol versions differ	Reported fitting-time reduction 72.2%; 20/22 non-inferior endpoints. Full intervals not extracted; excludes total research cost.

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Table 16 (continued)

Study and comparison	Run unit	Confirmation	Uncertainty and inference
MolRSI / antibacterial memory U21 *Positive direction* Prior memory versus no prior memory	1 search per arm per target; 6 targets; 5 data partitions	Test scores excluded from search feedback	Six positive AUPRC directions. Partition SD does not quantify search variance; source-memory formation cost excluded.
MolRSI / pMHC memory U22 *Positive direction* Miniprotein memory transferred to target search	One selected workflow per arm; reruns undetermined	gp100 development followed by MART-1 test	AUPRC 0.0125 to 0.0172; between-search uncertainty undetermined. Same target allowance is not total-cost matching.
SwarmRSI / graph ablation U23 *Positive direction* Shared-graph organization/reuse ablation	90 tasks; 3 seeds; inference pairs tasks	NatureBench task scores; separate final audit not established here	Mean-g difference 4.7 points; task-bootstrap 95% CI [0.3,9.3]. Organization and reuse change together.
MASS / workflow transfer U24 *Positive direction* Generation workflows, same final executor	12 tasks, 11 iterations; 1 execution per pair	Same task identities; independent task transfer not established	Mean win share 0.64. Six ratings per pair are not executions; generation changes several roles, so the improver effect is unresolved.
DiscoveryRSI / generations U25 *Positive direction* Sequentially trained checkpoints	241 questions; independent training reruns not extracted	Automated rubric; not experimental discovery confirmation	4.86 to 5.18 to 5.39, plotted 95% intervals. Novelty/importance decline; numeric CI endpoints not extracted.
GNoME / active-learning rounds U27 *Positive direction* Predictor plus expanding active-learning dataset	6 rounds; independent retraining reruns not extracted	Composition-disjoint computational reference	MAE 21 to 11 meV/atom; intervals not extracted. Retained benefit is a data/model package, not proof of synthesis or a better updater.
RoboCat / self-generated data U28 *Positive direction* Demonstrations with versus without self-generated data	1 model; 100 evaluation episodes per task	Four hardware tasks; training uses learned success detection	Higher success on four tasks; intervals absent in inspected extraction. Episodes do not measure training-run variance.
Adaptive domain randomization / large policies U29 *Positive direction* Adaptive versus manual domain randomization	1 training run per arm; 10 real trials each	Hardware block reorientation after simulation training	2.7 (SE 1.1) versus 13.3 (3.6) successes. Large arm matches compute/time; XXL result does not share that match.

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Table 16 (continued)

Study and comparison	Run unit	Confirmation	Uncertainty and inference
VLAW / two rounds U30 *Positive direction* Two rounds with matched real input, extra synthetic data	50 evaluations per task/method; training-rerun variance not extracted	Five hardware task categories	0.868 versus 0.752 after round 2. Interval not extracted; synthetic exposure/compute differ despite equal real input.
ASPIRE / frozen library U31 *Positive direction* Frozen repair library of increasing size	Held-out tasks; repeated training/search runs not established	No debugging or retries during held-out execution	4.7% to 30.5% as source tasks increase. No uncertainty in extracted record; frozen execution does not test a return path.
ScienceRSI / transfer recipe U34 *Positive direction* Selective versus dense transfer inside the same RL regime	One selected 4B checkpoint evaluated per arm	Internal-dev checkpoint selection; external benchmark readout	Science groups 55.0/59.3 versus 50.1/56.4. No seed SD or paired CI; a retained model package, not a repeated-cycle return-path effect.

Note. Study names and row IDs link to Table 13 and its source-located cell records. Outcome direction is distinct from design, repetition, confirmation and uncertainty.

10.2 Cross-Case Contrasts: Timing, Selection, and Coupled Inheritance

Across the selected configurations, the transferable object is an evaluation question rather than a common scientific score. Retrospective biological replay tests the use of supplied outcomes, whereas interactive laboratory and field policies also change which observations become available (Gupta et al., 2025; Wainrib et al., 2026; Szymanski et al., 2023; Berild et al., 2024). Early feedback can guide selection before a device-level endpoint is known, and a later measurement correction can revise support for a reported outcome (Attia et al., 2020; Dave et al., 2022; Szymanski et al., 2026). These conditions distinguish evidence use, acquisition, retained capability, and improvement procedures. The common framework aligns claim, consumer, comparator, and validation scope; endpoints and scientific assumptions remain domain-specific. Table 17 connects each contrast to a practical control. The cases motivate prospective tests; they do not identify causal moderators across disciplines.

Controls must match the claimed stage. The battery search's 16 days exclude predictor training and later validation; Clio's conductivity loop does not consume subsequent pouch-cell outcomes, and Koscher's later partition-coefficient underprediction limits extrapolation from local predictive gains (Attia et al., 2020; Dave et al., 2022; Koscher et al., 2023a). Reporting both proxy and delayed endpoint, including unresolved outcomes at the deadline, addresses this timing distinction. Fixing acquired data instead isolates evidence use while removing acquisition's effect; allowing trajectories to diverge estimates the interactive policy package. Neither supplies unobserved physical outcomes without additional assumptions (Section 10.6.4).

Matching also has a stage boundary. MolRSI inherits solutions, refinements, and execution experience together; SwarmRSI preserves content while changing organization and reuse; MASS fixes final execution while varying discovery roles during formation (Zhang et al., 2026c; Meng et al., 2026b; Lee et al., 2026). These are different package effects. Holding raw history and compatible interfaces common can isolate how a procedure turns evidence into updates; removing required knowledge would instead test a broken interface.

Heterogeneous evidence motivates conditional tests. STOP, MetaRSI, AIDE^2, and AlphaEvolve include positive, degraded, and inconclusive process comparisons, without defining a common effect size (Zelikman et al., 2024; Tan et al., 2026; Srikanth et al., 2026; Novikov et al., 2025). Appendix E, S9 distinguishes executed source-dependence accounting from an interpretive map of broader synthesis claims. The joined evidence trail in Table 2 is inspectable, but its superiority over alternative review frameworks remains untested. H1–H4 are consequently proposed studies rather than conclusions estimated from these effects.

Table 17. Evidence trails for cross-case conclusions.

Contrast	Observed configuration	Boundary or counterexample	Inference and next control
Feedback channel versus information benefit	U10 tests genuine versus permuted outcomes in Cell Painting replay.	U09 reports similar means in another configuration; this is not equivalence or proof of non-use.	Benefit is configuration-specific. Match supplied information and perturb its content or availability; cross-paper differences do not identify moderators (Section 10.6.4).
Delayed endpoint versus fast proxy	U15 uses early-life predictions; U16 optimizes conductivity before cell testing.	U12's partition predictions are systematically biased in a later exploitation study despite local predictive gains.	Prespecify confirmation and delay rules; report pending endpoints alongside proxy gains (H1, Section 10.3).
Execution versus usable evidence	U14 separates recipe attempts, assigned targets, and corrected phase support.	U17 acquires selected field locations; neither case supplies missing outcomes at unobserved interventions.	Separate attempts, completion, valid observations, and target outcomes; compare policies prospectively when acquisition changes (Section 10.6.4).
Information versus inherited procedure	U21–U22 compare source-memory packages; U23 changes graph organization and reuse with content preserved.	U24 matches final execution but changes several roles during workflow formation.	Match raw history and compatible starts when comparing update procedures; final-executor matching alone is insufficient (Section 11.2).
Process editing versus improver efficacy	U01 and U04 report positive bounded improver contrasts; U33 tests meta-prompt co-evolution as a package.	U02 reports degradation and U05 an inconclusive common-start test.	Cross earlier/later procedures with common starts, retain harms and failures, then intervene during formation for origin claims (Section 11.2.2).
Correction versus capability repair	U14 has an external correction; the review-artifact case executes a recoding dependency.	The affected A-Lab assignment-to-policy edges remain unknown.	Trace actual consumption and test fresh-task utility after repair, including independently supported controls (H3; Section 10.7).

Note. "Boundary" includes a counterexample, an unresolved comparison, or a different endpoint; it is not uniformly a negative result.

10.3 Four Falsifiable Design Hypotheses

These proposals concern resource allocation (H1), reliable inheritance (H2–H3), and editable scope (H4). None alone identifies recursive origin; that requires the formation-stage intervention in Section 11.2.2. For each study, prespecify population, moderators, intervention magnitude, endpoint, resource rule, and analysis before target evaluation.

H1: Acquisition adaptation yields greater confirmed utility than generator enlargement at specified equal incremental investments when useful candidates are available and measurement is expensive. Battery optimization and adaptive sampling motivate the question, not its answer (Attia et al., 2020; Berild et al., 2024). Use development-only checks of quality, feasibility, diversity, and coverage to define candidate-sufficient and candidate-limited strata; define measurement cost and delay independently of later policy outcomes. At each point in a preregistered resource grid δB, compare a named acquisition intervention with a named generator intervention from the same base system. Include intervention training and use, charge failures, and share confirmation and deadline rules. If resources cannot be matched componentwise, declare conversion weights or a Pareto comparison. A common ceiling alone does not match intervention magnitude. H1 predicts a positive acquisition-minus-generation utility contrast in the candidate-sufficient,

expensive-measurement stratum and a smaller or reversed contrast in the candidate-limited stratum. Report absolute gains and the interaction; contrary contrasts refute H1 within the tested range.

H2: Evidence-grounded applicability conditions improve useful method transfer under a declared shift. Compare one procedure with absent, source-grounded, or irrelevant conditions matched for length and retrieval cost (Jin et al., 2025a; Meng et al., 2026b). Freeze conditions developed on source evidence, then assess confirmed utility and harmful misapplications in both in-scope and shifted contexts. Improved refusal alone is insufficient: prespecify acceptable loss of useful coverage and costs. This tests condition content. A separate comparison of fixed and adaptive condition-construction procedures, given the same evidence and interfaces and charged for construction, tests adaptive management. Sufficiently precise results excluding the prespecified gain over an information-matched fixed procedure within the coverage and cost bounds would count against the latter claim; wide uncertainty remains inconclusive.

H3: Dependency-aware correction reduces invalid carryover while preserving independently supported capability. Supply the same adjudicated correction to source-record repair, dependency-aware review, and an information-matched maintained-library or retraining control (Szymanski et al., 2023, 2026). Prespecify affected dependencies, unaffected controls, and maintenance actions; charge the human reference trace. On fresh tasks, compare obsolete recommendations, retained utility, recovery time, and total cost. A descendant warrants review, not automatic deletion. Missing provenance, entangled parameters, or indiscriminate withdrawal may defeat the predicted advantage. Sufficiently precise results excluding the prespecified carryover reduction within the capability-loss and cost bounds would count against H3 in that setting; wide uncertainty remains inconclusive. Successful record repair alone does not establish an improved correction procedure.

H4: Broader editable scope helps more when the bottleneck lies outside the restricted editor's interface. Compare compatible broad and restricted editors at equal search and verification budgets (Zhang et al., 2026a; Ahmed et al., 2026). Define inside/outside-interface strata by known interventions or development-only diagnosis; both editors must retain a meaningful response in each stratum. H4 predicts a larger broad-editor advantage outside the restricted interface. Report absolute utility, regressions, failed edits, and verification costs: a positive interaction with negative absolute gains does not warrant deployment. Coherent component interventions can locate useful edits; otherwise the effect concerns the package. To test recursive origin, intervene on the named return path during editor formation rather than only comparing selected final editors.

10.4 Formal and Physical Sciences as Comparative Tests

10.4.1 Formal mathematics and algorithms: strong verifiers, bounded claims

Formal domains offer inspectable feedback. AlphaTensor learns search policies for algebraically checkable tensor decompositions; AlphaDev discovers computational routines incorporated into a software library (Fawzi et al., 2022; Mankowitz et al., 2023). Their infrastructure gains do not alone identify acceleration of the research systems that produced them.

FunSearch retains executable programs to discover mathematical constructions; valid constructions need not be globally optimal (Romera-Paredes et al., 2024). AlphaEvolve additionally co-evolves meta-prompts. Its tensor-decomposition ablation is now extracted as U33, a bounded process-package comparison; it does not establish unlimited improvement or identify every component's contribution (Novikov et al., 2025).

Theorem proving contributes a complementary form of closure. AlphaGeometry learns auxiliary constructions from synthetic theorems and proofs, with symbolic deduction constraining the result. Its achievements apply to a representable geometry fragment, rather than every kind of olympiad mathematics (Trinh et al., 2024). AlphaProof combines formal verification, reinforcement learning, and test-time adaptation over generated problem variants (Hubert et al., 2026). These systems demonstrate why self-generated training material can be productive when checking is strong. However, learning from generated exercises, learning a transferable search policy, and learning a better procedure for generating and selecting exercises remain analytically distinct.

A proof checker validates a formal statement, not its fidelity to the intended question or its mathematical importance. Machine-guided conjecture development with human proof illustrates the complementary collaborative unit (Davies et al., 2021).

10.4.2 Physics: equations, representations, designs, and control

SINDy and symbolic equation discovery exploit specified function spaces or structural assumptions (Brunton et al., 2016; Udrescu and Tegmark, 2020; Schmidt and Lipson, 2009). Recovering an equation under those conditions does not alone identify a new law, a causal effect, or revision of the search assumptions.

Symbolic distillation from structured neural models includes relations obtained from cosmological simulation; SciNet recovers useful variables in toy settings (Cranmer et al., 2020; Iten et al., 2020). These results separate interpretable representation from external empirical adequacy.

There are also important precedents for persistent scientific memory. Wu and Tegmark's AI physicist stores and unifies theories in a theory hub and reuses them across toy environments (Wu and Tegmark, 2019). MELVIN reports learning from simpler quantum-experiment solutions to accelerate searches for more complex experiments (Krenn et al., 2016). These mechanisms matter because the retained object is useful to future inference or search, rather than merely documenting a completed result. Their restricted environments bound the claim, but do not make the evidence irrelevant to scientific self-improvement.

PyTheus produces graph-based quantum-optics designs (Ruiz-Gonzalez et al., 2023); physical realization is a separate outcome. Tokamak policies supply external hardware evidence after simulation training (Degrave et al., 2022). Neither design validity nor runtime control alone measures improvement of the scientific research procedure.

10.5 Robot Learning as a Comparative Test: Self-Generated Experience and Costly Trials

Robot learning separates four editable objects: policy, success criterion, training distribution, and evaluation protocol. Policy is capability C_t; rewards and curricula can function within U_t when consumed by later learning. Their role depends on stage and consumer. The selected configurations illustrate four mechanisms and their evaluation boundaries, not a field-wide census. Table 18 provides phase-level navigation; Appendix C.6 retains the detailed comparisons, trial counts, and source-specific qualifications.

10.5.1 Self-generated experience under a fixed update procedure

A return path through data can improve capability without changing the update rule. RoboCat retrains on demonstrations and its own collected episodes under a fixed adaptation recipe; its reported one-iteration comparison concerns retained capability, not old/new improver performance (Bousmalis et al., 2023). SIME shows that which self-generated segments are retained matters, while AutoRT separates collection from the control policy (Jin et al., 2025b; Ahn et al., 2024). RECAP restarts model weights but retains data across offline rounds; RISE holds its dynamics and progress models fixed; VLAW updates both world model and policy while fitting its reward model separately (Physical Intelligence et al., 2025; Yang et al., 2026a; Guo et al., 2026). These are different information paths, not evidence that the rules producing updates have themselves improved.

When a robot also chooses tasks, successive success rates mix capability with task difficulty and coverage. A retained-capability test needs a task distribution fixed before the loop and an account of rejected tasks, failed collections, and reset failures. Bongard's self-modeling robot offers an earlier controlled example of action selection for model discrimination, but its advantage is conditional on the supplied model class and random-action baselines (Bongard et al., 2006).

10.5.2 Editing rewards and training distributions

Eureka revises reward code using task fitness and reward-component feedback; its reflection ablation tests that channel, while the benchmark fitness, language model, and outer search procedure remain fixed (Ma et al., 2024a). DrEureka and Eurekaverse extend the process to randomization and curricula, exposing hardware selection costs and failures that belong in the evaluation (Ma et al., 2024b; Liang et al., 2024). AURA retains configurations with human feedback, but its curriculum-level launch rate cannot be compared directly with environment-level rates (Zhu et al., 2025). Fixed-rule domain randomization can also improve outcomes without revising the adaptation rule; RoboGen and DreamGen generate training experiences without an inspected earlier/later proposal-procedure comparison (Akkaya et al., 2019; Wang et al., 2024; Jang et al., 2025). Editing a downstream learning input and improving its generator are distinct claims.

Table 18. Robot-learning configurations by phase and consumer.

Configuration	Retained object and stage	Reported comparison and inference boundary
RoboCat (Bousmalis et al., 2023)	Policy/data; fixed adaptation recipe.	Demonstrations versus demonstrations plus self-generated data. Retained-capability evidence; no recorded old/new-improver snapshot contrast.
RECAP (Physical Intelligence et al., 2025)	Data, value function, and policy; iterated offline updates.	One/two rounds versus supervised fine-tuning. Package benefit under the reported feedback and human-input conditions.
Eureka (Ma et al., 2024a)	Reward code consumed by policy training.	Human versus searched reward and reflection ablation. Five iterations per search, 16 candidates per iteration, five independent searches. Reward revision is not itself a comparison of reward generators.
DrEureka (Ma et al., 2024b)	Reward and randomization configuration.	Sim-to-real outcomes under learned versus human-designed configurations. Hardware-based selection of the best policy is part of the procedure and cost.
ADR (Akkaya et al., 2019)	Randomization state under a fixed threshold rule.	Equal-wall-clock comparison against manual randomization. Evidence for adaptive training, not revision of the threshold rule.
ENPIRE–construction (Xiao et al., 2026)	Environment, reset, safety, verifier; human-assisted setup.	Editable procedures are finalized before policy improvement. A construction phase, not evidence of continual online verifier revision.
ENPIRE–policy (Xiao et al., 2026)	Policy-development code; immutable Gym interface.	Executed training-process revisions. A fixed verifier does not rule out improver gain; the relevant comparison must concern the revised procedures.
VLAW (Guo et al., 2026)	Policy/world model; reward model fitted separately.	Two-round package comparison against filtered behavioral cloning. Distinguish world-model and policy effects from verifier calibration.
ASPIRE (Lu et al., 2026b)	Frozen library of repair guidance at transfer.	Later-task comparison with/without retained guidance. Its role is determined by the consuming execution or improvement process.

Note. "No recorded snapshot contrast" concerns the inspected comparison, not the entire robotics literature.

10.5.3 Construction and policy improvement are separate phases

ENPIRE first uses human feedback to develop safety, reset, and verification procedures, then freezes them as a Gym interface while agents edit policy-development code and hyperparameters (Xiao et al., 2026, §§2.1–2.2). The first phase is human-assisted infrastructure construction; the second is process editing under a fixed interface. Pooling them would imply unsupported autonomous online verifier evolution. Conversely, a fixed interface or final audit does not preclude improver gain: compare the audited gains produced by earlier and later procedures from compatible starts.

10.5.4 Physical trials constrain verification

Physical evaluation is scarce and noisy. Studies often use fewer than 50 rollouts per condition (Vincent et al., 2024); for illustration, 16 successes in 20 independent Bernoulli trials give an exact 95% interval of $[0.56, 0.94]$. Appendix C.6 gives the assumptions, additional precision calculations, and the larger paired TRI study (TRI LBM Team et al., 2025). More trials alone do not resolve simultaneous changes in data normalization, tasks, or evaluation conditions.

Cheaper verifiers need setting-specific calibration. SIMPLER, AutoEval, and RoboArena evaluate different simulation, classifier, and ranking instruments; strong agreement in one setting does not establish interchangeable endpoints elsewhere (Li et al., 2024; Zhou et al., 2025; Atreya et al., 2025). When a learned verifier both labels training data and scores the resulting policy, policy improvement and verifier drift may be confounded.

Independent human counts on an audit subsample are one safeguard, not the only validation design. Training and final audit compete for robot hours, motivating explicit allocation; H1 additionally requires evidence about candidate supply and the bottleneck.

10.6 Embodied Discovery: Execution, Observation Validity, and Inheritance

A laboratory robot affects both the intervention and whether its outcome becomes usable evidence. Improved execution, acquisition, and scientific interpretation therefore require separate endpoints, even when they coexist in one platform.

10.6.1 A-Lab: target realization is not scientific novelty

A-Lab's original report, correction, and external criticism concern distinct propositions. Table 9 preserves the materials evidence chain; Table 19 separates the denominators needed here to evaluate execution, target realization, and scientific novelty. In particular, Leeman and colleagues' three potentially successful predicted syntheses are previously reported compounds, not "three new materials" (Leeman et al., 2024, §§IV.H–V).

Table 19. Distinct A-Lab propositions and denominators.

Source / record	Proposition and unit	Interpretation permitted / not permitted
Original campaign (Szymanski et al., 2023)	41/58 target compounds under the original characterization.	Historical reported target realization; not the current corrected estimate and not a novelty rate.
Corrected campaign (Szymanski et al., 2026)	36/57 targets in the revised account; 36 supported and four inconclusive among 40 reassessed assignments.	Revised target-identity assessment. Inconclusive does not mean disproved, and target identity does not establish literature-relative novelty.
External criticism (Leeman et al., 2024)	Three potentially successful predicted syntheses; all previously reported.	A source-specific synthesis interpretation and a separate rejection of new-discovery status. Not a lower bound on the same quantity measured by 36/57.
Recipe-level campaign log (Szymanski et al., 2023)	353 recipe attempts, not 353 independent compound targets.	Workload and attempted interventions. Valid-observation and confirmation rates require additional event-level denominators.

Note. These entries preserve source-specific claims and denominators; they are not independent replications or endpoints on one success-rate scale.

We do not adjudicate every crystallographic disagreement. The supported inference is narrower: target realization, phase assignment, purity, and novelty relative to a declared database or literature date are different propositions. A range formed by combining 3/58 and 36/57 as a "new-material rate" would be invalid. Record a novelty predicate $N_{\mathcal{R},d}$ only after specifying reference collection $\mathcal{R}$, date d, and equivalence criterion. Database absence and a newly executed synthesis are not substitutes for that predicate.

The separate inheritance question is which retained artifacts actually consumed revised assignments. Section 10.7 traces the supported and unknown edges; a correction triggers dependency review, not automatic invalidation of every descendant or of all nine active-learning-associated targets.

10.6.2 Informative observations and research methods

Robot-scientist and mobile-chemistry systems connect experimental decisions to laboratory execution (King et al., 2009; Burger et al., 2020). Hypothesis-learning microscopy instead uses competing physical models to guide acquisition, and can reveal inadequacy in the supplied hypothesis set (Ziatdinov et al., 2022). These are different scientific aims, not an ordering in which only explicit hypothesis discrimination counts as a method. Improved prediction, calibration, feasible sampling, or fault handling can be methodologically valuable when evaluated against the corresponding endpoint.

Likewise, an updated Gaussian-process posterior remains a retained information state when the fitting and acquisition map is fixed. It does not become a changed update rule merely because the measured quantity concerns competing explanations. Conversely, a learned acquisition rule can be a component of U even when

its objective is predictive accuracy rather than explanation selection. Apply the interface-and-consumer procedure in Section 3.1.4 to both microscopy and adaptive environmental observation (Berild et al., 2024). Evidence content changes what can be learned; it does not by itself change the functional definition of an improver.

10.6.3 What transfers across campaigns and platforms

Executable chemistry illustrates several inheritance objects. XDL separates a procedure from its hardware implementation, while compatibility checks can reveal unsupported operations (Mehr et al., 2020). RoboChem demonstrates optimized recipes and scale-related testing within its experimental setting (Slattery et al., 2024). RoboChem-Flex transfers data into a multi-task surrogate for a subsequent substrate campaign (Pilon et al., 2026). Procedure reuse, dataset transfer, and a revised transfer rule are distinct interventions. A short second campaign shows that reuse occurred; without the corresponding no-transfer comparison it does not quantify the saved experiments. A compatible instruction set establishes executability, not automatically equivalent chemistry.

Manipulation can reduce sample loss while selection remains fixed; changed calibration can alter observation validity. Transferred artifacts should retain instrument, operating range, sample conditions, calibration version, and consumer, with both execution and scientific endpoint tested after a shift.

10.6.4 An event-based account of observation selection

Fix a history and proposal policy. Let ξ denote a candidate intervention and its declared context, and let $A = 1$ indicate an attempted intervention. Distinguish E (the protocol reaches its prespecified completion point), V (the target observation passes validity checks), and Y (the target criterion is met). For the target-observation analysis, define $V = 0$ when the required measurement is unavailable; other failure logs can still be useful for execution learning. A valid negative result has $E = V = 1$ and $Y = 0$. Missing or invalid target measurements do not imply $Y = 0$.

Let $q(\xi)$ be the candidate distribution among attempts and define

$$s_C(\xi) = \Pr(E = 1, V = 1 \mid A = 1, \xi, C), \qquad q_C^{\mathrm{obs}}(\xi) = \frac{q(\xi) s_C(\xi)}{\int q(u) s_C(u)\, du}, \tag{2}$$

provided the denominator is positive. This is a conditioning identity for a fixed proposal distribution, not a model identifying why observations are missing. If $s_{C'}(\xi) = k s_C(\xi)$ on the relevant support, with no clipping above one, $q_{C'}^{\mathrm{obs}} = q_C^{\mathrm{obs}}$. Equal additive increases generally do not preserve it: for two equally proposed candidates, changing completion-and-validity probabilities from $(0.2, 0.8)$ to $(0.3, 0.9)$ changes the first candidate's observed share from 0.20 to 0.25. These numbers are an analytical illustration, not experimental results.

Consequently, recipe attempts are draws from q; usable target observations are selected by E, V; confirmed positives are additionally selected by Y; and novelty requires a further $N_{\mathcal{R},d}$ judgment. A-Lab's 353 recipes cannot all be labeled draws from q_C^{obs} without per-attempt validity information. Its unachieved targets may include valid negative observations. The mobile chemist's discarded results are not usable observations for the discarded endpoint, although their failure logs may inform execution (Szymanski et al., 2023; Burger et al., 2020). Target counts cannot be substituted for recipe-level denominators. Report n_A, n_E, n_V, n_Y with the unit, nesting rule, and reasons for missingness; report novelty only when it is assessed.

A support gap with $s_C(\xi) = 0$ is not repaired by further fitting only on completed observations. Recovering a target-population estimand from selected observations would additionally require appropriate support, measured selection variables, and identifying assumptions. The identity alone does not establish ignorability, identify missing outcomes, or justify inverse weighting. When C also changes the proposal policy, both q and s_C change; their effects cannot be assigned to execution from completed-case accuracy alone.

10.6.5 Proxy validity and the limits of label-correction replay

Proxy quality is not one universal "survival rate." Correlation of simulated and physical scores, precision among screened candidates, target-confirmation fractions, and calibration error answer different questions. SIMPLER reports a mean of per-task Pearson correlations of 0.924 for its visually matched Google-robot comparisons, not a novelty fraction or a rank correlation (Li et al., 2024). A-Lab's corrected target count

estimates another object. Preserve the metric, population, and confirmation procedure instead of pooling them.

A useful **fixed-history sensitivity analysis** can substitute revised labels into the same logged state and recompute a recipe ranking. It identifies how the decision function responds to the label change under that history. It does not reconstruct a full counterfactual campaign: after the first newly selected, previously unexecuted recipe, its physical outcome is unavailable in the log. Few changed rankings do not establish unchanged scientific utility; many changed rankings do not establish that all nine reported active-learning-associated targets depended on invalid labels.

A stronger **offline policy evaluation** needs action support, a defined logging policy, appropriate assumptions about assignment and outcomes, and uncertainty that reflects those assumptions. Where deterministic logs lack overlap, the corresponding counterfactual is not identified by replay. A **prospective comparison** can instead randomize compatible campaigns to outcome-only versus provenance-aware policies, fixing initial candidate spaces, time and resource ceilings, and independent confirmation rules. Include downtime, resets, invalid measurements, and human adjudication. This compares information use or acquisition policies, not necessarily improved improvers. Compare two policy-generation procedures from common starts; add a formation-stage intervention when attributing their difference to a recursive return path (Section 11.2). These are proposed analyses and experiments; none was executed for this survey.

10.7 A Published Correction and an Executed Review-Artifact Dependency Case

A changed source does not by itself establish a changed research capability. Two cases distinguish publication-level correction from an executed review-artifact dependency. The companion graph contains 17 versioned nodes and 15 typed edges, with support, revalidation status, and edge locators. This is an application of provenance and truth maintenance, not a new representation theory (Moreau and Missier, 2013; Doyle, 1979; Buneman et al., 2001).

A-Lab: a real correction with an unresolved execution lineage. The January 2026 amendment reports manual reassessment of diffraction patterns associated with 40 positive assignments: 36 were supported and four remained inconclusive. It also removes $Zn_2Cr_3FeO_8$ because of its inclusion in training data and qualifies novelty as relative to the prediction platform (Szymanski et al., 2026). The corrected article reports 36 realized targets out of 57 (Szymanski et al., 2023). These are distinct operations: reassessing assignments, changing target scope, and clarifying novelty. Neither inconclusive identity nor reference-relative novelty means that every synthesis result is invalid.

Table 20 separates verified publication relations from unknown execution links. We did not reconstruct specimen identities or reanalyse diffraction data. The sources warrant replacing the summary, but do not identify which revised assignments entered which ARROWS3 state or later recipe. They therefore establish neither damage to nor repair of active learning, including the nine active-learning-associated outcomes.

RoboChem-Flex: executed repair of a review interpretation. The preserved R2 card `EMBODIED-ROBOCHEMFLEX-TRANSFER` classified prior-campaign data reuse and a later 90% isolated yield after eight experiments as supported retained-capability benefit. The inspected source describes reuse without a matched no-transfer comparison (Pilon et al., 2026). R3 corrected the type to *mechanism*; R4 preserves that corrected card and the reported yield.

An identical query selects cards with retained-capability type and supported status. Applied to the preserved and corrected cards, its answer changes from true to false. Two dependent review views therefore require recomputation. They were constructed for the R3 demonstrator, not recovered as pre-existing R2 outputs. This repairs our interpretation, not the experimental paper; descriptive yield retains source support independent of the erroneous benefit inference, not independent laboratory replication.

What the replay establishes. The supplied `tools/replay_dependencies.py` follows declared verified dependencies, stops at unknown links, and preserves version replacements. It checks graph references, hashes, expected review sets, and the actual before/after card query. The result is one source-summary review target, two derived review targets, and two unresolved A-Lab execution edges. Traversal creates a review obligation; it

Table 20. Correction dependencies and permitted conclusions.

Dependency	Verification	Permitted conclusion
Reported XRD set → manual reassessment → revised assignments	Published-source relation	The reported support status changed; raw-data reanalysis by this review is absent.
Prior positive assignments → published campaign summary	Published-source aggregation	Review and replace the affected summary, preserving the corrected numerator and target scope.
Affected assignment ⇢ ARROWS3 state ⇢ later recipe	Unknown event-level edges	Stop propagation here. Generic feedback use does not establish this particular consumption path or its effect.
R2 claim card → benefit-selection view → downstream eligibility summary	Executed review-artifact derivation	Recompute the two derived review objects after correcting the claim type; preserve the source's descriptive outcome.

Note. Verified means supported within the stated source or artifact scope; it does not mean independently replicated scientific truth. Full node and edge IDs are supplied in the dependency artifact.

does not automatically invalidate every dependent claim. Remaining support can justify retention, restriction, revalidation, or replacement.

The executed example establishes reproducible repair within a review artifact. A scientific-agent repair result would additionally require versioned measurement-to-update consumption, an affected retained artifact, and downstream assessment before and after a justified repair. Those records and experiments are absent here. Consequently, neither behavioral recovery nor superiority over existing provenance frameworks is claimed.

10.8 Transfer Boundaries and Additional Scientific Domains

Name the object crossing the boundary: a representation, answer, executable tool, diagnostic method, or updater. Whole-system transfer does not isolate methodology; removing knowledge or interfaces needed to execute a method can create an incoherent comparison. Preserve compatibility and state which component effects remain unidentified.

The five domain chapters share mechanisms and evidence records. Their primary placement follows the scientific question: electrode composition and microstructure primarily concern materials, whereas charging and device degradation concern energy; molecular function primarily concerns life sciences, whereas representation, reactivity, and synthesis concern chemistry. A study may inform several comparisons while retaining one canonical evidence record.

Formal mathematics and computational physics provide unusually inspectable verification or simulation interfaces. Their local tests neither require physical experiments in every field nor automatically validate empirical claims outside those interfaces. Astronomy, geoscience, and agricultural science offer future tests of scarce interventions, inverse problems, and seasonal variation. These are expansion directions, not comprehensive coverage here. A new main chapter requires a distinct evidence regime, reviewed configurations, and specialist synthesis; nominal breadth alone adds little.

11 Evaluating Scientific and Recursive Progress

11.1 Define the estimand before choosing the benchmark

Select the claim family first: scientific output, feedback use, retained-capability benefit, improver gain, or recursive attribution. Figure 3 connects each claim to its controls, endpoint, independent unit, and inferential limit. These are alternative audit questions, not successive levels. Enter the chosen question in the update contract (Eq. 1) before selecting a benchmark or inspecting the result.

Name the intervention, target population, and resource constraint. For a frozen-checkpoint audit, suppress persistent updating but allow ordinary reasoning and working memory. Let Z denote a research instance with distinct permitted development information and protected assessment evidence. For expected audited utility

J at budget B, compare later capability and knowledge (C_t, K_t) with their initial versions:

$$\Delta_{\text{checkpoint}} = \mathbb{E}_Z[J(C_t, K_t; Z, B) - J(C_0, K_0; Z, B)].$$

This conditions on selected checkpoints and can combine method gains with additional knowledge. For that conditional comparison, independent target instances or independently assigned experimental batches can estimate performance uncertainty, subject to their sampling and shared dependencies. They do not replicate checkpoint construction. Evaluating the algorithm that produced the checkpoints instead requires independent complete adaptation runs or campaigns and a fixed checkpoint rule; assessment instances are nested within, or crossed with, those runs. If adaptation remains enabled, include its updater in the evaluated system. Method-transfer controls must preserve procedural content while withholding target solutions: clearing all memory erases the treatment, whereas carrying solved targets contaminates the comparison.

11.2 Compare improvers from a common start

Let $\mathcal{P}(U, A; Z, B, r)$ apply improver U to learner artifact A on instance Z, under resource constraint B and randomness r. The process accesses only the permitted information within Z; the audit uses its protected evidence. Compare earlier and later improver snapshots from the same start:

$$\Delta_U = \mathbb{E}_{Z,r}[J_{\text{audit}}(\mathcal{P}(U_t, A_0; Z, B, r); Z, B_{\text{eval}}) - J_{\text{audit}}(\mathcal{P}(U_0, A_0; Z, B, r); Z, B_{\text{eval}})]\,. \tag{3}$$

Pair randomness where meaningful; otherwise average separately by arm. Improvement and audit budgets, access, and assessment conditions are common. This operationalizes an existing common-start principle (Wu et al., 2026; Chen et al., 2026c). It conditions on the selected snapshots and does not establish their recursive origin or indefinite acceleration. A positive relative advantage can also mean less degradation. Report both improvers' absolute gains:

$$G(U; A_0) = \mathbb{E}_{Z,r}[J_{\text{audit}}(\mathcal{P}(U, A_0; Z, B, r); Z, B_{\text{eval}}) - J_{\text{audit}}(A_0; Z, B_{\text{eval}})]\,. \tag{4}$$

For initial, old, and new scores of 90, 70, and 85, respectively, the new procedure has relative advantage 15 but absolute loss 5.

A snapshot fixes initial code, parameters, and persistent state; within-process adaptation can remain part of $\mathcal{P}$. Freezing changes to the improver tests a fixed procedure's proposal or search quality, which can still involve multiple steps. One-step proposal quality requires an explicit single-step design. Inseparable research and improvement components require compatible package comparisons.

11.2.1 A common start does not establish method transfer

Repeated tasks at one start cannot test portability to another researcher. Declare a target distribution Q over initial systems and an interface contract before inspecting comparison outcomes:

$$\Delta_U^Q = \mathbb{E}_{A\sim Q,Z,r}[J_{\text{audit}}(\mathcal{P}(U_t, A; Z, B, r); Z, B_{\text{eval}}) - J_{\text{audit}}(\mathcal{P}(U_0, A; Z, B, r); Z, B_{\text{eval}})]\,. \tag{5}$$

Cross both improvers with the original and independently constructed starts, repeat complete processes, and block task families. Report start-specific contrasts as well as the average: benefit to a weak start can conceal harm to a strong one. Compatibility repairs must be prespecified, charged, and treated as part of the intervention. Selecting only starts on which the final improver executes would change Q after observing treatment outcomes. Use the failure-aware rule below rather than that exclusion. A laboratory-specific method may retain its narrower scope.

11.2.2 Recursive origin: intervene while the improver is formed

This proposed two-stage protocol tests whether enabling a named return path produces more effective improvers. It is a research design, not an experiment performed by this survey. Here subscripts $z \in \{0, 1\}$ denote formation arms rather than earlier and later snapshots.

Before either stage. Register the source and target populations, target-start distribution Q, interface schema, tool permissions, protected audit, resource ceilings, checkpoint-selection rule, and primary endpoint. Construct or sample target starts independently of formation outcomes; keep target audit tasks inaccessible during formation. Compatibility is a requirement of the registered interface, not a filter applied to successful final procedures. Fix fallback behavior, failure penalties if used, and treatment of missing measurements before randomization.

Stage 1: randomized formation. From initial state H_0, randomize independent complete formation runs to a named return path enabled ($z = 1$) or disabled ($z = 0$):

$$(A_z^{\mathrm{f}}, U_z) = \mathcal{F}(z; H_0, B_f, r_f). \tag{6}$$

Specify admissible edits, observations, and evaluator protection. A DGM-inspired control keeps the initial modifier responsible for all descendant edits; the enabled arm permits descendants to become modifiers. Archive format, parent eligibility, task-access rules, and descendant retention remain common. This changes a role rather than deleting all memory. Failed formation returns a declared failure state $U_\perp$ and the prespecified safe retained artifact, and remains in the assigned arm.

Match ex ante resource constraints and accounting, not realized costs or post-treatment histories. Interactive arms may collect different experience under common permissions; that effect includes acquisition. A separate coherent common-data intervention can isolate use of supplied experience. Charge unsuccessful edits, verification, and meta-training. Sham computation cannot compensate for lost information.

Stage 2: independent target assessment. Save the initial snapshot of *every* formed improver and apply it to the same registered target-start and task design at budget B_t. Let $\mathcal{P}^\dagger$ include a prespecified fallback: an unusable improver, incompatible output, crash, or exhausted update budget returns the unchanged target start, records a failure indicator, and retains all spent costs. If that fallback is unsuitable, specify an alternative safe action and its loss before formation. Do not drop failed runs or redefine Q around their survivors. The target effect is

$$\Delta^U_{\mathrm{origin}} = \mathbb{E}_{r_f, A\sim Q, Z, r_t}\left[J_{\mathrm{audit}}\{\mathcal{P}^\dagger(U_1, A; Z, B_t, r_t)\} - J_{\mathrm{audit}}\{\mathcal{P}^\dagger(U_0, A; Z, B_t, r_t)\}\right], \tag{7}$$

with common audit arguments suppressed. Report audited utility, failure frequency, and cost separately; any scalar failure penalty or utility weighting is fixed in advance. The fallback gives a well-defined deployment-policy effect, not an estimate restricted to successful formation. An invalid assay remains invalid rather than being relabeled a negative scientific result. For an unobserved endpoint, use the registered missingness analysis and sensitivity bounds; a numerical administrative failure score is not a measured scientific outcome.

Also audit the whole formation-stage systems under a common target assessment:

$$\Delta^{\mathrm{sys}}_{\mathrm{origin}} = \mathbb{E}_{r_f}\left[J_{\mathrm{audit}}(A_1^{\mathrm{f}}) - J_{\mathrm{audit}}(A_0^{\mathrm{f}})\right]. \tag{8}$$

Here A_z^{f} is the formation output, not the common target start. Equation (7) averages over generated improvers; Equation (3) conditions on selected snapshots. The system and improver-origin effects can disagree. Neither is a natural mediation effect or proof of sustained acceleration.

The independent assignment and analysis unit is a formation run. Target starts, tasks, and repeated target processes are nested or crossed observations, not new formation replicates. Use inference at the formation level, preserving any randomized blocks and shared-task structure; justify formation-run counts from pilot variability and a prespecified meaningful effect. Report arm-specific absolute gains, failures, target productivity, and total formation-plus-use cost. Identification additionally requires coherent interventions, a valid common audit, and controlled interference between runs.

Digital instantiation. Form modifiers on source code tasks through a versioned editor API; isolate archives and withhold target audits. Before formation, fix independently constructed starter agents satisfying that API and a no-edit fallback. Evaluate all modifiers on these starts, auditing correctness and resources. Task repetitions refine each modifier's assessment; only new randomized formations increase formation-level replication. This is a DGM-inspired design, not a description of all controls in its published experiment.

Physical instantiation. Randomize independent laboratory campaigns to a recipe-proposal procedure that may revise itself from adjudicated discrepancies or the initial fixed procedure. Both retain the same categories of completed, invalid, and valid-negative records, with common safety, human approval, and scientific confirmation rules. Block batch and instrument effects under equal time and material ceilings. Predefine new-batch targets and an initial-procedure fallback, then assess all formed procedures using independent

samples and blinded endpoints. A common-record study can separately test use of supplied corrections; it cannot replay unexecuted experiments. If assay state and procedure cannot be separated coherently, compare the compatible package and leave the separable improver-origin effect unidentified.

11.3 Baselines should target plausible explanations

Choose controls by claim: fixed agent plus extra search, no persistent state, fixed external improver, and compatible knowledge or method ablations. No memory tests total retained-state value, not method transfer alone; a stronger fixed model tests an external upgrade.

Where interfaces permit, cross knowledge and method inheritance. Their interaction may be essential: deleting required knowledge can break a method rather than identify its contribution. Report each arm's information and compatibility. Digital comparisons can pair tasks and random numbers, although identical seeds do not guarantee determinism. Physical comparisons require randomized allocation, batch or instrument blocking, reference samples, and measurement replication. An irreversible state cannot be cloned; use independent samples or a justified crossover design.

11.4 Separate development, confirmation, and transfer

Development selects updates; confirmation assesses a frozen claim; transfer changes its scope. An access ledger records these roles. Once aggregate results guide repeated selection, nominally held-out data become development feedback even if their contents remain hidden.

Specify the transfer axis: instances, datasets, instruments, laboratories, scientific questions, or methodological regimes. Random compound splits do not establish scaffold, temporal, or cross-assay transfer. Test regressions, calibration, and recognition of invalid experiments as well as positive-candidate yield. Report capability tradeoffs; useful improvement within one laboratory need not be represented as universal transport.

11.5 Repeat the complete process

A complete run includes research, feedback, updating, selection, and inheritance. Repeated outputs or judge ratings from one final model do not replicate this process. Preserve the hierarchy of projects, runs, tasks, and measurements in uncertainty estimates.

Retain failures, regressions, rollbacks, and exhausted budgets. Best-so-far archives are nondecreasing by construction, unlike the current researcher or improver. Report current and best states, discovery times, failure rates, and resources. Justify complete-run counts by the intended effect and precision, not a universal minimum.

11.6 Measure scientific validity jointly with process validity

Scientific audits inspect experiments, provenance, controls, analyses, and conclusion scope; process audits inspect inheritance, later use, comparators, resources, and repeated runs. A validated molecule can arise from an unimproved search procedure, and an efficient improver can optimize a misleading biological proxy.

Keep a fixed reference track for common-distribution comparisons and an open research track for agenda selection. The latter must record alternatives, difficulty, expected value, and all proposed projects. Higher completion among self-selected projects does not alone show better question choice.

11.7 Existing benchmarks as reusable components

These benchmarks can supply tasks, validators, or transfer settings; process controls and longitudinal updates supply the additional claim identification. Near-ceiling scores may conceal differences in cost, reliability, and harder decisions, so preserve those outcomes rather than interpreting saturation as general research competence.

11.8 Resource accounting and the minimum report

Use a resource vector rather than token counts alone:

$$B = (B_{\text{compute}}, B_{\text{training}}, B_{\text{experiments}}, B_{\text{materials}}, B_{\text{time}}, B_{\text{human}}).$$

Choose the claim, then choose the audit

Claim	Control or evidence	Endpoint and independent unit	Inference limit
S Scientific output	Claim-matched proof, computation, observation, intervention, or replication.	**Endpoint:** the scientific result. **Unit:** follows the measurement, experimental design, or formal claim.	Does not establish an improved research process.
F Feedback use	Consumption trace; for influence, a valid intervention on feedback content, admission, availability, or targeting.	**Endpoint:** consumption, decision, or outcome. **Influence unit:** independently assigned feedback exposure; retain clustering.	Availability is not use; use is not retained benefit.
C Retained capability	Retained versus compatible no-retention or information-matched control; protect target assessment.	**Endpoint:** later utility. **Selected artifacts:** independent target instances or batches. **Producing procedure:** independent full adaptation runs or campaigns; assessments nested or crossed.	Target repetitions do not replicate artifact construction. Benefit is not improver gain.
U Improver gain	Old and new improvers on common compatible starts and matched constraints; report both absolute gains.	**Endpoint:** quality of generated updates. **Unit:** complete update processes; vary starts for method-transfer claims.	A better improver does not establish how it became better.
R Recursive attribution	Intervene on the return path during formation; fix the target beforehand and assess every formed procedure, including failures.	**Endpoint:** the specified later benefit. **Unit:** independent formation runs; later tests nested or crossed.	Attribution to one path does not establish indefinite acceleration.

Shared requirements: claim-relative scientific validation, protected evaluation, explicit resources, provenance, failures, and uncertainty.
Read across one row. These five claim families are not levels or a cumulative score.

Figure 3. From claim to audit design. Each row identifies a comparison, endpoint and independent unit, and the inference it cannot supply. The five rows are not a hierarchy. For retained capability, uncertainty conditional on selected artifacts differs from uncertainty about the process that produced them. Improver gain compares update procedures; recursive attribution intervenes on the return path during their formation. All process outcomes still require claim-relative scientific validation.

Table 21. Reusable benchmark components and missing longitudinal designs.

Benchmark	Useful measurement component	Additional design needed for scientific RSI
ScienceAgentBench (Chen et al., 2024)	Executing data-driven scientific tasks	Persistent update episodes, process controls, and repeated full runs
DiscoveryBench (Majumder et al., 2024)	Data-driven discovery and hypothesis evaluation	Separation of discovery quality, retained method, and selection feedback
PaperBench (Starace et al., 2025)	Replicating specified AI research	New-project transfer and improver comparison; replication is not novelty
RSIBench-Data (Meng et al., 2026a)	Data-centric AI research under a defined training setup	Explicit deployment of learned artifacts into the researcher for a claimed return path

Distinguish construction from use, include failed proposals and confirmation, and identify the actually matched constraints. If training is counted within compute, disclose that overlap instead of double-counting it. Compute–laboratory tradeoffs may require Pareto comparisons rather than arbitrary conversion to one cost.

Use the twelve-item protocol in Appendix E, S6 to report the audit, and the nine-field contract to connect its evidence to one claim. The filled RoboChem-Flex example and companion `evidence/update_contract_example_r6.json` demonstrate the mapping, missingness reasons, and version citation. Copy the field definitions, replace the example's evidence and locators, and report each unavailable comparison or consumer trace. This yields a reusable report without treating field completion as an empirical result or requiring every system to establish recursive origin.

11.9 Statistical design for adaptive research systems

Repeated selection can overfit validation feedback without direct training on test labels (Cawley and Talbot, 2010; Dwork et al., 2015a). The Ladder and information-theoretic analyses address adaptive exposure under specific assumptions (Blum and Hardt, 2015; Russo and Zou, 2016); an unrestricted scientific agent does not acquire their guarantees merely through a private leaderboard. Selective and sequential inference address different consequences of data-dependent hypotheses, analyses, and stopping (Fithian et al., 2014; Howard et al., 2021; Johari et al., 2022). Choose fixed-horizon confirmation or an appropriate sequential design before interpreting a favorable stopping point. A confidence sequence does not repair an invalid assay or an undisclosed change of estimand.

Variance arises from complete runs, splits, tuning, implementation, and tasks (Henderson et al., 2018; Bouthillier et al., 2021). Preserve task heterogeneity and hierarchical uncertainty rather than reporting only a benchmark mean (Agarwal et al., 2021). Numerous measurements from few laboratories, patients, or batches remain few independent scientific units; cross-validation can be highly uncertain in such settings (Varoquaux, 2018). Documentation improves inspectability (Pineau et al., 2021), whereas leakage checks and new-distribution evaluation address validity and transport (Kapoor and Narayanan, 2023; Recht et al., 2019; Koh et al., 2021). Track both the evolving method and its evaluation environment.

11.9.1 Control errors across the stream of scientific claims

A valid single-claim test does not control a whole research program. For 100 true-null tests each having type-I error exactly 0.05, the expected number of false rejections is five by linearity, without requiring independence. Increasing attempted claims can therefore increase apparent discoveries without improving quality. This is separate from testing whether the agent outperforms a comparator.

Record claim identifiers, estimands, endpoints, selection and reuse history, initiation and completion times, rules, and status, including abandonment and endpoint changes. Define the reported claim family and whether the target is family-wise error, false-discovery rate, or another decision loss. For a fixed family, valid per-claim levels prespecified to sum to at most α give a conservative family-wise bound by the union bound. Adaptive selection additionally requires validity under the relevant selection history; error allocation cannot repair invalid tests.

For asynchronous programs, Zrnic et al. (2021) address overlapping tests and local dependence through conflict sets, with distinct assumptions and guarantees for modified FDR and FDR. Shared controls, revised hypotheses, and adaptive test initiation must satisfy the chosen procedure's conditions; otherwise the guarantee remains unresolved. Report the actual family and procedure, not just "multiple-testing correction." Within-task policy comparisons do not automatically control errors across tasks.

These requirements concern stochastic claims and comparisons. Attempting many formally checked proofs does not itself require FDR correction; specification, checker validity, and relevance are the pertinent risks. Statistical control likewise cannot repair mislabeled samples or establish observational causality.

11.10 A Common Protocol with Domain-Specific Confirmation

The common design fixes the intervention, available evidence, resources, target population, and audit; the scientific endpoint remains domain-specific. Life-science studies reserve new biological contexts and distinguish assay validity from biological effect. Chemistry preserves reagent, instrument, and execution metadata under declared reaction-family shifts. Materials separate computed properties from physical identification and corrected labels. Energy separates early predictions from final device or lifetime outcomes. Environment uses temporal and geographic tests while recording observing-system changes.

Robotics separates an improved executed policy from a better policy learning or evaluation procedure, with reserved physical conditions. Discovery pipelines link conclusions to executed operations and measurements before testing inherited workflow revisions (Sections 10.5–10.6). Retain execution failures and human interventions; test a claimed recursive return path separately. These are proposed instantiations, not a newly executed benchmark. Every claim record links configuration, comparison, outcome, and scope. Evidence access and result polarity remain distinct: an unread or unreported comparison is not a negative experiment.

12 Resources, Infrastructure, and Collective Scientific Learning

12.1 Improvement Under Finite Resources

12.1.1 Improvement is an allocation problem

A research system allocates resources among immediate discovery, reusable capabilities, and better improvement procedures. The value of each investment depends on its reuse horizon.

If construction costs C_{build} and saves $\delta c > 0$ per comparable task at unchanged utility, nominal break-even is $C_{\text{build}}/\delta c$. This accounting identity excludes uncertainty, discounting, shift, and maintenance. When quality also changes, compare resource–quality frontiers.

Bayesian experimental design and amortized design policies connect information objectives to repeated decisions (Chaloner and Verdinelli, 1995; Foster et al., 2019, 2021). Their gains remain conditional on the generative model and task distribution; efficient simulated design does not establish that the modeled uncertainty is scientifically relevant.

12.1.2 Computation, experimentation, and confirmation run on different clocks

Computation, evidence generation, and independent confirmation have different clocks. Faster code generation does not shorten cell growth, instrument queues, or replication. Parallel equipment also introduces capacity and coordination constraints.

Battery optimization and electrolyte search separate rapid selection feedback from later endpoints (Attia et al., 2020; Dave et al., 2022). Measure validated outcomes, pending claims, and the location of the bottleneck; increased candidate production may only lengthen the confirmation queue.

12.1.3 A finite-horizon allocation formulation

An analytical extension makes the opportunity cost explicit. Index decision events by i, with calendar time s_i and a research deadline T. An action a_i can initiate a discovery experiment, diagnostic measurement, capability update, or confirmation, or wait for a pending result. Its completion time is $c_i = s_i + d_i$, where duration d_i can be uncertain. History $h(s_i)$ contains only information available by s_i, including which jobs are still pending. Let $\mathcal{K}^{\text{confirmed}}(T)$ denote claims that have met their prespecified confirmation criteria by the deadline, and let $b(a_i)$ be the vector of resources committed by action a_i. A finite-horizon policy can be

written as

$$\max_{\pi} \mathbb{E}_{\pi}\left[U_T\left(\mathcal{K}^{\text{confirmed}}(T)\right)\right], \qquad \sum_{i:s_i \le T} b(a_i) \preceq B,$$

subject to shared-capacity constraints such as

$$\sum_{i:s_i \le s < c_i} r_\ell(a_i) \le q_\ell \quad \text{for each resource } \ell \text{ and calendar time } s.$$

Here B is the finite resource budget, $r_\ell(a_i)$ is the amount of resource ℓ occupied while an action is active, and q_ℓ is its available parallel capacity. The budget counts resources committed to actions initiated by T, including work still pending at the deadline; the value function counts only claims confirmed by T. Equipment-hours are additive resource use; they are not calendar duration when jobs run concurrently. The policy satisfies $a_i \sim \pi(\cdot \mid h(s_i))$, so it cannot use results that have not yet arrived. Waiting is an admissible action and consumes no new experimental resource. The state retains pending and censored outcomes without treating them as negative evidence.

Utility U_T and confirmation criteria encode scientific judgments. Report their components and sensitivity to weights; use resource frontiers or stated shadow prices rather than silently pooling incomparable resources. This is an accounting formulation, not a fitted transition model or convergence theorem. Retractions can reduce the value of the confirmed set.

A portable diagnostic method can lower immediate output while improving future work; retraining can instead consume the batch needed to expose a false result. Varying horizon and resource budget makes this opportunity cost testable.

12.1.4 An illustrative decision threshold, not an empirical scaling law

Consider a deliberately simplified laboratory with one sequential resource. Each investigation, including confirmation, consumes three time units and three budget units. A fixed researcher produces a confirmed useful result with probability $p_0 = 0.5$ per investigation. A capability update consumes three units of each resource and, if effective, raises that probability to $p_1 = 0.8$. The update is retained for the remaining investigations; no further learning occurs. Set budget and deadline both to H, give each confirmed result unit value, and assign no residual value to the final capability. These are illustrative assumptions, not fitted parameters from a reviewed system.

$$V_{\text{fixed}}(H) = 0.5\lfloor H/3 \rfloor, \qquad V_{\text{update}}(H) = 0.8 \max\{0, \lfloor (H-3)/3 \rfloor\}.$$

At $H = 6$, updating yields expected value 0.8 versus 1.0; at $H = 12$, 2.4 versus 2.0; at $H = 24$, 5.6 versus 4.0. A better final researcher is therefore not sufficient for a better finite campaign. If the update succeeds with probability q and otherwise leaves p_0 unchanged, the $H = 12$ advantage requires $3[0.5 + 0.3q] > 2$, or $q > 5/9$. Regressions would raise that threshold. Parallel equipment, longer confirmation delays, or value assigned to reuse after the deadline would change the calculation.

This example concerns capability investment. Improver investment additionally requires an estimated distribution of later update benefits and regressions, with its acquisition cost. Complete repeated runs and prospective allocation tests are needed to supply those transitions.

12.2 Changing Environments and Limits to Accumulation

12.2.1 Cumulative gain, acceleration, and open-endedness

Endpoint gain, positive average improvement rate, and acceleration are different observations. Diminishing gains, isolated breakthroughs, and best-so-far archives can produce misleading short trends.

Specify the quantity, resource denominator, baseline, and window for acceleration. Ceiling effects and changing tasks complicate finite differences; gains per token, experiment, and day may disagree. Repeated improvement on a bounded suite does not establish open-endedness.

Updates can increase complexity, break interfaces, amplify evaluator errors, or exhaust validation budgets. Fixed outer constraints may preserve comparability while allowing useful internal adaptation.

12.2.2 Endogenous distribution shift and transport

Research policies create endogenous shift by changing questions, experiments, and instruments. Searching rare chemistry or simulator edge cases can invalidate feedback that was calibrated on familiar inputs.

Transportability, data-fusion, and invariance frameworks motivate asking which mechanisms remain stable, which variables change, and what target evidence is needed (Pearl and Bareinboim, 2014; Bareinboim and Pearl, 2016; Peters et al., 2016). They do not automatically identify transfer effects for scientific agents.

Document positive and negative applicability conditions. Transfer of a batch-effect mechanism is different from reuse of a numerical threshold, and artifact transfer does not show that an improver can recreate the gain.

12.2.3 Forgetting, homogenization, and external evidence

Retain regression tests and explicit tradeoff rules: shared updates may improve diagnosis while damaging execution or calibration. SEAL makes sequential retention an empirical question (Zweiger et al., 2025).

Model-collapse results concern particular recursive data-generation and training conditions (Shumailov et al., 2024). They do not condemn all RSI. External evidence quality, rare-case retention, and the use of evaluator errors as training targets determine the relevant risk.

Agents sharing models, sources, and evaluators can repeat one error while appearing to corroborate it. Independent measurements and competing methods matter more than wording diversity; their contribution to discovery remains testable.

12.3 Infrastructure, Provenance, and Human Roles

12.3.1 Provenance should connect claims to capabilities

Provenance and dependency-directed revision are established ideas. PROV-DM represents entities, activities, usage, derivation, revision, and invalidation (Moreau and Missier, 2013); truth maintenance tracks reasons for revising beliefs (Doyle, 1979); data provenance distinguishes source location from the data supporting a derived result (Buneman et al., 2001). Adaptive-workflow provenance is also an explicit research topic (Stage et al., 2025). Alnasir's scientific computing review already covers persistent model state and the provenance of model-mediated decisions (Alnasir, 2026). PROV-AGENT already connects model invocations, prompts, responses, and agent tools to broader workflow provenance and supplies a runtime implementation (Souza et al., 2025). We therefore claim neither a new provenance theory nor priority for tracing adaptive scientific workflows.

The narrower reporting proposal is to join those records to the five claim types in Table 3. A retained artifact should identify its producer, actual consumer, version, evidence status, and the benefit or process claim that remains to be revalidated. Usage, derivation, and mere citation must remain distinct. After a correction, traversal over verified dependencies identifies a review set; it does not identify an effect size or dictate that every descendant be deleted. Each affected object needs a disposition: retain with independent support, lower confidence, revalidate, restrict use, or withdraw. Unknown consumption edges stay unknown. PROV invalidation denotes cessation of an entity's availability in that model; we separately record evidential status and revalidation need rather than treating a provenance relation as a truth judgment.

Section 10.7 implements this distinction at the published-source and review-artifact levels. The A-Lab case lacks the measurement-label-to-policy-state trace needed for actual capability repair. Parameter-level influence may require targeted evaluation or retraining; removing a document from retrieval does not demonstrate that repair. Appendix A specifies the inherited concepts and the limited addition.

A hereditary-disease laboratory reclassified 94,453 of 2,051,736 observed variants (4.60%) over 8.5 years, with 105,172 events; 61 events (0.06%) crossed the pathogenic–benign spectrum (Metz et al., 2024). Most changes resolved uncertainty rather than reversing interpretation. This motivates review of cached assertions and their descendants, but does not show that any surveyed agent used an outdated variant. Correcting a record also cannot restore consumed samples or elapsed opportunities.

12.3.2 Editable science requires protected evaluation

Internal evaluators and analysis methods may change, while failed-trial records and final audit criteria remain protected and versioned. This separation preserves an interpretable comparison without banning evaluator learning.

Table 22. Failure propagation, controls, and verification tests.

Failure path	How it can propagate	Control and corresponding test
A provisional explanation becomes a remembered fact	Retrieval, synthetic data, and later decisions inherit false certainty	Typed evidence and provenance; test revision after contradictory evidence
An analysis exploits a simulator or scoring artifact	The artifact is reinforced as a successful research method	External or perturbed validation; test performance when the artifact is removed
An evolving evaluator favors its own outputs	Selection and training amplify a shared preference	Common protected audit; cross-evaluate old and new outputs
A tool or model update breaks a retained skill	Incompatibility appears as scientific failure	Versioned interfaces; replay validated tasks after updates
A corrected result remains embedded in learned behavior	Downstream studies reuse discredited evidence	Dependency inspection and capability retesting after correction
Agents converge on the same unsupported explanation	Shared memory turns agreement into apparent corroboration	Track evidence ancestry; test independent measurements and dissenting procedures

These controls should be tested through deliberately introduced faults and realistic recovery tasks. A provenance system that records every action but cannot locate affected capabilities after a correction has not solved the operational problem.

12.3.3 Computational rollback and physical irreversibility

Software checkpoints can often be restored; samples, interventions, and elapsed opportunities cannot. Laboratory rollback is therefore recovery for future actions. Charge its physical resources and preserve the interventions already made.

Record human input by content and time: priority setting, evidence adjudication, authorization, repair, or confirmation. A generic "human in the loop" label obscures attribution even when collaboration is useful.

12.3.4 Collective improvement and institutional independence

Shared laboratory or agent methods expand the unit of analysis beyond one run. Trace common components and evidence so that redistribution is not counted as independent confirmation.

Acceptance, promotion, and reversal of a method involve judgments about importance and uncertainty. Automation should expose their evidence and authority rather than conceal them in one reward.

12.3.5 From a demonstrator to a maintained research service

Organizational reuse requires an executable interface, operating conditions, versioned evidence, resource expectations, and a failure detector. This design recommendation complements SDL concerns about scalability, generalization, and provenance (Canty and Abolhasani, 2026); it is not a capability established for all reviewed platforms.

Keep three promotions inspectable: observation to knowledge, procedure to reusable capability, and capability to update mechanism. Shared infrastructure can reduce review costs but correlate errors across laboratories. Report evidence ancestry and validation independence alongside deployment count.

13 A Research Agenda for Cumulative Scientific Intelligence

The completed work comprises configuration-level extraction, explicit claim and interface rules, cross-domain synthesis, and scoped dependency replay. These support an inspectable review, not established coding

reliability or improved scientific-agent performance. Two validation programs follow: testing the review instrument and testing the scientific mechanisms it motivates.

13.1 Validate the Review Instrument on Independent Cases

Freeze the codebook and source versions, then have independent domain readers code cases not used to develop the rules. Preserve original disagreements, alternative boundaries, and adjudication reasons. Compare the complete contract with existing RSI coding and provenance-based analysis on the same cases: which warranted conclusions change, which errors are detected, and at what reading cost? Reconstruct actual measurement-to-memory or training-to-decision consumption paths where records permit. These tasks test reliability and added analytical value; they cannot establish recursive gains in the surveyed systems.

13.2 Establish When Reliable Inheritance Helps

H2 tests whether source-grounded applicability conditions preserve useful method transfer under a declared shift. For diagnosis, vary disturbances requiring identical or incompatible repairs and compare explicit diagnosis, direct learning, and uncertainty-aware updates at matched information and cost. The outcome is useful subsequent decisions, not explanatory plausibility.

H3 provides a direct test of dependency-aware maintenance after a scientific correction. Give each arm the same adjudicated correction and compare source-record repair, dependency-aware review, and an information-matched maintained-library or retraining baseline. Prespecify affected artifacts, independently supported controls, fresh target tasks, and acceptable capability-loss and resource bounds. Measure obsolete recommendations, retained scientific utility, recovery time, and total maintenance cost, including the human effort needed to establish reference dependencies. An advantage requires reduced invalid carryover without unacceptable loss of supported capability; record repair alone is insufficient. Sufficiently precise results excluding the prespecified advantage would count against the proposed benefit in the tested setting, whereas wide uncertainty would remain inconclusive. This study tests reliable inheritance. Recursive-origin claims additionally require intervention during formation of the maintenance procedure.

13.3 Identify Effective Improvers and Their Origin

Cross earlier and later improvers with independently constructed compatible starts; report absolute gains and start-specific harms. STOP's positive and degrading variants, MetaRSI's common-start advantage, and AIDE^2's inconclusive ignition test motivate these conditional comparisons (Zelikman et al., 2024; Tan et al., 2026; Srikanth et al., 2026). H1 and H4 test allocation and editable-scope moderators. Common-data and interactive studies answer different questions about evidence use and acquisition.

For recursive origin, randomize the named return path during formation and evaluate every formed procedure on targets fixed beforehand (Section 11.2.2). Preserve failed formations and compatibility failures. A bounded origin effect is distinct from sustained acceleration, which requires a rate estimand and an adequate time/resource horizon.

13.4 Connect Process Gains to Long-Term Scientific Value

Track confirmed utility, latency, retention, and total formation-plus-use cost across proposed, abandoned, completed, and confirmed investigations. Pair fixed-reference evaluation with open question selection, and define the claim family and error criterion. Delayed endpoints and environmental drift test whether an initial advantage survives confirmation and change. Appendix D supplies complementary falsification designs.

14 Conclusion

Scientific improvement claims require both a valid outcome and an interpretable inheritance comparison. Across the selected configurations, feedback consumption, retained benefit, improver effectiveness, and recursive attribution receive different kinds of support. Positive, degrading, and inconclusive comparisons therefore cannot be compressed into one verdict on whether RSI exists.

The survey connects these evaluation distinctions to a further question: what remains warranted when supporting evidence or operating conditions change? The update contract and versioned records link evidence to its consumer, comparison, and correction obligations. Worked cases locate supported revisions and the missing execution edges beyond which effects remain unknown. Cross-domain synthesis identifies acquisition-dependent observations, context-sensitive validity, compatibility, and delayed confirmation as constraints on interpreting inherited gains.

The survey makes scientific improvement claims inspectable across changes in models, tasks, and evidence. Its claim distinctions, update contract, and comparison designs provide a shared language for asking what improved, under which conditions, and what remains supported after revision. The current contribution is a critical synthesis and proposed methodological resource; independent application and prospective tests are needed to establish its reliability and usefulness. Progress should be assessed through confirmed scientific value, retained benefits and harms, and complete resource accounting, without treating bounded gains as evidence of indefinite recursive acceleration.

A Intellectual Lineage

Scientific RSI draws on intellectual traditions that should be connected without being conflated. Good's argument that a sufficiently capable machine designer could produce better machines establishes a conceptual route to recursive improvement; it does not specify a realizable search process or demonstrate sustained growth (Good, 1965). The Gödel machine supplies a different kind of claim: self-modification is licensed by a proof relative to the machine's formal assumptions and utility specification (Schmidhuber, 2003). Neither argument establishes that a practical language agent can improve scientific judgment indefinitely. The former is a possibility argument; the latter is a conditional formal construction. Empirical improvement of a bounded research workflow is a third category requiring measured outcomes.

A complementary lineage concerns how inquiry corrects itself. Platt's strong inference emphasizes competing hypotheses and experiments whose outcomes discriminate among them (Platt, 1964). Box treats scientific learning as repeated confrontation between models and observations, with discrepancies motivating revision (Box, 1976). These accounts illuminate why unsuccessful predictions and negative results can be valuable. They do not prescribe one universally optimal scientific workflow. In complex observational settings, several explanations may survive the same evidence, and an apparent refutation may instead implicate a measurement procedure or auxiliary assumption. For scientific agents, the relevant design principle is therefore to preserve alternatives and expose the inferential consequences of evidence, rather than to convert every unsuccessful experiment into a generic instruction to try again.

Active learning makes part of this principle computational. MacKay derives different selection rules for different targets of information and explicitly identifies the dependence on an appropriate hypothesis space (MacKay, 1992). Cohn and colleagues likewise study informative data selection through particular statistical models (Cohn et al., 1996). These formulations make an essential distinction visible: an experiment can improve knowledge without improving a task score, while an apparently valuable observation can be informative only under a misspecified model. An agent that repeatedly selects data to increase its own confidence may consequently become more certain without becoming more accurate. Updating beliefs within a fixed hypothesis space should be distinguished from detecting that the space itself needs revision.

Meta-learning offers another precise connection. A learned optimizer makes parameter updates the output of a trained model (Andrychowicz et al., 2016). RL^2 and learning-to-reinforcement-learn systems place rapid adaptation in recurrent dynamics while a slower outer procedure trains those dynamics (Duan et al., 2016; Wang et al., 2016). MAML and Reptile learn initial conditions that facilitate subsequent adaptation (Finn et al., 2017; Nichol et al., 2018). These works demonstrate different ways of learning *how to adapt*, but the outer training algorithm need not become an object of autonomous revision. Multiple optimization timescales therefore do not, by themselves, establish recursive self-improvement. The substantive questions are what state changes, what procedure produces that change, and whether feedback can also change that procedure. The claim and interface rules in Section 3.1.4 and the comparisons in Section 11.2 specify those questions without treating meta-learning as a lower rung of a recursion hierarchy.

This distinction also prevents an overly narrow definition based on source-code editing. A system may acquire a transferable experimental strategy through memory or parameter updates without editing its program. Conversely, extensive code changes may merely fit a fixed benchmark. Recursion concerns the causal organization of adaptation; beneficial improvement concerns its measured consequences. Neither entails the other. A useful scientific claim must therefore specify the scope of the learned capability, the conditions under which it transfers, and the resources required to obtain it.

The resulting foundations support three separable propositions. First, a system can learn from scientific observations. Second, it can learn a more effective procedure for acquiring or interpreting those observations. Third, it can improve the process that produces such procedural changes. Evidence for one proposition should not be silently promoted into evidence for the next. This layered reading preserves the contributions of active learning, meta-learning, and autonomous experimentation while reserving stronger RSI claims for demonstrations that identify an altered improvement mechanism and establish its downstream scientific value.

Provenance and revisable support. Doyle's truth-maintenance formulation records reasons for beliefs and revises them when assumptions change (Doyle, 1979). Buneman et al. distinguish why-provenance, concerning contributing data, from where-provenance, concerning source locations (Buneman et al., 2001).

Neither a bibliographic citation nor a dependency path is a causal estimate of scientific benefit. PROV-DM supplies a general vocabulary for entities, activities, derivations, revisions, and provenance bundles (Moreau and Missier, 2013). Our record types reuse these ideas; the manuscript does not introduce a rival formal provenance semantics or implement a complete truth-maintenance engine.

Nearest workflow accounts and the remaining distinction. Stage et al. review prospective, retrospective, and evolution provenance, and examine recording runtime adaptations (Stage et al., 2025). Alnasir explicitly treats persistent scientific decision state and the model version, inputs, rule, and control action behind an adaptive branch (Alnasir, 2026). PROV-AGENT extends W3C PROV to link agent interactions and downstream workflow objects in an implemented system (Souza et al., 2025). Controlled knowledge updating further links evidential strength, scope, persistence, and the choice of update substrate, and distinguishes update selection, execution, and maintenance (Chavira Juárez et al., 2026). Its attention to delayed contradiction and revisable support overlaps the maintenance problem considered here. These accounts preclude claiming that capability provenance or evidence-sensitive persistent updating is new. This survey applies them to a more specific evidence-audit use: connect a documented dependency to the scientific output, feedback-use, retained-benefit, improver-gain, or recursive-attribution claim that may need reassessment. A changed source can trigger review without showing that downstream behavior changed. The source-level A-Lab graph and executed review-artifact recoding in Section 10.7 illustrate that limited use. They do not establish a general scientific-agent repair algorithm, its completeness, or its efficiency.

B Scientific Validity and Measurement

B.1 Negative evidence and unsuccessful execution

A useful scientific agent should distinguish at least four outcomes: evidence supporting a claim, evidence contradicting it, evidence insufficient to discriminate, and an execution whose validity is unresolved. These categories refer to a specified claim and measurement design. An effect can be smaller than hypothesized without being zero; an inconclusive estimate does not demonstrate equivalence. Likewise, an invalid experiment should not be encoded as a negative scientific result.

ADMET-EvO makes this distinction operational by retaining endpoint constraints, diagnostic state, and intervention history, with separate evidential outcomes. Deterministic evaluation and state-transition components record these outcomes, while the retained history can condition later planning. The reported configuration benefits do not by themselves isolate an improvement in the method used to select later interventions (Zhou et al., 2026). More generally, the design illustrates how epistemic status can influence an agent's next action rather than appearing only in its final report.

Learning from negative evidence requires an incentive compatible with scientific discrimination. A policy rewarded only for supported hypotheses may select easy confirmations or avoid risky measurements. A policy rewarded only for surprise may overvalue instrument faults. A useful objective can combine expected information, experimental validity, scientific relevance, and cost. The precise weights depend on the research program; no universal scalar can encode all scientific values. The evaluation should consequently preserve component outcomes rather than hiding disagreement inside one score.

B.2 Measurement changes can masquerade as intelligence gains

An autonomous scientist can change both its research decisions and the means used to evaluate them. Improved assay sensitivity, revised preprocessing, or a new judge can make outputs appear better without changing underlying capability. Such changes may be legitimate scientific advances, but they alter the measurement instrument. Evaluation must therefore distinguish improving the researcher from improving the ruler used to measure the researcher.

A defensible design retains measurement versions and re-evaluates old and new artifacts under common standards where feasible. Physical instruments require calibration and reference samples; computational pipelines require controlled versions and preserved raw data. If a new measurement exposes errors in the old one, the correct result may be a revision of the performance history rather than a smooth upward trajectory. Correcting inherited scientific records is part of progress.

A-Lab provides a concrete reason to retain this possibility. Its original article and subsequent correction should be read as a linked record, including revised synthesis and novelty assessments (Szymanski et al., 2023, 2026). The general implication is not that autonomous experimentation is unreliable. It is that validation and correction remain necessary even when an experimental workflow is highly automated. In a learning system, the correction must also reach any capabilities learned from the original assessment.

B.3 Evaluator independence has several meanings

An evaluator may be independent in organization, instrumentation, model family, data access, or selection history. These are different properties. A measurement from another laboratory can still be a development signal if the agent repeatedly uses it to choose modifications. A second language model can share systematic errors with the first. Agreement among generated reports is not equivalent to independent replication of an empirical finding.

For this reason, development validation and final confirmation must have distinct roles. Development evidence may be rich and interactive; its function is to improve candidates. A final audit estimates a declared claim about frozen artifacts or complete processes. Repeatedly tuning to audit results turns the audit into development, even if its inputs are not directly inserted into training. Adaptive data-analysis research provides the relevant statistical warning and, under specific conditions, methods for controlled reuse (Dwork et al., 2015b). In expensive science, the solution may be a reserved confirmation batch rather than a large conventional test set.

B.4 Scientific value is not exhausted by correctness

Correctness, novelty, explanatory value, practical usefulness, and research importance are related but non-equivalent. Reproducing a known result can be valuable infrastructure; a novel correlation can be scientifically fragile; and an elegant proof of a minor lemma may be less useful than a robust experimental protocol. A survey of scientific RSI should avoid treating benchmark success as a complete measure of discovery.

The distinction is especially important when agents choose their own problems. Higher success rates can result from better judgment or from retreat to easier questions. Evaluation should therefore combine a fixed reference track, which identifies capability change, with an open research track, which assesses question selection and discovery value. The latter requires records of rejected proposals and failed investigations, not merely polished successful outputs. Scientific significance may still require expert judgment, but that judgment should be attached to explicit evidence and uncertainty rather than absorbed into an opaque reward.

C Extended Case Analyses and Adaptive-System Profiles

C.1 Autonomous experimentation: several kinds of closure

The early Robot Scientist work coupled hypothesis generation, experiment selection, physical execution, and elimination of inconsistent hypotheses in yeast functional genomics. Its significance lies in making scientific inference operational: measurements could change which explanation the machine considered plausible and which experiment it conducted next. This was a substantive autonomous inquiry loop, although changing biological hypotheses is different from autonomously changing the hypothesis-generation method (King et al., 2004, 2009). Eve subsequently connected screening, hit confirmation, and lead generation through quantitative structure–activity learning. Its economic analysis concerns the selection of compounds under experimental costs; target inhibition provides a different kind of evidence from clinical efficacy or improvement of the learning algorithm (Williams et al., 2015).

Modern laboratories extend this pattern through more flexible hardware and richer optimization. A mobile robotic chemist demonstrates prolonged physical execution and adaptive experiment selection (Burger et al., 2020). Thin-film experimentation connects a model-based optimizer to changes in composition and processing (MacLeod et al., 2020). CAMEO combines active learning, phase mapping, and property optimization at a synchrotron, including a mechanism for requesting human contributions. Its implemented measurement loop should be distinguished from autonomous synthesis and simulation described as future extensions (Kusne et al., 2020). These systems support valuable claims about experimental throughput, decision quality, and local learning without requiring an assertion of unrestricted autonomy.

Chemical discovery also contains several feedback boundaries. Granda and colleagues use a pretrained spectral classifier to label observations and refit an LDA reactivity model that guides later selections, while

chemists perform product follow-up (Granda et al., 2018, 2019). Koscher and colleagues connect synthesis and characterization to property-model updates and subsequent discovery (Koscher et al., 2023a). The published article documents automatic retraining on accumulated measurements, reuse of updated models in later design, and reduced cross-validated prediction errors in the explored scaffold families. These cases establish retained predictor updates and operational return paths; they distinguish improvement of a chemical predictor from improvement of the procedure that develops it. Section 6 distinguishes this mechanism evidence from predictive, chemical-output, and improver comparisons.

Reusable representations make these loops easier to extend. Coley and colleagues connect AI synthesis planning to a robotic flow platform through recipe files, with expert chemists supplying additional implementation details (Coley et al., 2019). Steiner and colleagues express synthesis through a chemical programming language and modular operations (Steiner et al., 2019). These artifacts are potential units of scientific inheritance: a procedure can be versioned, reused, diagnosed, and eventually modified. The papers establish enabling infrastructure and execution capability; demonstrating autonomous procedural improvement would additionally require identifying which revisions were learned, how they were retained, and whether they improved later experiments.

C.2 Materials discovery: computational and experimental evidence

The materials chapter compares computational stability, synthesis, characterization, and corrected labels directly. Its central cases are GNoME and A-Lab (Merchant et al., 2023; Szymanski et al., 2023, 2026). The correction-dependent maintenance question remains open: revising a scientific record should trigger review of affected predictors and procedures, but the survey does not claim that a complete downstream retraining or revalidation study has already been performed.

C.3 Biology: validated products and changing research capabilities

Biological discovery provides strong examples of experimentally useful AI outputs whose evidential scope should remain explicit. Deep-learning-guided antibiotic screening led to experimentally characterized antibacterial candidates (Stokes et al., 2020). Explainable modeling later connected predicted activity to chemical substructures and supported discovery of a structural class with experimental follow-up (Wong et al., 2024). These studies demonstrate consequential scientific assistance. Yet a model explanation is not automatically a biological mechanism, and activity in experimental systems is not clinical benefit. Neither result, on its own, measures whether the system subsequently became more effective at discovering antibiotics.

Protein research further separates the quality of research tools from the autonomy of research organizations. AlphaFold's blind structure-prediction evaluation offers a useful model for protecting tests from development feedback (Jumper et al., 2021). AlphaFold 3 extends prediction across biomolecular interaction types (Abramson et al., 2024). Such capabilities can substantially strengthen an agent's toolkit, but accurate structures are intermediate scientific representations. The downstream question may concern binding, catalytic activity, expression, toxicity, or behavior in a cellular environment, each requiring its own evidence.

Design studies make this hierarchy visible. ProteinMPNN experimentally rescues previously unsuccessful designs and emphasizes that sequence-recovery metrics do not guarantee correct folding or function (Dauparas et al., 2022). RFdiffusion produces protein designs whose structures and functions are experimentally characterized (Watson et al., 2023). ProGen generates functional protein sequences after family-specific fine-tuning; the reported experiments do not establish equivalent functionality without that fine-tuning (Madani et al., 2023). The shared lesson is that an internal score is useful for selecting candidates, while product validation requires an assay appropriate to the claimed property. Updating a design model under a human-developed protocol should also be distinguished from an agent learning how to develop better protocols.

Language-model research collaborators operate at a broader organizational level. Co-Scientist supports hypothesis generation, debate, and refinement, with selected proposals examined through human-guided experimental work (Gottweis et al., 2026). The Virtual Lab connects interacting AI roles to experimentally assessed nanobody design (Swanson et al., 2025). These studies make coordination and hypothesis quality empirically relevant, but successful selected hypotheses cannot alone identify a persistent improvement in the collaborators. Evaluation must track how retained experience changes later research under comparable starting conditions.

C.4 Research agents and benchmarks: measuring the work being claimed

Tool-using language models change the breadth of tasks that can be orchestrated, but they do not remove the need for domain-specific checks. Coscientist connects language-based planning to chemical research tools and laboratory execution (Boiko et al., 2023). ChemCrow demonstrates chemistry tool integration and exposes a particularly useful evaluation problem: LLM-based assessment can favor fluent answers that expert chemists judge less correct (Bran et al., 2024). A scientific agent evaluated primarily by another language model can therefore optimize the presentation of research more effectively than its substance.

The AI Scientist line extends automation toward experiment implementation, reporting, and review (Lu et al., 2024; Yamada et al., 2025; Lu et al., 2026a). These contributions make end-to-end research workflows testable. A convincing paper or workshop acceptance is an outcome of that workflow, whereas lasting research capability requires comparisons over subsequent tasks. The scientific merit of individual outputs, completeness of the automation, and quality of inherited research procedures should consequently be reported separately.

Benchmarks cover complementary pieces of this problem. SciCode examines scientific programming through problems combining domain knowledge, reasoning, and integrated code generation (Tian et al., 2024). ScienceAgentBench evaluates scientific workflow execution, DiscoveryBench emphasizes data-driven discovery, and PaperBench examines replication of machine-learning research (Chen et al., 2024; Majumder et al., 2024; Starace et al., 2025). CORE-Bench instead focuses on computational reproducibility using provided code and data, including practical execution and interpretation (Siegel et al., 2025). These are useful, differing constructs; reproducing a computational result is not the same as independently replicating a scientific claim with new observations.

A longitudinal benchmark must additionally measure what changes between evaluations. Later work on CORE-Bench explicitly examines shortcuts, out-of-distribution behavior, efficiency, reliability, model and scaffold contributions, and human collaboration after headline accuracy becomes high (Nadgir et al., 2026). This is especially relevant to RSI: a stable success rate can conceal lower cost or greater reliability, while a rising score can conceal exploitation of a benchmark artifact. Evaluating improvement requires specifying the intended capability and the permitted evidence pathway, not simply maximizing the number of solved tasks.

C.5 Scientific agents and recursive mechanisms

The recent literature turns recursive scientific improvement into several different experimental objects. Some systems rewrite the software that conducts research; others train a model from research trajectories, preserve an evolving record of evidence, or improve a predictor that guides experiments. These mechanisms can coexist, but they answer different questions. A useful empirical comparison therefore follows the path from acquired evidence to the retained change and then to its subsequent use. The decisive issue is what crosses the boundary between iterations: a better answer, a reusable procedure, a trained policy, an improved judge, or a combination. This approach recognizes substantive progress without treating every feedback loop as the same intervention.

Editable research machinery. Section 10's configuration panel includes positive and degraded STOP results, DGM's path-related control, and MetaRSI's common-start result alongside quantitative common-start and transfer comparisons for AIDE[2] and Hyperagents, and an execution-level record for Organon (Srikanth et al., 2026; Zhang et al., 2026a; Ahmed et al., 2026). These cases are retained as separate claims: attempted improver gain, transferred package benefit, and activated process revision. A patch that exists, a patch that executes, and a patch that improves later learning should not be conflated.

Coupled model and workflow learning. ScienceBuddy's fixed-model harness comparison and coupled-cycle configuration are discussed separately in the life-sciences chapter and Section 10 (Xue et al., 2026). Their separation prevents a component-specific result from being attributed to the entire return path.

MASS assigns execution, evaluation and workflow optimization to a shared model, then trains from selected multi-agent trajectories. Its fixed-base execution comparison is informative: workflows discovered with the updated generation can remain useful when executed by the original model. Yet the discovery processes used different executors and evaluators, so this comparison does not isolate optimizer changes alone. Moreover, agreement with external model judges is a judgment-consistency measure, not a direct measurement of

scientific truth. The reported cycles support coupled capability transfer with uneven generalization, including regressions on some coding evaluations (Lee et al., 2026).

The common scientific opportunity is to learn from the organization of research, not only its final outputs. A trajectory contains failed tool calls, rejected assumptions, delegation decisions and revisions that disappear from a polished report. Learning from this information could improve later investigations. But trajectory selection creates an attribution problem: successful episodes can contain accidental or scientifically irrelevant actions, while unsuccessful episodes can contain sound decisions under adverse conditions. Thus, better imitation of selected histories and better experimental judgment are related hypotheses that require separate measurements. The relevant counterfactual asks which retained behavior changed decisions when fresh evidence became available.

Persistent scientific state and hypothesis development. DiscoveryRSI connects hypothesis-development records to contribution-weighted learning and harness replay. In the audited v2, assessed hypothesis quality improves over two rounds, and expert assessment provides another view of proposal quality. The work also evaluates a battery-domain extension. Its conclusion nevertheless distinguishes concrete hypotheses and compiled investigation plans from a completed discovery loop. Retrospective correspondence with post-cutoff publications is sensitive to retrieval and proposal format; it does not establish that later scientists adopted the generated ideas. The strongest supported interpretation concerns learning to formulate and refine investigations, with empirical execution and wet-laboratory closure still outstanding (Zhong et al., 2026).

SwarmRSI retains hypotheses, programs, evidence and unresolved implementation needs in a shared research state. Its v2 cross-task experiment compares inherited and empty states on twelve predetermined held-out tasks with matched budgets and three seeds. This is evidence that accumulated state can improve subsequent search. Because that state includes program components, and successive generations have seen more source tasks, it is not a clean estimate of improvement in a transferable research method alone. Its biological studies reanalyze existing datasets; their associations and expert follow-ups do not establish new causal mechanisms or clinical validity (Meng et al., 2026b).

ADMET-EvO adds a useful alternative emphasis: evidence ownership. A language model proposes actions, while deterministic evaluation and state-update components assign verdicts and retain supported, refuted, failed and inconclusive outcomes. Model, representation and data interventions are investigated under endpoint-specific contracts. This supports cumulative, evidence-guided computational research, rather than demonstrated self-modification of the updater. The distinction also applies to task expansion: the forty-three endpoint results presented as positive cases should not be read as an unconditional success rate for arbitrary new research questions (Zhou et al., 2026).

These systems suggest that scientific memory should be analyzed as an intervention with content, provenance and costs. A stored program may transfer implementation work; a stored failure may prevent redundant experiments; a stored methodological rule may change how new problems are approached. Those benefits are all valuable, but they are not interchangeable. Removing all inherited state would erase the mechanism that cumulative science is meant to exploit. Conversely, transferring everything and attributing the resulting advantage exclusively to a better improver would overstate identification. Component-specific transfer studies can estimate narrower effects, while whole-state comparisons estimate the practical value of the complete research history.

Predictive feedback and physical-domain evidence. RSI-Anything makes predictions of action consequences part of candidate selection and later adaptation. Its experimental domains instantiate different portions of this architecture: code repair holds the coder fixed while examining predictor adaptation; laboratory planning follows five retained policy versions from one process; chemistry compares feedback-guided reaction search at matched experiment counts. These provide several kinds of evidence, not repeated demonstrations of one fully activated recursive mechanism. In particular, improved reaction yields support experimental selection efficiency without isolating the enhanced capacity of a successor improvement procedure (Cui et al., 2026).

Sun and colleagues' fluid-control system jointly retains controller code and design guidance while keeping its language model, execution framework and computational-fluid-dynamics evaluator fixed. Frozen-controller perturbation tests address artifact generalization. Continuing three-dimensional search from a retained two-

dimensional controller addresses warm-start reuse. Neither is a comparison of old and new research improvers acting on an identical starting controller. All experiments remain simulations; the scientific significance of their physically reasoned controllers should therefore be separated from claims of demonstrated real-world transfer (Sun et al., 2026).

Across these studies, the strongest warranted conclusion is that several mechanisms for cumulative scientific adaptation now have concrete implementations and bounded empirical support. Their results do not form a single scalar ranking of recursion strength. A system can expose broad self-editing while providing limited causal attribution; another can retain a fixed updater while furnishing carefully separated predictive evidence. Likewise, a physically measured result can validate a proposed action without demonstrating improved scientific methodology, while a computational experiment can directly test a methodological intervention. The evaluation axis and the scientific-evidence axis must therefore be reported together.

Entries summarize claim boundaries rather than assign a maturity rank. "Inconclusive" denotes an attempted comparison; limited isolation denotes a narrower attribution problem.

Table 23. Selected evidence profiles for recent adaptive systems.

System	Retained change	Scientific or task feedback	Process-evidence boundary
STOP (Zelikman et al., 2024)	Executable improvement procedures	Held-out algorithmic tasks	Positive and degraded variants; selected transfer is conditional
DGM (Zhang et al., 2025b)	Executable agent programs	Coding tasks	Frozen-initial-modifier control; unequal realized costs limit attribution
MetaRSI (Tan et al., 2026)	Meta-updated improver	Four digital task domains	Common-start contrast; separate meta-training cost and benchmark scope
Hyperagents (Zhang et al., 2026a)	Task and meta-agent programs	Digital task families	Transfer carries both components
AIDE^2 (Srikanth et al., 2026)	Research-agent implementation	AI research and transfer tasks	Common-start ignition test is inconclusive
ScienceBuddy (Xue et al., 2026)	Harness and model updates	Interactive scientific tasks	Coupled adaptation; reflector gain not isolated
MASS (Lee et al., 2026)	Shared model and workflows	Research-task execution and judgments	Coupled roles limit isolated attribution
DiscoveryRSI (Zhong et al., 2026)	Hypothesis histories and learned policy	Proposal assessment and replay	Plans remain distinct from completed experiments
SwarmRSI (Meng et al., 2026b)	Shared facts and program components	Computational research tasks	Whole-state transfer combines information and methods
Organon (Ahmed et al., 2026)	Tools, search, memory, coordination	Computational discovery	Co-occurring edits limit individual causal effects
ADMET-EvO (Zhou et al., 2026)	Evidence state and predictive configurations	Held-out predictive outcomes	Evidence-conditioned planning; improver gain not isolated
RSI-Anything (Cui et al., 2026)	Domain-dependent predictors and policies	Computational and experimental feedback	Activated return paths differ across domains

C.6 Detailed Robot-Learning Comparisons

These details support the four mechanisms in Section 10.5. Reported quantities retain their source-specific units and evaluation conditions; they are not pooled effects or additional configurations in the audited core panel.

Self-modeling and action selection. Bongard's four-legged robot maintained fifteen candidate self-models and selected, at each of sixteen action cycles, the actuation on which predicted sensor readings disagreed most (Bongard et al., 2006). Across thirty experiments per condition, it recovered a topologically correct model in 13/30 runs versus 7/30 and 8/30 for two random-action baselines; mean model error was 7.31 versus 9.62 cm. After a leg was broken, inferred limb-length error was 2.17 versus 5.60 cm, and the revised model generated a compensating gait. The admissible class was fixed to planar arrangements of eight limbs; the robot could not know whether a model was correct, and the authors report a sim-to-real gait discrepancy. The evidence concerns the specified action rule and model class, not superiority over every task-progress policy.

Demonstrations and self-generated experience. RoboCat's 1.18B-parameter policy was pretrained on 240 tasks, fine-tuned with 500–1000 teleoperated demonstrations of a new task, deployed for collection, then retrained on the collected and original data (Bousmalis et al., 2023). A ResNet-18 detector with reported accuracy around 90% admitted successful episodes; final evaluation was counted by humans (source Appendix F.2). Source Table 8 lists 1,081 successful and 11,155 failed self-generated inverted-pyramid episodes, and 195 successful and 2,171 failed strawberry-lifting episodes. The fine-tuning, detector, and admission rules were human-fixed. One successful iteration was demonstrated, with one model evaluated over 100 episodes per task and no seeds or intervals; before/after numeric values appear only in source Figure 10.

SIME perturbs a diffusion policy's conditioning latent and retains successful portions of attempts in scenarios with less than 50% policy success (Jin et al., 2025b). Over five simulated rounds, selected data continued improving the policy while baseline gains vanished; equal-volume random selection gained 2.1 percentage points versus 11.6 for selection. Within-episode annotation could be manual; automation was exploratory. AutoRT collected 77,000 episodes with 53 robots over seven months, including 73,293 scripted episodes at 21% success and 3,060 teleoperated episodes at 82%, while explicitly decoupling collection from the control policy (Ahn et al., 2024). Downstream RT-1 co-fine-tuning changed picking from 0/24 to 3/24 and wiping from 1/10 to 3/10, providing limited precision about collection value.

Iterated policy and world-model updates. RECAP fits a distributional value function offline, conditions the policy on binarized advantage, and collects human success/quality labels plus occasional teleoperated corrections (Physical Intelligence et al., 2025). Each round restarts value-function and policy fitting from pretrained checkpoints. Two iterations are reported: strict laundry folding reached 97% after two rounds of 600 trajectories each without intervention data; box assembly used 600 autonomous and 360 intervention trials per round on three robots. Reward and objective remain fixed, and the authors describe iterated offline updates (source Section VII). RISE freezes dynamics and progress-value models while updating policy; three bimanual tasks changed from 35/30/35% to 85/85/95%, with 20 autonomous trials each (Yang et al., 2026a). VLAW alternates two rounds of 50 real rollouts per task, world-model fine-tuning, and policy fine-tuning on 500 synthetic rollouts filtered by a reward model fitted once (Guo et al., 2026). Mean score across five DROID categories changed from 0.460 to 0.868, versus 0.752 for filtered behavior cloning over the same two rounds. Its reward threshold was chosen on 40 hand-labeled trajectories, trading eight false positives for twelve false negatives (source Appendix C). Data may persist when weights restart; these one/two-round comparisons do not establish revision of the update rule, and their labels remain human-supplied or fallible.

Reward search and physical selection costs. Eureka samples $K = 16$ reward programs per iteration, trains PPO policies, and returns reward-component values and task fitness to the next prompt; each search has five iterations and each task five independent searches (Ma et al., 2024a). Its 83% headline means exceeding or matching the human reward on 24 of 29 tasks. The 52% average normalized gain clips task scores to $[0, 3]$ and takes the maximum over five searches. Removing reflection reduces the average normalized Isaac score by 28.6%. Benchmark authors define candidate fitness; the outer loop, language model, and PPO hyperparameters (tuned for human rewards, source Appendix F) remain fixed. No hardware experiment appears in the inspected version.

DrEureka proposes randomization ranges under a reward-aware physics prior (Ma et al., 2024b). The best of sixteen configurations reaches 1.83 ± 0.07 m/s versus 1.32 ± 0.44 m/s for the human pipeline; the configuration mean is 1.66 ± 0.25 m/s. Selection requires running all 48 policies on the robot, a limitation identified by the authors. Eurekaverse evolves terrain programs; its human baseline triggered motor-protection faults and

damaged hardware in the ramp condition (Liang et al., 2024). AURA stores curriculum specifications, reward, randomization, and training configurations with two to four sentences of human feedback per run in a vector database (Zhu et al., 2025). Its 99% launch success is per curriculum, whereas the quoted Eureka rates of 12% and 49% are per environment. These are different units; human feedback still closes each AURA iteration.

Fixed adaptation and generative training pipelines. Automatic domain randomization changes each simulator parameter range by a fixed increment, expanding above twenty successes and shrinking below ten in a 240-episode buffer (Akkaya et al., 2019). At equal 13.8-day training time, real-hand consecutive block reorientations were 13.3 ± 3.6 versus 2.7 ± 1.1 for manual randomization, over ten trials each. RoboGen uses GPT-4 to propose tasks, scenes, and reward code, but has no training-outcome return path to proposals; humans judge skill-learning success from video (source Appendix B.2) (Wang et al., 2024). DreamGen fine-tunes a video generator, produces trajectories and pseudo-actions, then co-trains policy (Jang et al., 2025). One pass changed success from 11.2% to 43.2% on fourteen novel behaviors over ten rollouts per checkpoint; the simulated study generated 240,000 trajectories in 54 hours on 1,500 GPUs. These comparisons concern adaptive training or generated data, not an earlier/later generator contrast.

Trial precision and multiplicity. The cited 2026 studies report 15, 20, 20, and 50 trials per condition (Chen et al., 2026a; Yang et al., 2026a; Liang et al., 2026; Guo et al., 2026). For independent Bernoulli outcomes, exact Clopper–Pearson 95% intervals are $[0.56, 0.94]$ for 16/20 and $[0.66, 0.90]$ for 40/50. A half-width of 0.10 near success probability one half needs about 100 trials; distinguishing 70% from 80% under an independent equal-size two-group normal approximation, two-sided $\alpha = 0.05$, and 80% power needs about 293 trials per arm. These are illustrative calculations without multiplicity adjustment, not survey experiments or prescriptions for paired, clustered, multi-task, or sequential designs.

The TRI study used 1,800 blind, randomized, paired real-world rollouts, with fifty per task, policy, and condition (TRI LBM Team et al., 2025). Sequential comparisons with Bonferroni adjustment within each task found the fine-tuned model better than single-task baselines in two of three nominal real-world tasks and three of sixteen simulated tasks. This does not establish global error control across the displayed tasks. Data-normalization choices often dominated architectural choices, illustrating why concurrent changes need separation.

Calibration of cheaper verifiers. SIMPLER's visually matched simulation compares six Google-robot checkpoints and reports mean per-task Pearson correlation 0.924 and mean maximum rank violation 0.056 against roughly 1,500 paired real trials (Li et al., 2024). In AutoEval's WidowX drawer-closing task, that simulator scores OpenVLA 2/50 versus 46/50 on the physical robot (Zhou et al., 2025). AutoEval deploys a vision-language success classifier only above 95% held-out accuracy with a learned reset policy, reports agreement $r = 0.942$ with human-run evaluation and about 850 episodes/day, and misjudges 3/50 episodes in its audited run. RoboArena conducts double-blind pairwise comparisons across seven institutions with a task-aware Bradley–Terry model; under simulated distribution shift, agreement with the oracle is $r = 0.838$ versus 0.692 for a conventional fixed-task protocol (Atreya et al., 2025). These measured calibration results are conditional on their settings. RoboCat's human counts and the TRI design illustrate independent audit safeguards; they do not make every learned training/evaluation verifier interchangeable.

D Detailed Research Design Questions

These proposed experiments complement the main hypotheses rather than report completed validation. Questions 1–5 test prerequisites or reliability mechanisms; Question 6 directly tests improved improvers and recursive origin; Question 7 tests research allocation. A result at one level cannot establish another.

D.1 Can agents distinguish a false hypothesis from a failed experiment?

Vary documented hypothesis errors, implementation faults, instrument failures, and uninformative designs. Compare explicit attribution, direct updating, and uncertainty-aware decisions with equal observations and budgets. The prediction is better follow-up decisions and calibrated uncertainty on new cases, not more plausible explanations. Include regimes where different causes imply the same corrective action. No decision

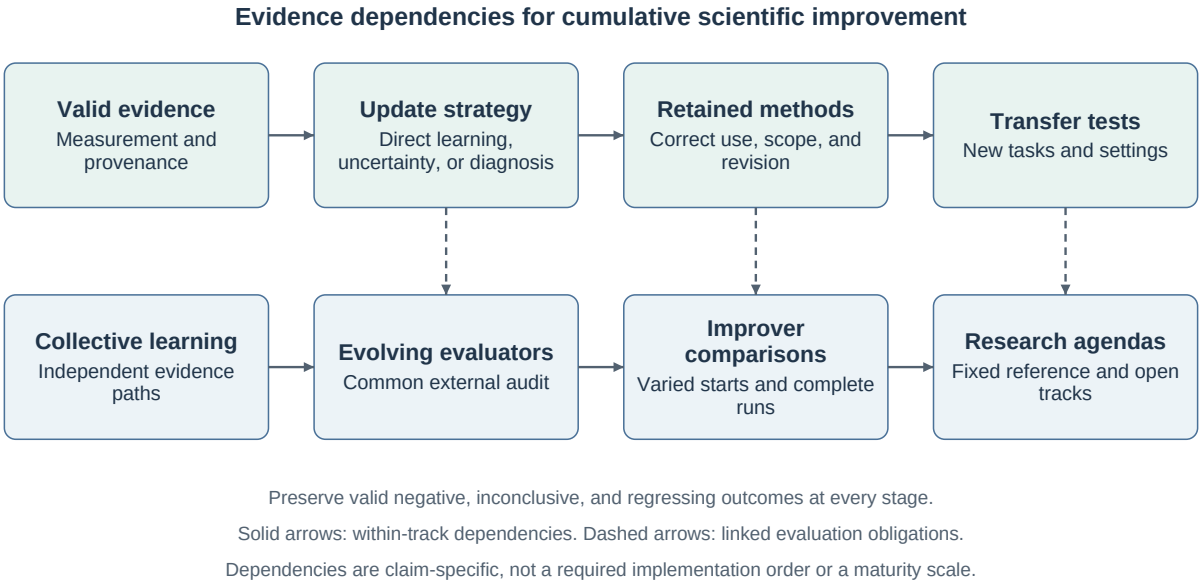


Figure 4. Evidential dependencies among research questions. Valid evidence supports interpretable inheritance and process comparisons; evolving evaluators and collective learning affect both. Agenda learning requires a fixed reference track and an open track. Arrows indicate evidential dependencies, not a prescribed implementation sequence.

benefit, gains requiring privileged fault labels, or resource diversion from useful measurements would constrain or refute the diagnostic advantage. This tests a reliability mechanism, not recursion.

D.2 Can negative results improve learning without inducing conservatism?

In environments with known competing mechanisms, compare objectives rewarding validated discrimination and positive-result frequency. Match exposure to negative evidence; measure false claims, information gained, coverage of valuable risky questions, and cost. A policy that selects predictable low-value experiments fails the intended test even if its apparent error rate falls. An equal or better calibrated positive-result policy defeats the claimed objective advantage within that setting.

D.3 Which research methods transfer across topics and laboratories?

Use H2's information controls and independently varied topics, instruments, datasets, or laboratories. Where compatible, cross factual and procedural inheritance. Measure useful execution, harmful misapplication, and correction after a premise changes. Target-solution reuse is a separate treatment. Conditions that are too narrow, costly, or no better than simpler records constrain the representation claim. A transfer result establishes retained utility; old/new common-start comparisons are additionally needed to claim that the method-construction procedure improved.

D.4 Can an evaluator improve while remaining scientifically calibrated?

Freeze old and revised evaluators learned from externally confirmed errors. Apply them to common candidates and compare the research processes they induce under one protected external audit. Include artifacts exploiting either evaluator and cross-evaluate outputs. Higher internal scores without better confirmed outcomes, or worse selected-case vulnerability despite average gains, refute the intended advantage. A fixed final audit can detect improved evaluator construction; it does not prevent the internal evaluator from adapting.

D.5 Does collective learning improve reliability or accelerate shared error?

Vary shared models, data, memories, and independence of confirmation. Compare selective sharing with evidence ancestry against unrestricted sharing, including an introduced plausible false source. Measure confirmed utility, duplicated work, error spread, correction latency, and total cost. Shared wording changes are not independent evidence. If unrestricted sharing is equally reliable and cheaper, ancestry maintenance

has not justified its cost in that regime. To claim improver gain, compare the learned sharing procedures from common network starts, rather than only their final collective knowledge.

D.6 Can systems become better at producing further improvements?

First perform the common-start comparisons in Section 11.2; distinguish absolute benefit, relative benefit, one-step quality, and transfer across starts. Then use Section 11.2.2's randomized formation protocol if recursive origin is claimed. Keep the target population and failure rule fixed before formation and analyze independent formations, not their target repetitions. Reproducible harm or a sufficiently precise interval excluding a prespecified meaningful gain counts against the tested hypothesis; wide uncertainty is inconclusive. Target leakage or stronger starting systems defeats the attribution. Acceleration further requires a rate comparison across an adequate window, not a positive final difference.

D.7 Can agents learn what is worth investigating?

Run a common reference track alongside open question selection. Assess proposals before their outcomes are known, retain rejected ideas and failed projects, and report difficulty, opportunity costs, and eventual confirmation. Compare learned allocation with fixed policies using the same opportunity set. Easier selected projects or missed prior literature can explain higher completion or apparent novelty. Short-term value judgments remain proxies; bounded research communities with inspectable outcomes offer a tractable initial test.

E Review Methods, Evidence Codebook, and Reporting

Literature inclusion cutoff: 2 November 2026. R4 targeted checks: 9 November 2026. This is a purposive critical survey, with configuration-specific extraction and an AI-assisted internal consistency audit. The source registers preserve earlier access assertions; they are not equivalent to fresh verification. The claim-level `verification_ledger.csv` separates inherited assertions from passages actually checked in the R4 evidence revision. R6 preserves those verification scopes. "Supported" denotes a bounded source claim, not independent replication. No independent human coding or experimental reproduction is reported.

S1. Review design and retrieval

The manuscript was developed as a seed-driven critical review using reference tracing and targeted primary-source inspection. The memory survey cited here is a literature-discovery source, not an empirical demonstration of scientific RSI (Hu et al., 2025). A complete timestamped record of the original search and screening process was not retained in the available project. Consequently, the table below is an inherited representative retrieval register, not a reconstructed exhaustive search log. Earlier review materials recorded targeted checks of A-Lab, STOP, DGM, MetaRSI, Eureka, ENPIRE, SIMPLER, and nearby RSI/physical-AI frameworks. These are retained as historical assertions. The ledger identifies the additional checks performed in the R4 evidence revision, including their restricted field and passage scope.

Retrieval covered arXiv, publisher websites, primary conference proceedings, author and institutional repositories, and PubMed metadata or abstracts. Search-engine results located primary papers and, where direct access failed, exposed indexed passages from those papers. Citation expansion added foundations in learning and experimental design, scientific applications, and evaluation methodology. Inclusion served conceptual coverage and claim checking, rather than estimation of the prevalence of RSI capabilities across a formally sampled population.

The following query families and entry routes were recorded in the earlier manuscript. Their completeness and individual execution dates are not independently verifiable from the retained files.

Search family	Inherited query families / entry routes
General scientific RSI	*"recursive self-improvement" "science" scientific agents*; *"self-evolving" "scientific" agents discovery* (the latter restricted to arXiv)
Experiment selection and feedback	*"learning to experiment" "scientific discovery"*, restricted to arXiv/PMLR; *"autonomous scientific discovery" "negative results" feedback*, restricted to arXiv/Nature
Experimental platforms	*"The Automation of Science" King 2009 Adam*; *"A mobile robotic chemist" Burger 2020 Nature*; exact A-Lab title
Corrections and publication versions	*"s41586-025-09992-y"*; *"Author Correction" "autonomous laboratory" 2026*; *"s41586-026-10644-y"*; arXiv version histories
Memory and adaptation	*"Self-Adapting Language Models" SEAL*; *"MemEvolve" "Meta-Evolution"*; *"Agent Workflow Memory"*; direct MAML and Gödel-machine records
Historical methodology	*"Science and Statistics" "Box" "iteration"*; *"Strong Inference" Platt 1964 science*; *"Advances in Computers" "Volume 6" "1965" Good*
Frontier systems	Direct versioned arXiv HTML for Hyperagents, AIDE^2, ScienceBuddy, MASS, DiscoveryRSI, SwarmRSI, Organon, RSI-Anything, and the evaluation audit

Within retrieved full texts, targeted searches included *reflector*, *Ignition*, *evaluator*, *wet*, *causal*, *temporal*, *execution*, *Cross-task research state inheritance*, and *Reaction optimization*. These searches guided examination of mechanisms, controls, limitations, and appendices. They did not establish that a term absent from a search was absent conceptually from the work.

No exhaustive bibliographic-database screening, independent human dual coding, formal inter-rater reliability estimation, or experimental replication was performed. Executed scripts validate and propagate this review's records; they do not rerun the surveyed scientific systems. The search-engine entries logged in the R4 evidence revision below are not bibliographic-database study counts or an exhaustive exclusion flow. No agreement coefficient is reported. Literature counts describe the assembled bibliography, not the size of the field.

Logged supplementary retrieval and configuration selection. The R4 register (`revision/r4_retrieval_screening.json`) records six executed search-engine queries on 9 November 2026, with 112 returned URL entries and 112 exact-URL-unique entries. The query families address cross-science inheritance, life-science feedback, chemical/material transfer and correction, energy optimization, environmental sampling, and workflow provenance. These are URL entries, not 112 studies: versions and multiple routes to one work remain distinct. Initial dispositions use titles and indexed passages; they are not 112 full-text eligibility decisions. The register preserves each actual query, URL, access route, disposition, and reason. Forty-one entries were not selected for configuration extraction, 26 were already represented, 22 were outside the selected question, 12 were secondary routes, three supplied methodological context, one was screened as adjacent boundary background without a new manuscript claim, one resolved a primary route, two remained access limited, two had unresolved version dates, and two fell outside the cutoff. Access-limited candidates remain unresolved rather than becoming negative evidence. This supplement does not reconstruct the missing original search or establish completeness.

A separate targeted-check log (`revision/r4_primary_checks.json`) records source-specific searches and primary passage inspection, including failed routes. It supports the AlphaEvolve meta-prompt and ScienceRSI transfer/feedback configurations added in R4 and that revision's additional extraction for STOP, MetaRSI, and AIDE^2. The earlier three-query supplement remains historical in `history_R3/retrieval_log.json`; it is not relabeled as an R4 search. The current configuration map contains 35 rows supported by 39 of 53 claim cards; 14 cards remain domain-only. The `coverage_register.json` records these selections. Positive, degraded, inconclusive, component-package, and mechanism-only cases were selected to challenge distinctions among the five claims, not to estimate field prevalence.

S2. Access levels and version handling

Access was recorded at source level. **Targeted full text** means that the paper was opened and the relevant sections inspected; it does not mean that every appendix, artifact, or result was audited. **Abstract and**

metadata support only claims visible at that level. **Indexed primary passages** identify original-paper text recovered through search when direct retrieval was restricted. An abstract-only inspection cannot establish that an experiment, limitation, or control is absent from the full paper.

Version identifiers were retained because titles, author lists, results, and interpretations can change. The A-Lab correction informed the account of its outcomes and novelty scope (Szymanski et al., 2023, 2026). The formal Co-Scientist article was distinguished from its earlier preprint (Gottweis et al., 2026). DiscoveryRSI and SwarmRSI were updated to their inspected second versions. When metadata and document author lists differed, the bibliography used an accurately verified abbreviated author list rather than inventing a reconciliation. Document-generation dates were not substituted for repository submission dates.

The current verification ledger contains eight cards with fresh, targeted R4 primary-field checks and 45 whose earlier verification scope is retained. The R3 ledger and semantic audit are preserved separately. A current internal consistency check of all 53 cards is not fresh primary verification of all 53.

Historical records also require qualification: Good's original reprint carries a 1965 copyright date, whereas the current publisher page labels the volume 1966 (Good, 1965). Such discrepancies are documented rather than concealed. Publication or preprint status indicates source provenance, not automatic validity or invalidity of the scientific claim.

S3. Evidence codebook

The configuration panel applies the following codebook to selected claims; the broader bibliography has not received exhaustive claim-level coding. **Mechanism evidence** identifies what is modified, retained, and actually reused. **Scientific evidence** identifies the proposition about the world or a formal object, together with the measurement, computation, proof, or experiment used to assess it. **Capability and process evidence** asks whether inherited changes improve later research and whether a changed improver generates better subsequent changes. These dimensions can receive different judgments for the same system.

Persistence is considered separately from demonstrated benefit. A stored artifact establishes retention; a later read or execution establishes reuse; a controlled comparison can establish benefit. None alone establishes that the improvement procedure improved. A shared starting state on fresh instances within a continuing project can test persistent benefit. Cross-project transfer provides stronger generalization evidence but is not a definitional requirement for persistence.

Coding begins with the declared system boundary, phase, inherited artifact, and next consumer (Section 3.1.4). The assessor distinguishes changed input or fitted state from a changed action-selection or update rule under that interface. Coupled roles are retained as a package rather than forced into an isolated component effect. Phase changes require distinct configurations: ENPIRE's environment construction and later immutable-interface policy search are separate records. A fixed independent audit is compatible with a positive improver comparison.

Each record has one required claim type. Boundary-case task performance is coded as `capability_performance`, separately from a retained-benefit claim. The other types are: `mechanism`, `scientific_output`, `feedback_use`, `retained_capability`, `improver_gain`, or `recursive_origin`. The last concerns the contribution of the return path, rather than merely its existence. Operational routing of observations is mechanism evidence; feedback-use records in this panel concern interventions on feedback content, admission, or targeting. An F contrast is interpreted only against its stated endpoint: altered benchmark supervision does not by itself establish validity of scientific evidence or a new discovery. A mechanism record can include an associated reported score without reclassifying it as an identified effect of that mechanism.

Claim status is `supported`, `contrary_or_degraded`, `tested_inconclusive`, `not_reported_in_inspected_scope`, `not_assessed`, or `not_applicable`. Supported means that the inspected evidence supports the bounded claim, not independent replication. Each null field separately records whether it was not extracted, not reported in the inspected passages, inaccessible, or inapplicable. These distinctions prevent a missing entry from becoming an unsupported assertion that the paper omitted an experiment.

The matrix adds a separate record for each configuration–claim cell. Each cell links claim IDs and primary locators and separates *reported direction*, *design and effect identification*, *uncertainty*, *source access*, and *interpretation*. A positive direction is not a certificate of causal identification. The compact table uses `ND` when the claim is undetermined in this extraction; it is not a finding that the full paper omitted a test. `NA`

means not assessed for that row. Mechanism-only and joint-package interventions are retained explicitly. `P` denotes a positive process-package or component contrast with unresolved attribution: in U it is not an identified old/new-improver gain; in R it is not an isolated, cost-matched return-path effect. AlphaEvolve's U and R package readings derive from the same ablation and are not independent evidence. The three design panels accompanying the map expose comparison, complete-run unit, confirmation, and uncertainty for all 24 rows with directional F/C/U/R entries. The detailed legend and JSON retain cell-level distinctions; the five-column symbols are a navigation aid.

An unsuccessful significance test is not automatically evidence of equivalence or regression. Invalid measurements, tool failures, and unresolved sample provenance are interpreted as limitations of evidence, rather than scientific refutations. A well-controlled negative experiment may improve knowledge while failing to increase an optimization score.

Evaluation roles are also distinguished. Feedback used to generate, select, or retain candidates is **development feedback**, even when obtained from physical experiments. A **final audit** is protected from that selection process. Agreement among model judges is not treated as experimental replication. AIDE^2's ignition comparison illustrates why "tested but inconclusive" must remain distinct from "not tested" (Srikanth et al., 2026). Comparisons that transfer both task state and an improver establish an effect of the combined intervention unless additional controls isolate the components (Zhang et al., 2026a; Meng et al., 2026b; Chen et al., 2026c).

S4. Evidence-card schema

The accompanying schema requires the fields below. Records are stored by domain, with earlier additions in `revision_claims.json` and `r3_claims.json`, and three R4 additions in `r4_claims.json`; source keys link them to the bibliography. Validation checks required types, status values, source keys, locator presence, and a reason for every null field. The R4 missingness register additionally indexes explicitly incomplete subfields even when a partially informative field is non-null. Its entries describe extraction scope, not a new paper-wide absence audit. These checks concern record integrity, not the truth of the underlying research.

Field	Recorded content
Source identity	Citation key, title, version, publication status, access date
Claim	Unique record ID, configuration, required claim type, and bounded proposition
System boundary	Models, software, memory, tools, instruments, people, external services
Update target	Modified components and components held fixed
Feedback	Origin, validity checks, frequency, and whether available during selection
Inheritance	Retained state and evidence of its subsequent use
Comparison	Starting states, controls, repetitions, resource constraints
Scientific validation	Claim-specific computational, observational, experimental, or formal test
Result	Status, metric, direction, estimate, unit, uncertainty, and supported scope
Interpretation	Limitations and typed reasons for missing fields; not extracted is distinct from not reported
Traceability	Exact section, figure, appendix, artifact, and source link

S5. Compact source-locator matrix

The matrix supplies navigation for selected central claims. It does not replace the papers' experimental descriptions. "Targeted" retains the access meaning defined in S2; repository papers retain their preprint status unless the bibliography separately verifies publication. Unless explicitly marked R4 below, the passage scope is inherited and is not a claim of new retrieval. Short source links open the recorded resource; version suffixes are ordinary text, and the bibliography and evidence records retain the full identifiers.

Source	Source access	Evidence locator
A-Lab (Szymanski et al., 2023, 2026)	Paper; Correction	Revised results, Figs. 2–3; correction passages; current dependency audit distinguished from inherited access
Co-Scientist (Gottweis et al., 2026)	Paper, published	System analysis, real-world validation; targeted passages and earlier v1
AlphaEvolve (Novikov et al., 2025)	Paper, v1	R4: §2.2; §4; Fig. 8 left/caption. Three independent searches; intra-target SD; component-package contrast
DGM (Zhang et al., 2025b)	Paper, v3	§§4.3–4.4; Apps. C.4, E.1; control and cost passages
STOP (Zelikman et al., 2024)	Paper, v3	R4: §§5.1, 5.3; Fig. 4, PDF pp. 5–7. Five GPT-4 searches; 25 per smaller-model condition; one-SE bands
MetaRSI (Tan et al., 2026)	Paper, v1	R4: §5.4; Fig. 11(c), PDF p. 24; App. F. Common start, five seeds; contrast dispersion not reported in these passages
Eureka (Ma et al., 2024a)	Paper, author-hosted	§3.2: five searches, five iterations, sixteen candidates; final policy repeats separate
ENPIRE (Xiao et al., 2026)	Paper, v2	§§2.1–2.2; construction and immutable-interface phases
Leeman et al. (Leeman et al., 2024)	Paper, published	Abstract and concluding assessment; three previously reported compounds
SIMPLER (Li et al., 2024)	Paper, HTML	Table I and evaluation definition; mean per-task Pearson correlation
Luo et al. (Luo et al., 2026)	Paper, v1	Learner/improver roles, common starts, inheritance, and physical costs
Han et al. (Han et al., 2026)	Paper, v1	Functional boundaries and evaluation protocols; targeted full text
Hyperagents (Zhang et al., 2026a)	Paper, v1	§§5.2, 7; Apps. D.3–D.4; targeted
AIDE^2 (Srikanth et al., 2026)	Paper, v1	R4: §3.6; Fig. 7. Three 50-step runs per arm; min–max bands; private grade used in selection
ScienceRSI (Xu et al., 2026)	Paper, v1, posted 22 October	R4: §4.3; Apps. A.6, B.1–B.2, C.1; Tables A4–A5. One selected checkpoint per arm; no across-seed SD/paired CI
ScienceBuddy (Xue et al., 2026)	Paper, v1	§§4.2–4.4, 5; targeted
MASS (Lee et al., 2026)	Paper, v1	§4.2; Apps. C.1, C.3, C.5, E.3; targeted
DiscoveryRSI (Zhong et al., 2026)	Paper, v2	§§3.2–3.5, 6, 7.1, 7.4–7.5, 9; Fig. 5; App. F; targeted
SwarmRSI (Meng et al., 2026b)	Paper, v2	Table 4; Apps. B.4–B.5, B.7/Table 6, C.2, E, F; targeted
Organon (Ahmed et al., 2026)	Paper, v1	§3.2; Apps. A.1, A.11, A.13; targeted
Granda et al. (Granda et al., 2018, 2019)	Corrected article; Correction, 2019	Methods; Figs. 3–5; corrected SI S10–S17, S21–S31, S80–S84; author simulation code

Continued on next page

Source	Source access	Evidence locator
Koscher et al. (Koscher et al., 2023a,b)	Published article; Preprint, ChemRxiv v1 and supplement	Inherited extraction: published pp. 1–12, Figs. 1, 5–6 and p. 11; preprint and SI separately recorded. R3 check: institutional abstract; publisher full text restricted; journal SI not obtained
RSI-Anything (Cui et al., 2026; GenVerse, 2026)	Paper, v1; chemical release dated 12 October	§§2, 3.4, 5–6; Fig. 6/Table 4; release README, selector, and evaluation record; separate configurations
ADMET-EvO (Zhou et al., 2026; OTeam-AI4S, 2026b,a)	Paper, v2; versioned author-release artifacts	§§2.1–2.5, 3.1–3.5; Fig. 2C; SI S1–S2; recipe tables and earlier controls distinguished below
MolRSI (Zhang et al., 2026c)	Paper, v2	Methods 4.1–4.3; SI S1–S4; Fig. 2, Fig. 5, Table S13; no independent raw-data or trace audit
Fluid-control agent (Sun et al., 2026)	Paper, v4	Results; Methods/Algorithm 1; Discussion; App. B; targeted
RSI evaluation audit (Chen et al., 2026c)	Paper, v1	§§3–5, 6.2/Table 5, 7.3; targeted

Chemistry version notes. The inherited extraction from Koscher's formal article supplies the mechanism and numerical results used in Section 6: its reported totals are 294 exploration molecules and nine exploitation molecules, and Figure 5 evaluates predictor versions by cross-validation (Koscher et al., 2023a). The preprint's different totals and its statement of retraining from random initialization remain attributed to that earlier version (Koscher et al., 2023b). Canty's thesis is a further account adapted from the same work, not an independent replication (Canty, 2024). The R3 check reached the institutional abstract, while publisher full-text retrieval was restricted. The earlier formal-article extraction is therefore not represented as freshly reverified. The separate journal supplement has not been inspected; no claim about additional controls in that supplement is made.

For ADMET-EvO, the inherited R3 extraction records agreement between the arXiv v1/v2 scientific text and tables, but the author release contains several evaluation protocols (Zhou et al., 2026; OTeam-AI4S, 2026a). The numerical counterpart of Figure 2C selects on development seeds 1–3 and confirms on fresh validation seeds 4–5 without loading the official test set. Numerical agreement does not establish complete run identity with the paper. The public counterpart of Supplementary Table S1 reports feature/model values as four-seed certification means and data-axis values as best-seed exploratory results. The paper separately reports held-out data-axis confirmation; these results are not interchangeable. Earlier predictive controls exclude external data and use a different seed protocol, whereas 16 of 22 final recipes retain external data, so those controls do not directly evaluate the final configuration (OTeam-AI4S, 2026b; Zhou et al., 2026). These distinctions are preserved in the accompanying versioned source and claim records.

S6. Twelve-item reporting protocol

The protocol synthesizes the survey's scientific-evidence requirements with existing distinctions between product and process evaluation and the risks of adaptive validation (Chen et al., 2026c; Wu et al., 2026; Dwork et al., 2015b). It is a reporting proposal, not a claim that every reviewed system satisfies a common standard. "Not implemented" and "not applicable" are legitimate responses when their consequences are explained.

1. **Specify the claim.** Select scientific output, feedback use, retained-capability benefit, improver gain, or recursive attribution; identify an observable outcome for each claim being tested.
2. **Define the boundary.** Identify fixed, editable, externally upgraded, and human-operated components, including instruments and evaluation services.
3. **Document inheritance.** Record saved states, their versions, subsequent reads or execution, and the exact point where they affect another research cycle.
4. **Preserve scientific evidence.** Connect claims to measurements, conditions, raw data, analysis, and validation; retain negative and uncertain outcomes.

5. **Separate error types.** Distinguish hypothesis failure, implementation failure, instrument malfunction, biased data, and inadequate design; describe their different update consequences.
6. **Provide relevant controls.** Compare against appropriate fixed-system search, absent persistence, or fixed-improver conditions; for feedback claims, intervene on content, admission, or targeting and name the endpoint. Disclose changes in models, tools, and budgets.
7. **Test the improver.** Apply inherited and revised improvers to common compatible starting objects under comparable resources; vary starting objects for method-transfer claims and use a common external assessment.
8. **Measure retention and transfer.** Distinguish fresh instances within a project from new tasks, time periods, datasets, or laboratories; report interference with earlier capabilities.
9. **Protect final evaluation.** Separate development, confirmation, and transfer. Declare the scientific claim family and error-control procedure; track adaptive selection and reuse of audit feedback.
10. **Report complete trajectories and the independent unit.** For effects conditional on selected artifacts, distinguish target-instance or experimental-batch uncertainty from construction uncertainty. For effects of an adaptation or formation procedure, repeat complete runs or campaigns and retain the hierarchy of later assessments. Include failures, reversals, peak and final outcomes, and uncertainty.
11. **Account for resources.** Report training, inference, search, physical experiments, elapsed time, and human work separately; assess whether later gains repay adaptation costs.
12. **Enable retraction.** Version conclusions, evidence, memory, programs, models, and evaluators together, so that invalidated findings trigger examination of dependent capabilities.

Using the protocol: a filled update contract. The following contract evaluates the proposed retained-capability benefit of RoboChem-Flex's cross-substrate reuse (Pilon et al., 2026). It is a worked extraction from the existing review record, not a new laboratory audit. The source-reported consumer and the executed consumer in the review demonstrator are kept separate.

RoboChem-Flex: one claim, nine fields

$\mathcal{B}$ **Boundary.** The RoboChem-Flex platform and its optimizer; humans define the search parameters and perform reported supporting laboratory work. A predictor-only boundary excludes the platform that generated the inherited data.

τ **Phase.** Prior-campaign data are retained into a second-substrate campaign. A longer retention horizon is not established by this case.

E **Feedback.** The prior reaction dataset and reported yield observations; inherited extraction from case study 5 and Fig. 6, pp. 9–10. This edition does not newly audit the raw measurements.

V **Validity.** The source reports an isolated-yield endpoint. The review does not independently reproduce the chemical result or establish a causal transfer benefit.

I **Inheritance.** The retained prior-campaign dataset enters a later multi-task GP; the fitting/acquisition procedure is not thereby shown to be revised.

W **Consumer.** The source reports the later multi-task GP as the data consumer. Record-level transfer identifiers and laboratory read-event logs were not extracted. At review level, the dependency replay consumes the corrected claim card when rebuilding its declared views.

Π **Comparison and unit.** No matched no-transfer comparison was identified in the inspected main text. Four initial plus four further experiments describe one target campaign, not eight independent transfer-effect replicates. Comparative uncertainty and full resource matching are unresolved.

J **Endpoint and scope.** The reported 90% isolated yield after eight experiments is descriptive for the named substrate transfer. It does not identify improvement over a protected no-transfer arm.

$\mathcal{D}$ **Correction.** The review card is coded as mechanism, not demonstrated benefit. Replaying the declared review dependencies changes the benefit views while preserving the yield description; no laboratory retraining or behavioral repair is claimed.

Data dictionary and reuse. The machine-readable companion `evidence/update_contract_example_r6.json` contains this example, the nine field definitions, and explicit missingness reasons; the blank `evidence/update_contract_template_r6.json` supports new reports. Each field has a `value`, `support_scope`, and `source_refs`; unavailable details are `null` with a reason rather than inferred events. `support_scope` distinguishes source reports, review interpretation, executed review-level replay, and unresolved evidence. The file links the unchanged card `EMBODIED-ROBOCHEMFLEX-TRANSFER` and the case `ROBOCHEMFLEX-REVIEW-RECODING`; it adds neither a new claim record nor an independent observation. To reuse the contract, choose a claim, fill its provenance and consumer fields, then specify the comparison, unit, protected endpoint, and correction consequences before assigning an interpretation. A missing comparator leaves the benefit claim open even when reuse is documented.

Version citation. Cite the underlying primary source for the scientific result and this review's frozen R6 companion release (9 November 2026) for the extraction: include the file path, `example_id`, linked claim/case IDs, and the artifact hash in `source_manifest.json`. This identifies an inspectable local release; no public repository or DOI is implied. Independent human coding, raw-data reproduction, and a prospective evaluation of the protocol's usefulness remain outstanding.

S7. Artifact access and executable checks

The PDF embeds `evidence_bundle.zip`. The companion archive contains 53 claim records, 35 configuration rows, and 175 cell judgments, with source registers, schemas, a coverage register, and 24 compact design-panel rows. Thirty-nine cards contribute to the map and 14 are retained only in domain discussions. The evidence README identifies the files; the printed map is generated from the JSON. Counts describe this frozen purposive resource. No public repository or DOI has been assigned.

Section 10.7 documents a published A-Lab correction and a RoboChem-Flex review-card correction with two derived views constructed for the demonstrator. Executed replay follows verified dependencies, preserves remaining support, and stops at unknown experimental links. It checks the declared review rule, not scientific-agent performance or complete experimental provenance.

Structural checks validate schemas, citation keys, IDs, locators, and record links; selected semantic checks flag claim/control mismatches. The build regenerates the PDF and checks embedded-bundle byte identity. Hashes identify the source release. These checks do not certify scientific truth or historical review page correspondence.

S8. Boundary sensitivity and outstanding validation

Section 3.1.4 provides three analytical alternative-boundary readings, not independent human ratings. Dependency examples establish representational feasibility, not superiority over PROV, truth-maintenance, or workflow frameworks.

Section E, S9 reports an executed mechanical source-dependence diagnostic for five explicit synthesis statements, including every single-source removal, a joint STOP/MetaRSI removal, and a documented access-limitation stress test. This checks dependence of the represented argument, not robustness of an independently recoded corpus or every inaccessible source. Independent coding and residual source verification remain outstanding. The companion `revision/external_coding_protocol.md` is explicitly PROPOSED, NOT EXECUTED: it specifies held-out configurations, blinded ratings, disagreement categories, source-based adjudication, and boundary sensitivity. Its `revision/independent_coding_template.csv` is header only. A paired usefulness study should test error detection and decision time against a simpler claim ledger. Until then, this edition claims neither validated superiority nor field prevalence, priority, or completed reliability.

S9. Source-dependence diagnostic

We executed a bounded support-removal analysis of five statements used in the synthesis. Each statement has an explicit set of sufficient row groups in `evidence/synthesis_sensitivity_r4.json`. Removing a bibliographic source conservatively suppresses every panel row citing that source. A statement retains represented support if at least one complete sufficient group remains. This rule tests dependence of this assembled argument; it does not re-estimate effects, re-adjudicate partially supported rows, or simulate an independent review.

Table 27. Dependence of five bounded synthesis statements. Losing represented support is not evidence that a statement is false.

ID	Statement	Supporting rows	Removal result
S1	At least one positive bounded old/new improver contrast is represented.	U01 or U04	Either source alone retains support; removing STOP and MetaRSI together removes it.
S2	A degraded improver contrast is represented.	U02	Removing STOP removes this evidence. Its two smaller-model conditions are not independent sources.
S3	The Gupta/Cell Painting pair has different feedback-result states.	U09 and U10	Removing either study removes this particular contrast; it never identifies a cross-study moderator.
S4	A-Lab supplies the scientific publication-correction example.	U14	Removing any source in this conservatively grouped row suppresses the example. The review-card replay still exists but is not a scientific-agent case.
S5	MolRSI supplies two prior-memory target contexts.	U21 and U22	Removing MolRSI removes both; two rows do not create two independent studies.

The script enumerates every single-source removal in the panel and two explicit stress tests. Removing both positive-improver sources leaves no `U+` row; the added AlphaEvolve process-package contrast does not replace that estimand. Removing Koscher's historically extracted full-text source leaves these five statements supported, but removes that example from the delayed-endpoint/proxy argument. This second test addresses one documented access limitation, not all inaccessible literature.

The results expose concentrated evidence, especially the degradation and external-correction examples. They do not establish robustness of the taxonomy, completeness of retrieval, absence of publication bias, or independence between different source keys. The five propositions and their support sets are review-authored rather than preregistered or independently rated. The executable contribution is transparent dependency accounting; held-out human coding and comparison with simpler review representations remain necessary to evaluate reliability and usefulness.

Interpretive map of cross-domain propositions. Table 28 separates three broader synthesis judgments from the executed S1–S5 diagnostic. It records their source support, inferential limits, and proposed controls. It is an analysis map, not an additional source-removal result, an independent robustness assessment, or a newly executed experiment. The listed sources motivate the distinctions; their heterogeneous outcomes are not pooled to estimate a cross-domain effect.

Table 28. Support, boundaries, and proposed controls for cross-domain synthesis.

Proposition	Support and boundary	Proposed control
Evidence use and acquisition require different comparisons.	U09–U10 perturb supplied biological outcomes; U14 and U17 change the observations acquired. These contrasts do not identify missing physical outcomes or a universal benefit of adaptive acquisition.	Use common data to test evidence use; randomize compatible interactive policies to test acquisition. Record attempted, completed, valid, and confirmed outcomes (Section 10.6.4).
Proxy gains require endpoint-specific confirmation.	U15–U16 separate early selection feedback from later device outcomes; U12 exposes a boundary to extrapolating local prediction gains. These cases do not estimate a general proxy-failure rate.	Protect the claimed endpoint, match delay and resource rules, and retain pending confirmations at the deadline. Test the stated acquisition hypothesis prospectively (H1).
Matching the final consumer does not isolate the procedure that produced it.	Molecular-memory contrasts U21–U22, content-preserving organization U23, and workflow-formation contrast U24 match different stages. No shared effect size or moderator follows from their comparison.	Hold raw history and compatible starts common for a procedure comparison; intervene during formation for recursive origin (Sections 11.2 and 11.2.2).

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